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(19) **United States**(12) **Patent Application Publication****Kawaoka et al.**(10) **Pub. No.: US 2025/0277192 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **RECOMBINANT INFLUENZA VIRUSES
WITH STABILIZED NA**

(60) Provisional application No. 62/965,225, filed on Jan. 24, 2020.

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Seiya Yamayoshi, Kanagawa (JP)(21) Appl. No.: **19/213,713**(22) Filed: **May 20, 2025****Related U.S. Application Data**

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2039/552 (2013.01); **C12N 2760/16121**
(2013.01); **C12N 2760/16134** (2013.01); **C12N**
2760/16151 (2013.01); **C12N 2760/16163**
(2013.01)(57) **ABSTRACT**

Modified influenza virus neuraminidases are described herein that have stabilized NA tetramers which may improve vaccine production efficiency, thus improving the yield of vaccine viruses.

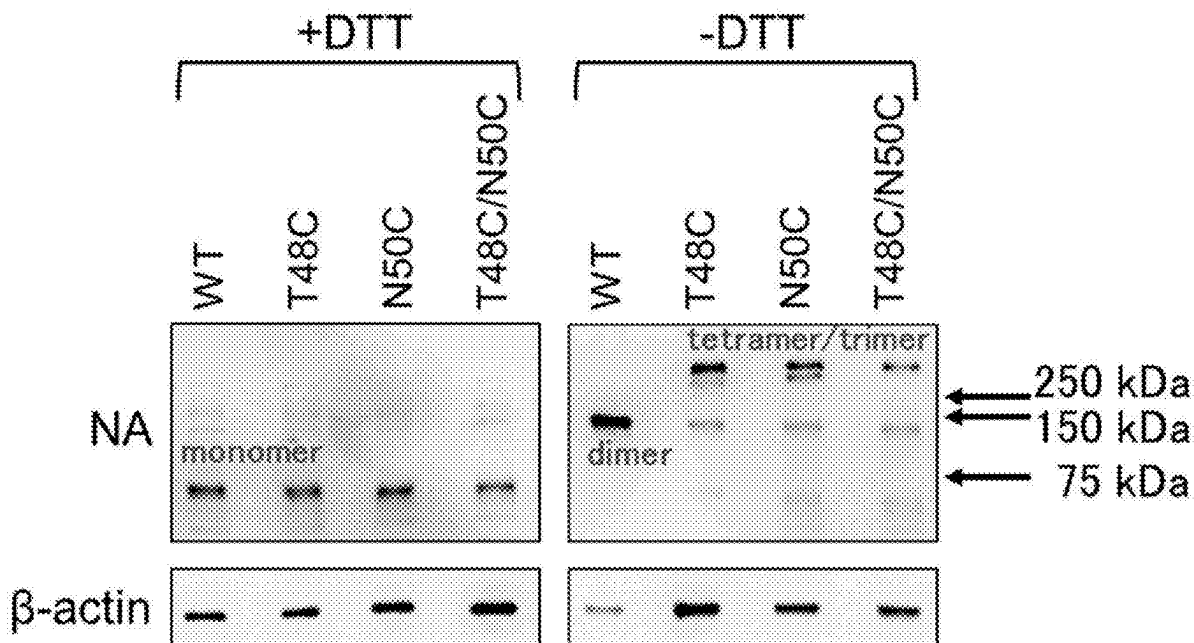
Specification includes a Sequence Listing.

Figure 1

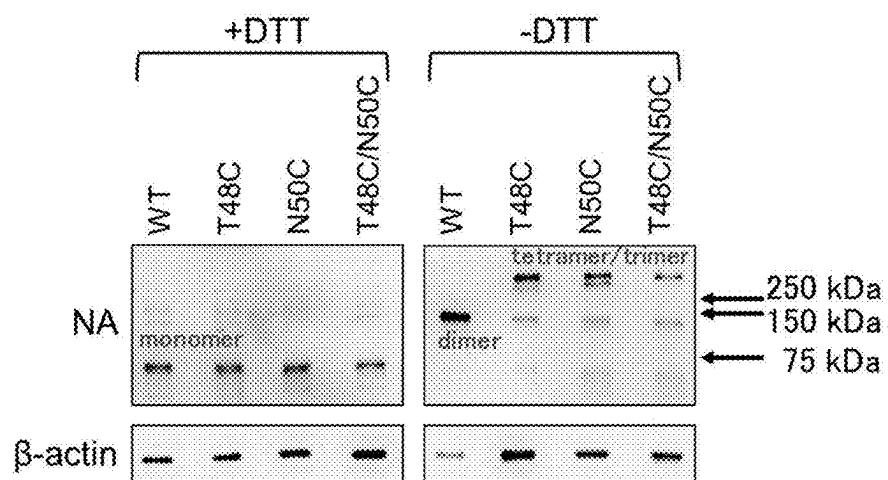


Figure 2

SEQ ID NO:1

MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNN
QVMLCEPTIERNITEIVYLTNTTIEKEICPKLAEYRNWSKPQCNI
GFAPFSKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTL
NNVHSNDIVHDRTPYRTLMMNELGVPPFHLGTKQVCIAWSSSSCHD
GKAWLHVCVTGDDENATASFIYNGRLADSIVSWSKKILRTOESEC
VCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVE
ECSCYPRYPGVRCVCRDNWKGSNRPIVDINIKDYSIVSSYVCSGLV
GDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRT
ISEKLRSYGYESFKVIEGWSNPNSKLQINRQVIVDRGNRSGYSGIFSV
EGKSCINRCFYVELIRGRKQETEVLTNSNSIVVFCGTSGTYGTGSW
PDGADINLMPI

A/Yokohama/2017/03

Figure 3

MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNVTEIVYLTNTTIEKEI
CPKPAEYRNWSKPQCGITGFAPFSKDNSIRLSAGGDIWVTREPVVSCDPDKCYQFALGQGTTLNNVHSNNTVRDRTP
YRTLLMNELGVPPFHLGTRKQVCIAWSSSSSCHDGKAWLHVCITGDDKNATASFIYNGRLVDSVVSWSKDILRTQESECV
CINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSKLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNRPIVIDINIK
DHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWVGRTINETSRLGYETFKVVEGWSN
PKSKLQINRQVIVDRGDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGA
DLNLMPI (SEQ ID NO:2)

Figure 4

Y2017M3L4-NA(32A, 147N, 329D, 347Q, del46--50aa)
(SEQ ID NO:3)

MNPNQKIITIGSVSLTISTICFFMQIAILITAVTLHFKQ
YEFNSPMLCEPTIHERNITEIVYLTNTTIEKEICPKLAE
YRNWSKPQC NITGFAPFSKD NSIRLSAGGDIWVTREP
YVSCDPDKCYQFALGQGTTLNNVHSNNIVHDRTPYR
TLLMNELGVPFHLGTKQVCIWSSSSSCHDGKAWLHV
CVTGDDENATASFIYNGRLADSIVSWSKKILRTQESE
CVCINGTCTVVM TDGSASGKADTKILFIEEGKIVHTS
TLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNRPI
VDINIKDYSIVSSYVCSGLVGDTPRKDDSSSSSHCLD
PNNEEGGQGVKGWAFDDGNDVWMGRTISEKLRSGY
ETFKVIEGWSNPNSKLQINRQVIVDRGNRSGYSGIFS
VEGKSCINRCFYVELIRGRKQETEV LWTSNSIVVFCG
TSGTYGTG SWPDGADINL MPI

Figure 5

N3 (Accession No. AAO62039.1)

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1 mnpnqkiiti gvvnttlsti alligvgnli fntvihekig dhgtvihptt ttpaipncsd
61 tiitynnntvi nnittiitea erlfkpplpl cpfrgffpfh kdnairlgen kdvivtrepy
121 vsodndncws falaqgallg tkhengtikd rtpyrsliaf pigtapvign ykeiciawss
181 sscfdgkewm hvcmtdndnd asaaiiyagr mtdsikswkr dilrtqesec qcidgtcvva
241 vtdgpaansa dhrvywireg rivkyenvpk tkighleecs cyvdidvyci crdnwkgsnr
301 pwmrinneti letgyvcskf hsdtprrpadp stvscdpspn vnggpgvkgf gfkvgndvwl
361 grtmstsgsr gfeilkvaeg winspnhaks vtqtlvsndd wsgysgsfiv ktkacfpqcf
421 yvelirgrpn knddvswtn sivtfcgldn epsgsnwpdg snigfmpk (SEQ ID NO:4)

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N4 (Accession No. AAO62043.1)

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1 mnpnqkiiti gsvsiiltti glllqitalc siwfsihynqv tqtheqpcsn nttynyntff
61 vnvtnvgnny ttviepsapd vvhysagrdl cpirgwapls kdngirigsr gevfivirepf
121 iscsisecrt fflttqgalln dkhsngtvkd rspfrtlmsc pigvapspn srfsesvawsa
181 tacsddpgwl tlgitgpadt avavikyngi itdtlkswkg nimrtqesec vcqdefcytl
241 itdgpsdaga fykilkirkg kivsmkdvda tgfhfeecsc ypsgtdiecv crdnwrqsnr
301 pwirfnsdld yqigyvcsgf fgdnprrpvdg tgschspvnn gkgrygvkgf sfrygdgvwi
361 grtklslesrs gfemvwdang wvstdkdsng vqdiidndnw sgysgsfsir gettgrnctv
421 pcfwvmerig qpkektiws gssiafcgvn sdtgswwpd gallpfidk (SEQ ID NO:5)

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N6 (Accession No. AAO62070.1)

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1 mnpnqkiici satgmtlsbv slligianlg lniglhykmg dtpdvnipnm netnstttii
61 nnhtqnnftn itnliivnkne egtflnltkp lcevnswil skdnairige dahilvtrep
121 ylsodpqgcr mfalsqgttl rgrhangtih drspfralis wemgqapsy nvrvecigws
181 stschedgisr msicmsgann nasavwygg rpvteipswa gniltqese cvchkgicpv
241 vmtdgpannr aatkliyfke gkiqkieela gntqhieecs cygavgvikc icrdnwkgan
301 rpvitidpem mthtskylcs kilttdsrpn dptngnodap itggspdpqv kgfafldren
361 swlgrtiskd srsgyemlkv pnaetdtqsg pishqvivnn qnwsygsgaf idywankecf
421 npefyvelir grpkessvlw tsnsivalcg skerlgswsw hdgaaiiyk (SEQ ID NO:6)

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N7 (Accession No. AIK26357.1)

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1 mnpnqklfal sgvaialsil nlligianvg lmvslhlkgs sdqdknwtct svtqnnttli
61 entyvntttv idketgtakp nylmlnkslc kvegwwvvak dnairfgese qiivtrepyv
121 scdplgckmy alhqgttirn khsngtihr tafrglistp lgsppvvsns dflcvgsst
181 schdgigrmt icvqgnndna tatvyydrll tttiktwagn ilrtqesecv chngtcvvim
241 tdgsassqay tkvlyfhkgf vikeealkgs arhieecscy ghnskvtovc rdnwqganrp
301 vieidmname htsqylctgv ldttsrpsdk smgdonnpit gspgapgvkg fgfldssntw
361 lgrtisprsr sgfemlkipn aetdpnskit ergeivdnnn wsgysgsfid ywdessecyn

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Figure 5 cont.

421 pofyvelirg rpeeakyvgw tsnslialog spisvsgsgsf pdgaqiqyfs (SEQ ID NO:7)

N8 (Accession No. AIK26315.1)

1 mnpnqkiitv gsvslglvvl nillhivsit vtvlvlpngg nakncnetvi reynetvrie
 61 kvtpqhntnv ieylekpesg hfmantealc dakgfapfsk dngirigsrg hvfvirepfv
 121 scsptecrtf fltqgslind khsngtvkdr spyrtlmsve igqspnvyqa rfeavawsat
 181 achdgkkwmt igvtgpdaka vavvhyggip tdvinswagd ilrtqessct ciqgecywvm
 241 tdgpanrqaq yrafkakqgk ivgqteisfn gshieecscy pnegkvecvc rdnwtgtnrp
 301 vlvispdlsy ragylcaglp sdtprgedsq ftgsetspvg nggygvkgfg frqgndvwmg
 361 rtisrtsrsg feilkvrngw vqnskeqikr qvvvdnlkws gysgsftlpv eltkrnculp
 421 cfwvemirgk peektiwtss ssivmcgvdh eiadswhdg ailpfdidkm (SEQ ID NO:8)

N9 (Accession No. ALH21371)

1 mnpnqkilct sataiigai avligianlg lniglhlkpg cncshsqpet tntsqtiinn
 61 yynetnitni qmeertsnf naltkgicti nswhiygkdn avrigessdv lvtrepvpsc
 121 dpdecrfyal sqgttirgkh sngtihdrsq yraliswpls spptvynsrvc ecigwsstsc
 181 hdgksrmsic isgpnnnasa vvwynrrpvt eintwarnil rtqesecvch ngvcpvvftd
 241 gsatgpadtr iyyfkegkil kwesltgtak hieecscyge rtgitctcrd nwqgsnrpvi
 301 qidpvamtht sqyicspvit dnprpndpni gkendpypgn nnnqvgkgsy ldgantwlgr
 361 tistasrsgy emlkvpnalt ddrskpiqqg tivlnadwsg ysgsfmdywa egdcyracfy
 421 velirgpkc dkvwtsnsi vsmcsstefl gqwnwpdgak icyfl (SEQ ID NO:9)

Figure 6

SEQ ID NO:10

MERIKELRNLMSQSRTREILTKTTVDHMAIIKKY
TSGRQEKNPALRMKWMAMKYPITADKRITEM
IPERNEQGQTLWSKMNDAGSDRVMVSPLAVTW
WNRNGPMTNTVHYPKIYKTYFERVERLKHGTF
GPVHFRNQVKIRRRVDINPGHADLSAKEAQDVI
MEVVFPNEVGARILTSESQLTITKEKKEELQDCK
ISPLMVAYMLERELVRKTRFLPVAGGTSSVYIE
VLHLTQGTCTWEQMYTPGGEVKNDDVDQSLIA
ARNIVRRAAVSADPLASLLEMCHSTQIGGIRMV
DILKQNPTEEQAVDICKAAMGLRISSSFSGGFT
FKRTSGSSVKREEEVLTGNLQTLKIRVHEGSEEF
TMVGRRATAILRKATRRLIQLIVSGRDEQSIAEA
IIVAMVFSQEDCMIKAVRGDLNFDVNANQRLNP
MHQLLRHFQKDAKVLFQNWGVEPIDNVMGMIG
ILPDMTPSIEMSMRGVRISKMGVDEYSSTERVV
VSDRFLRVRDQRGNVLLSPEEVSETQGTEKLT
TYSSSMWEINGPESVLVNTYQWIIRNWETVKI
QWSQNPTMLYNKMEFEPFQSLVPKAIRGQYSGF
VRTLFQQMRDVLGTFDTAQIIKLLPFAAAPPKQ
SRMQFSSFTVNVRGSGMRILVRGNSPVFNYNKA
TKRLTVLGKDAGTLTEDPDEGTAGVESAVLRGF
LILGKEDRRYGPALSINELSNLAKGEKANVLIGQ
GDVVLVMKRKRDRDSSILTDSQTATKRIRMAIN

SEQ ID NO:11

MDVNPTLLFLKVPAQNAISTTFPYTGDPPYSHG
TGTGYTMDTVNRTHQYSEKGRWTTNTETGAPQ
LNPIDGPLPEDNEPSGYAQTDCVLEAMAFLEES
HPGIFENSCIETMEVVQQTRVDKLTQGRQTYDW
TLNRNQPAATALANTIEVFRSNGLTANESGRLID
FLKDVME SMKKEEMGITTHFQRKRVRDNMTK
KMITQRTIGKRKQRLNKRGYLIRALTNTMTKD
AERGKLRRAIATPGMQIRGFVYFVETLARSICE
KLEQSGLPVGGNEKKAKLANVVRKMMTNSQDT
ELSFTITGDNTKWNENQNPRMFLAMITYMTRNQ
PEWFRNVLSIAPIMFSNKMARLGKGYMFESKSM

Figure 6 cont.

KLRTQIPAEMLASIDLK YFNDSTRKKIEKIRPLLI
EGTASLSPGMMMGMFNM LSTVLGV SILNLGQK
RYTKTTYWWDGLQSSDD FALIVNAPNHEGIQAG
VDRFYRTCKLLGINMSK KKS YINRTGTFEETSFF
YRYGFVANFSMELPSFG VSGINESADMSIGVTVI
KNNMINNDLGPATAQMA LQLFIKDYRYTYRCH
RGDTQIQTRRSFEIKKL WEQTRSKAGLLVSDGG
PNLYNIRNLHIPEVCLK WELMDEDYQGRLCNPL
NPFVSHKEIESMNNAVM MPAHGP AKNMEYDAV
ATTHSWIPKRNR SILNTSQRGVLEDEQMYQRCC
NLFEKFFPSSSYRRPVG ISSMVEAMVSRARIDAR
IDFESGRIKKEEFTEIM KICSTIEELRRQK

SEQ ID NO:12

MEDFVRQCFNPMIVELAE KTMKEYGEDLKIETN
KFAAICTHLEVCFMYSDF HFINEQGESIIVELGD
PNALLKHRFEIIEGRDRTMA WTVVNSICNTTGA
EKP KFLPDLYDYKENRFIEIGVTRREVHIYYLEK
ANKIKSEKTHIHIFSFTGEEMATR ADYTLDEESR
ARIKTRLFTIRQEMASRGL WDSFRQSERGEETIE
ERFEITGTMRKLADQSLPPNFSSLENFRAYVDGF
EPNGYIEGKLSQMSKEVNARIEPFLKTTPRPLRL
PNGPPCSQRSKFL LMDALKLSIEDPSHEGEGIPL
YDAIKCMRTFFGWKEPNVVKPHEKGINPNYLLS
WKQVLAELQDIENEEKIPKTKNMKKTSQLKWA
LGENMAPEKVDFDDCKDV GDLKQYDSDEPELR
SLASWIQNEFNKACELTDSSWIELDEIGEDVAPI
EHIASMR RNYFTSEVSHCRATEYIMKGVYINTA
LLNASCAAMDDFQLIPMISK CRTKEGRRKTNLY
GFIKGRSHLRNDTDVVNFVSM EFSLTDPRLPH
KWEKYCVLEIGDMLLRSAIGQVSRPMFLYVRTN
GTSKIKMKWGMEMRRCLLQSLQQIESMIEAESS
VKEKDMTKEFFENKSETWP IGESPKGVEESSIGK
VCRTLLAKSVFNSLYASPQLEGFSAESRKL LLI V
QALRDNLEPGTFDLGGLYEAIEECLINDPWVLL
NASWFNSFLTHALS

SEQ ID NO:13

Figure 6 cont.

MASQGTKRSYEQMETDGERQNATEIRASVGKMI
GGIGRFYIQMCTELKLSDYEGRLIQNSLTIERMV
LSAFDERRNKYLEEHPSAGKDPKKTGGPIYRRV
NGKWMRELILYDKEEIRRIWRQANNGDDATAG
LTHMMIWHSNLNDATYQRTRALVRTGMDPRMC
SLMQGSTLPRRSGAAGAAVKGVGTMMELVRM
IKRGINDRNFWRGENGRKTRIAYERM CNILK GK
FQTAAQKAMMDQVRESRNP GNAEFEDLTFLAR
SALILRGSVAHKSCLPACVYGP AVASGYDFERE
GYSLVGIDPFRL LQNSQVYSLIRPNENPAHKSQL
VWMACHSAAFEDLRVLSFIKGTKVVPRGKLSTR
GVQIASNENMETMESSTLELRSRYWAIRTRSGG
NTNQQRASAGQISIQPTFSVQRNLPFDRTTVMA
AFTGNTGRTSDMRTEIIRMMESARPEDVSFQG
RGVFELSDEKAASPIVPSFDMSNEGSYFFGDNA
EEYDN

SEQ ID NO:14

MSLLTEVETYVLSIIPSGPLKAEIAQRLEDVFAG
KNTDLEVLMEWLKTRPILSPLTKGILGFVFTLTV
PSE RGLQRRRFVQNALNGNGDPNNMDKAVKLY
RKLKREITFHGAKEISLSYSAGALASCMGLIYNR
MGAVTTEVAFGLVCATCEQIADSQHRSHRQMV
TTTNPLIRHENRMVLASTTAKAMEQ MAGSSEQA
AEAMEVASQARQM VQAMRTIGTHPSSSAGLKN
DLLENLQAYQKRMGVQMQRFK

Figure 7

N1

1 mnpnqkiiti gsvomtigma nlliqignii siwishsiql gnqnqietcn qsvityennt
 61 wvnqtyvnis ntnfaagqsv vsvklagnss lcpvsgwaiy skdsvrigrs kgdvfvirep
 121 fiscsplecr tffltqgall ndkhsngtik drspyrtilms cpigevpspy nsrfesvaws
 181 asachdginw ltigisgpdn gavavlkynq iitdtikswr nnilrtqese cacvngscft
 241 vmtdgpsngq asykifriek gkivksvemn apnyhyeecs cypdsseitc vcrdnwhgsn
 301 rpwvsfnqnl eyqigyicsg ifgdnpnpnd ktgscgpvss ngangvkgfs fkyngngwig
 361 rtkssissrng femiwdpngw tgtddnfsik qdivginews gysgsfvqhp eltglldcirp
 421 cfwvelirgr pkentiwtsg ssisfcgvns dtvgwswpdg aelpftidk (SEQ ID
 NO:15)

N7

1 mnpnqklifal sgvaiaalsil nlligisnvg lnvslhlkgs sdqdknwtct svtqnnntli
 61 entyvntttv idketgtakp nylmlnkslc kvegwwvvak dnairfgese qiivtrepyv
 121 scdplgckmy alhqgttirn khsngtihr tafrglistp lgsppvvsns dflcvgsst
 181 schdgigrmt icvqgnndna tatvyydrri tttiktwagn ilrtqesecv chngtcovim
 241 tdgsassqay tkvlyfhkgl vikeealkgs arhieecscy ghnskvtcvc rdnwqganrp
 301 vleidmname htsqylctgv ldtstrpsdk smgdcnnpit gspgapgvkg fgfldssntw
 361 lgrtisprsr sgfemlkipn aetdpnskit erqeivdnnn wsgysgsfid ywdessecyn
 421 pcfyvelirg rpeeakyvgw tsnslialcg spisvsggsf pdgaqiqyfs (SEQ ID
 NO:16)

N9

1 mnpnqkilct sataiiigai avligianlg lniglhlkpg cncshsqpet tntsqtiinn
 61 yynetnitni qmeertsrnf nntkglcti nswhiygkdn avrigessdv lvtrepvpsc
 121 dpdecrfyal sqgttirgkh sngtihrsq yrallswpls spptvynsrvc ecigwsstsc
 181 hdgksrmsic isgpnnnasa vvwynrrpva eintwarnil rtqesecvch ngvcppvftd
 241 gsatgpadtr iyyfkegkil kwesltgtak hieecscyge rtgitotord nwqgsnrpvi
 301 qidpvamtht sqyicspvlt dnprpndpni gkcndpypgn nngvkgfsy ldgantwlgr
 361 tistasrsgy emlkvpnalt ddrskpigqg tivlnadwsg ysgsfmdywa egdcyracfy
 421 velirgrpke dkvwwtensi vsmcsstefl gqwnwpdgak ieyfl (SEQ ID
 NO:17)

N2

1 mnpnqkiiti gsvsltisti cffmqiaili ttvtlhfkyq efnspnnqv mlceptiier
 61 niteivyltn ttiekeicpk laeyrnwskp qcnitgfapf skdsvirlsa ggdiwvtrep

Figure 7 cont.

121 yvsdcpdkcy qfalgggttl nnvhsndivh drtpyrtilm nelgvpfhlg tkqvciawss
181 sschdgkawl hvcvtgdden atasfiyngl ladsivswsk kilrtqesec vcingtctvv
241 mtdgsasgka dtkilfieeg kivhtstlsg saqhveecsc yprypgvrcv crdnwkgshr
301 pivdlnikdy sivssyvcsq lvgdtpnknd ssssshcldp nneeggghvk gwafddgndv
361 wmgrtisekl rsgyetfkvi egwsnpnsl qinrqvivdr gnrsqysgif svegkscinr
421 cfyvelirgr kqetevlwts nsivvfogts gtygtgswpd gadinlmpi (SEQ ID
NO:18)

Figure 8

>A/Hong Kong/4801/2014NA(T148K)

ATGAATCCAAATCAAAGATAATAACGATTGGCTCTGTTTCTCTCACCATTTCACAATATGCTTTTTCATGC
AAATTGCCATTTTGATAACTACTGTAACATTGCATTTCAAGCAATATGAATTCAACTCCCCCAAACAACC
AAGTGATGCTGTGTGAACCAACAATAATAGAAAGAAACATAACAGAGATAGTGATTTAACCAACACCACC
ATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACAG
GATTTGCACCTTTCTCTAAGGACAATTCGATCAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAGA
ACCTTATGTGTCATGCGATCCTGACAAGTGTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAACG
TGCATTCAAATAACAAAGTACGTGATAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCTT
TCCATCTGGGGACCAAGCAAGTGTGCATAGCATGGTCCAGCTCAAGTTGTCACGATGGAAAAGCATGGCT
GCATGTTTGTATAACGGGGGATGATAAAAATGCAACTGCTAGCTTCATTTACAATGGGAGGCTGTAGATA
GTGTTGTTTCATGGTCCAAAGATATTCTCAGGACCCAGGAGTCAGAATGCATTTGTATCAATGGAACCTTGT
ACAGTAGTAATGACTGATGGAAGTGCTTCAGGAAAAGCTGATACTAAAATACTATTTCATTGAGGAGGGGA
AAATCGTTCATACTAGCACATTGTCAGGAAGTGCTCAGCATGTCAAGAGTGCTCTTGCTATCCTCGATATC
CTGGTGTCTAGATGTGTCTGCAGAGACAACTGGAAGGGCTCCAATCGGCCCATCGTAGATATAAACATAAA
GGATCATAGCATTGTTTCCAGTTATGTGTGTTTCAGGACTTGTGGAGACACACCCAGAAAAACGACAGC
TCCAGCAGTAGCCATTGTTTGATCCTAACATGAAGAAGGTGGTCATGGAGTGAAAGGCTGGGCCTTT
GATGATGGAAATGACGTGTGGATGGGAAGAACATCAACGAGACGTACGCTTAGGGTATGAAACCTTC
AAAGTCATTGAAGGCTGGTCCAACCCTAAGTCCAAATTGCAGACAAATAGGCAAGTCATAGTTGACAGAG
GTGATAGGTCCGGTTATTCTGGTATTTCTCTGTGAAGGCAAAAGCTGCATAATCGGTGCTTTTATGTG
GAGTTGATTAGGGGAAGAAAAGAGGAACTGAAGTCTGTGGACCTCAAACAGTATTGTGTGTTTTGT
GGCACCTCAGGTACATATGGAACAGGCTCATGGCTGATGGGGCGGACCTCAATCTCATGCCTATATAAGC
TTTCGCAATTTTAGAAAAAACT (SEQ ID NO:19)

> A/Hong Kong/4801/2014NA(T148K, D151E, H347G, T369K)

ATGAATCCAAATCAAAGATAATAACGATTGGCTCTGTTTCTCTCACCATTTCACAATATGCTTTTTCATGC
AAATTGCCATTTTGATAACTACTGTAACATTGCATTTCAAGCAATATGAATTCAACTCCCCCAAACAACC
AAGTGATGCTGTGTGAACCAACAATAATAGAAAGAAACATAACAGAGATAGTGATTTAACCAACACCACC
ATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACAG
GATTTGCACCTTTCTCTAAGGACAATTCGATCAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAGA
ACCTTATGTGTCATGCGATCCTGACAAGTGTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAACG
TGCATTCAAATAACAAAGTACGTGAAAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCTT
TTCCATCTGGGGACCAAGCAAGTGTGCATAGCATGGTCCAGCTCAAGTTGTCACGATGGAAAAGCATGGC
TGCATGTTTGTATAACGGGGGATGATAAAAATGCAACTGCTAGCTTCATTTACAATGGGAGGCTGTAGAT
AGTGTGTTTCATGGTCCAAAGATATTCTCAGGACCCAGGAGTCAGAATGCATTTGTATCAATGGAACCTTG
TACAGTAGTAATGACTGATGGAAGTGCTTCAGGAAAAGCTGATACTAAAATACTATTTCATTGAGGAGGGG

Figure 8 cont.

AAAATCGTTCATACTAGCACATTGTCAGGAAGTGCTCAGCATGTGGAAGAGTGCTCTTGCTATCCTCGATAT
CCTGGTGTGAGATGTGTCTGCAGAGACAACCTGGAAGGGCTCCAATCGGCCCATCGTAGATATAAACATAA
AGGATCATAGCATTGTTTCCAGTTATGTGTGTTTCAGGACTTGTGGAGACACACCCAGAAAAAACGACAG
CTCCAGCAGTAGCCATTGTTTGGATCCTAACAAATGAAGAAGGTGGTGGCGGAGTGAAAGGCTGGGCCTT
TGATGATGGAAATGACGTGTGGATGGGAAGAACAATCAACGAGAAGTCACGCTTAGGGTATGAAACCTT
CAAAGTCATTGAAGGCTGGTCCAACCCTAAGTCCAAATTGCAGACAAATAGGCAAGTCATAGTTGACAGA
GGTGATAGGTCCGGTTATTCTGGTATTTCTCTGTTGAAGGCAAAAGCTGCATAAATCGGTGCTTTTATGT
GGAGTTGATTAGGGGAAGAAAAGAGGAAACTGAAGTCTTGTGGACCTCAAACAGTATTGTTGTGTTTTG
TGGCACCTCAGGTACATATGGAACAGGCTCATGGCCTGATGGGGCGGACCTCAATCTCATGCCTATATAAG
CTTTCGCAATTTTAGAAAAAACT (SEQ ID NO:20)

> A/Alaska/232/2015NA

ATGAATCCAAATCAAAGATAATAACGATTGGCTCTGTTTCTCTCACCATTCCACAATATGCTTCTTCATGC
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AAGTGATGCTGTGTGAACCAACAATAAGAAAGAAACATAACAGAGATAGTGATTGACCAACACCAC
CATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACA
GGATTTGCACCTTTCTAAGGACAATTCGATTAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAG
AACCTTATGTGTCATGCGATCCTGACAAGTGTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAAC
GTGCATTCAAATAACACAGTACGTGATAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCC
TTCCATCTGGGGACCAAGCAAGTGTGCATAGCATGGTCCAGCTCAAGTTGTCACGATGGAAGCATGG
CTGCATGTTTGATAACGGGGGATGATAAAATGCAACTGCTAGCTTCATTACAATGGGAGGCTTGATAGA
TAGTGTGTTTCATGGTCCAAAGATATTCTCAGGACCCAGGAGTCAGAAATGCGTTTGATCAATGGAACCTT
GTACAGTAGTAATGACTGATGGAAATGCTACAGGAAAAGCTGATACTAAAATACTATTCAATTGAGGAGGG
GAAATCGTTCATACTAGCAAATTGTCAGGAAGTGCTCAGCATGTGGAAGAGTGCTCTTGCTATCCTCGAT
ATCCTGGTGTGAGATGTGTCTGCAGAGACAACCTGGAAGGATCAACCGGCCCATCGTAGATATAAACATA
AAGGATCATAGCATTGTTTCCAGTTATGTGTGTTTCAGGACTTGTGGAGACACACCCAGAAAAAACGACA
GCTCCAGCAGTAGCCATTGTTTGAATCCTAACAAATGAAGAAGGTGGTCATGGAGTGAAAGGCTGGGCCT
TTGATGATGGAAATGACGTGTGGATGGGGAGAACAATCAACGAGACGTCACGCTTAGGGTATGAAACCT
TCAAAGTCGTTGAAGGCTGGTCCAACCCTAAGTCCAAATTGCAGATAAATAGGCAAGTCATAGTTGACAG
AGGTGATAGGTCCGGTTATTCTGGTATTTTCTCTGTTGAAGGCAAAAGCTGCATCAATCGGTGCTTTTATGT
GGAGTTGATTAGGGGAAGAAAAGAGGAAACTGAAGTCTTGTGGACCTCAAACAGTATTGTTGTGTTTTG
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(SEQ ID NO:21)

>A/Alaska/232/2015NA(T148K, D151E, N245S, G346V, T369K)

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Figure 8 cont.

AAATTGCCATCCTGATAACTACTGTAACATTGCATTTCAAGCAATATGAATTCAACTCCCCCCTAAACAACC
AAGTGATGCTGTGTGAACCAACAATAATAGAAAGAAACATAACAGAGATAGTGATTTGACCAACACCAC
CATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACA
GGATTTGCACCTTTCTCTAAGGACAATTCGATTAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAG
AACCTTATGTGTCATGCGATCCTGACAAGTGTTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAAC
GTGCATTCAAATAACAAAGTACGTGAGAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCC
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CTGCATGTTTGTATAACGGGGGATGATAAAAATGCAACTGCTAGCTTCATTTACAATGGGAGGCTTGTA
TAGTGTGTTTCATGGTCCAAAGATATTCTCAGGACCCAGGAGTCAGAATGCGTTTGATCAATGGAACCTT
GTACAGTAGTAATGACTGATGGAAGTGCTACAGGAAAAGCTGATACTAAAATACTATTTCATTGAGGAGGG
GAAAATCGTTCATACTAGCAAATTGTCAGGAAGTGCTCAGCATGTGGAAGAGTGCTCTTGCTATCCTCGAT
ATCCTGGTGTGATGTGTCTGCAGAGACAACCTGGAAAGGATCCAACCGGCCCATCGTAGATATAAACATA
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GCTCCAGCAGTAGCCATTGTTGAATCCTAACATGAAGAAGGTGTTTCATGGAGTGAAAGGCTGGGCCTT
TGATGATGGAAATGACGTGTGGATGGGGAGAACATCAACGAGAAGTCACGCTTAGGGTATGAAACCTT
CAAAGTCGTTGAAGGCTGGTCCAAACCTAAGTCCAAATTGCAGATAAATAGGCAAGTCATAGTTGACAGA
GGTGATAGGTCCGTTATTCTGGTATTTCTCTGTGTAAGGCAAAAGCTGCATCAATCGGTGCTTTTATGT
GGAGTTGATTAGGGGAAGAAAAGAGGAAACTGAAGTCTTGTGGACCTCAAACAGTATTGTTGTGTTTTG
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(SEQ ID NO:22)

A/Yokohama/147/2017NA

ATGAATCCAAATCAAAGATAATAACGATTGGCTCTGTTTCTCTACAATTTCCACAATATGCTTCTTCATGC
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GATTTGCACCTTTCTCTAAAGACAATTCGATTAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAGA
ACCTTATGTGTCATGCGATCTTGACAAGTGTTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAACG
TGCATTCAAATAACACAGTACGTGATAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCCCTT
TCCATCTGGGGACCAAGCAAGTGTGCATAGCATGGTCCAGCTCAAGTTGTCACGATGGAAAAGCATGGCT
GCATGTTTGTATAACGGGGGATGATAAAAATGCAACTGCTAGCTTCATTTACAWATGGGAGGCTTGATAGAT
AGTGTGTTTCATGGTCCAAACGATATTCTCAGGACCCAGGAGTCAGAATGCGTTTGATCAATGGAACCTG
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CCTGGTGTGATGTGTCTGCAGAGACAACCTGGAAAGGATCCAACCGGCCCATCATAGATATAAACATAA
AGGATCATAGCATTGTTTCCAGTTATGTGTGTTTCAGGACTTGTGGAGACACACCCAGAAAAAGCGACAG

Figure 8 cont.

CTCCAGCAGTAGCCATTGTTTGAATCCTAACAAATGAAGAAGGTGGTCATGGAGTGAAAGGCTGGGCCTTT
GATGATGGAAATGACGTGTGGATGGGGAGAACAATCAACGAGACGTACGCTTAGGGTATGAAACCTTC
AAAGTCGTTGAAGGCTGGTCCAACCTAAGTCCAAATTGCAGATAAATAGGCAAGTCATAGTTGACAGAG
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GAGTTGATCAGGGGAAGAAAAGAGGAACTGAAGTCTTGTGGACCTCAAACAGTATTGTTGTGTTTTGT
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(SEQ ID NO:23)

>A/Yokohama/48/2018NA

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ACCTTATGTGTATGCGATCCTGACAAGTGTATCAATTTGCCCTTGACAGGGAACAACACTAAACAACG
TGCATTCAAATAACACAGTACGTGATAGGACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCCTT
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GGAGTTGATTAGGGGAAGAAAAGAGGAACTGAAGTCTTGTGGACCTCAAACAGTATTGTTGTGTTTTG
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(SEQ ID NO:24)

>A/Delaware/33/2018NA

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AGTGATGCTGTGTGAACCAACAATAATAGAAAGAAACATAACAGAGATAGTGATTTTGACCAACACCACC
ATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACAG

Figure 8 cont.

GATTGACCTTTCTCTAAGGACAATTCGATTAGGCTTCCGCTGGTGGGGACATCTGGGTGACAAGAGA
ACCTTATGTGTCATGCGATCTTGACAAGTGTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAACG
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GTGTTGTCTCATGGTCCAATGATATTCTCAGGACCCAGGAATCAGAATGCGTTTGTATCAATGGAACCTGTA
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>A/Tokyo/UT-GR85/2019NA

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GATGATGGAAATGACGTGTGGATGGGGAGAACAAATCAACGAGACGTACGCTTAGGGTATGAAACCTTC
AAAGTCGTTGAAGGCTGGTCCAACCTAAGTCCAAATTGCAGATAAATAGGCAAGTCATAGTTGACAGAG

Figure 8 cont

GTGATAGGTCCGGTTATTCTGGTATTTCTCTGTTGAAGGCAAAAGCTGCATCAATCGGTGCTTTTATGTRG
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>A/Saint-Petersburg/RII-3245/2019NA

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GATGATGGAAATGACGTGTGGATGGGGAGAACAATCAACGAGACGTACGCTTAGGGTATGAAACCTTC
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GAGTTGATTAGGGGAAGAAAAGAGGAACTGAAGTCTTGTGGACCTCAAAACAGTATTGTTGTGTTTGTG
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TTTCGCAATTTTAGAAAAAACTCCTTGTTTCTACT (SEQ ID NO:27)

>A/Kanagawa/IC1820/2019NA

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AGTGATGCTGTGTGAACCAACAATAATAGAAAGAAACATAACAGAGATAGTGATTTTGACCAACACCACC
ATAGAGAAGGAAATATGCCCCAAACCAGCAGAATACAGAAATTGGTCAAAACCGCAATGTGGCATTACAG
GATTTGCACCTTTCTCTAAGGACAATTCGATTAGGCTTTCCGCTGGTGGGGACATCTGGGTGACAAGAGA
ACCTTATGTGTATGCGATCTTGACAAGTGTATCAATTTGCCCTTGGACAGGGAACAACACTAAACAACG
TGCATTCAAATAACACAGTACGTGATAGAACCCCTTATCGGACTCTATTGATGAATGAGTTGGGTGTTCTT

Figure 8 cont.

TCCATCTGGGGACCAAGCAAGTGTGCATAGCATGGTCCAGCTCAAGCTGTCACGATGGAAAAGCATGGC
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AGTGTGTTTCATGGTCCAACGATATTCTCAGGACCCAGGAGTCAGAATGCGTTTGATCAATGGAACCTG
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CTCCAGCAGTAGCCATTGTTTGAACCTAACAATGAAAAAGGTGATCATGGAGTGAAAGGCTGGGCCTTT
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GAGTTGATTAGGGGAAGAAAAGAGGAAACTGAAGTCTGTGGACCTCAAACAGTATTGTTGTGTTTTGT
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(SEQ ID NO:28)

>A/Kansas/14/2017NA

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GGAGTTGATTAGGGGAAGAAAAGAGGAAACTGAAGTCTGTGGACCTCAAACAGTATTGTTGTGTTTTG

Figure 8 cont.

TGGCACCTCAGGTACATATGGAACAGGCTCATGGCCTGATGGGGCGGACCTCAATCTCATGCATATATAAG
CTTTCGCAATTTTAGAAAAAACT (SEQ ID NO:29)

Figure 9. Exemplary influenza B neuraminidases

AGA18961

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mlpstiqtlit lfltsqgvll slyvsaslsy llysdillkf srteitapim pldcanasnv 61
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acgpkeckhf althyaapqg gyyngtredr nklrhliisvk lgkiptvens ifhmaawsgs 181
achdgrewty igvdgpdnsa llkikygeay tdtysyakn tlrtgesacn ciggdcyimi 241
tdgpasgise crflkiregr iikeifptgr vkhteectcg fasnktieca crdnsytakr 301
pfvklinvetd taeirlmcte tyltprpnd gsitgpcesn gdkgsggikg gfvhqrmask 361
igrwysrtms ktkrmgmgly vkydgdptwd sealalsgvm vsmeepgwys fgfeikdkkc 421
dvpcigiemv hdggkktwhs aataiyclmg sgqllwdtvt gvnmtl (SEQ ID NO:30)
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ACT85963.1

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mlpstiqtlit lfltsqgvll slyvsaslsy llysdillkf spteitaptm pldcanasnv 61
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acgpkeckhf althyaapqg gyyngtredk nklrhliisvk lgkiptvens ifhmaawsgs 181
achdgrewty igvdgpdnsa llkikygeay tdtysyann ilrtgesacn ciggdcyimi 241
tdgsasgise crflkiregr iikeifptgr vehteectcg fasnktieca crdnsytakr 301
pfvklinvetd taeirlmcte tyltprpdd gsitgpcesn gdkgsggikg gfvhqrmask 361
igrwysrtms ktkrmgmgly vkydgdptwd sdalalsgvm vsmeepgwys fgfeikdkkc 421
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ABL77391.1

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acgpkeckhf althyaapqg gyyngtrgdr nklrhliisvk lgkiptvens ifhmaawsgs 181
achdgkewty igvdgpdnsa llkikygeay tdtysyann ilrtgesacn ciggncylmi 241
tdgsasgvse crflkiregr iikeifptgr vkhteectcg fasnktieca crdnsytakr 301
pfvklinvetd taeirlmcte tyltprpdd gsitgpcesn gdkgsggikg gfvhqrmask 361
igrwysrtms ktkrmgmgly vkydgdptwd sdalalsgvm vsmeepgwys fgfeikdkkc 421
dvpcigiemv hdggketwhs aataiyclmg sgqllwdtvt gvdmal (SEQ ID NO:32)
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ABF21335.1

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cgpkeckhfa lthyaapqgg yyngtredrn klrhliisvkl gkiptvensi fhmaawsgsa 181
chdgrewtyi gvdgpdnsal ikikygeayt dtysyanni lrtgesacnc iggdcyimit 241
dgsasgiskc rflkiregri ikeifptgrv ehteectcgf asnktiecac rdnnytakrp 301
```

Figure 9 cont.

```
fvklinvetdt aeirlmctet yldtprpddg sitgpcesng dkgrggikkg fvhqrmaski 361
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AAA43743.1

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acgpkeckhf althyaagpg gyyngtrkdr nkrlrlisvk lgkiptvens ifhmaawsgs 181
achdgrewty igvdgpdnsa likikygeay tdyhsyann ilrtgesacn ciggdciymi 241
tdgsasgisk crflkiregr iikeifptgr vehteectcg fasnktieca crdnsytakr 301
pfvklinvetd taeirlmcte tyldtprpdd gsitgpcesn gdkglggikg gfvhqrmask 361
igrwysrtms ktermgmely vkydgdpwtd sealapsgvm vsmkepgwys fgfeikdkkc 421
dvpcigiemv hdggketwhs aataiyclmg sgqllwdvtv gvdmal (SEQ ID NO:34)
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AML44612.1 Influenza B virus (B/Victoria/3/2014)

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mlpstiqtlit ifltsggvil slyvsaslsy llysdillkf srtevtapim pldecanasnv 61
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acgpkeckhf althyaagpg gyyngtrredr nkrlrlisvk lgkiptvens ifhmaawsgs 181
achdgrewty igvdgpdnsa likikygeay tdyhsyakn ilrtgesacn ciggdciymi 241
tdgpasgise crflkiregr iikeifptgr vkhteectcg fasnktieca crdnsytakr 301
pfvklinvetd taeirlmctk tyldtprpnd gsitgpcesd gdegsggikg gfvhqrmask 361
igrwysrtms ktkrmgmgly vkydgdpwtd sealalsgvm vsmeepgwys fgfeikdkkc 421
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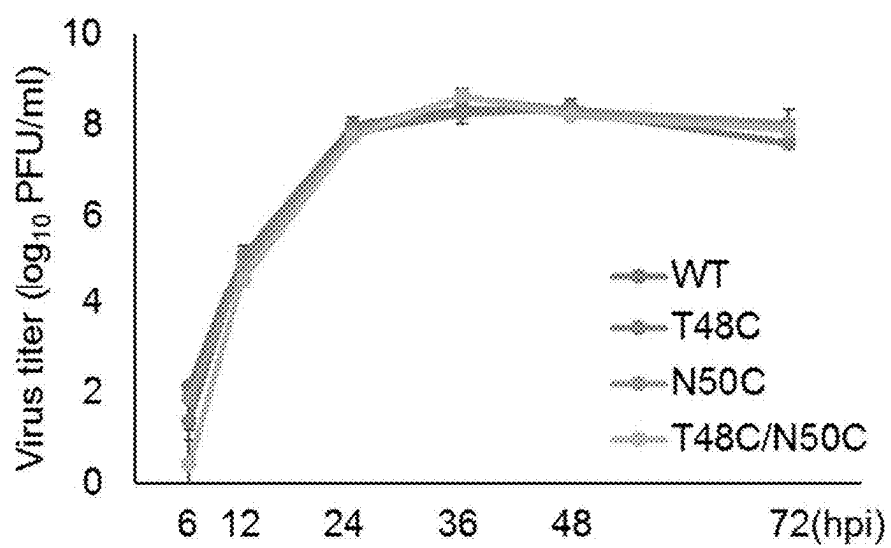


Figure 10

RECOMBINANT INFLUENZA VIRUSES WITH STABILIZED NA

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a divisional of U.S. application Ser. No. 18/365,082, filed Aug. 3, 2023, which is a divisional of U.S. application Ser. No. 17/155,625, filed Jan. 22, 2021, which claims the benefit of the filing date of U.S. application No. 62/965,225, filed on Jan. 24, 2020, the disclosures of which are incorporated by reference herein.

STATEMENT OF GOVERNMENT FUNDING

[0002] This invention was made with government support under HHSN272201400008C awarded by the National Institutes of Health. The government has certain rights in the invention.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING

[0003] A Sequence Listing is provided herewith as an xml file, “2356941.xml” created on Aug. 3, 2023, and having a size of 101,767 bytes. The content of the xml file is incorporated by reference herein in its entirety.

BACKGROUND

[0004] Influenza is a major respiratory disease in some mammals including horses and is responsible for substantial morbidity and economic losses each year. In addition, influenza virus infections can cause severe systemic disease in some avian species, leading to death. The segmented nature of the influenza virus genome allows for reassortment of segments during virus replication in cells infected with two or more influenza viruses. The reassortment of segments, combined with genetic mutation and drift, can give rise to a myriad of divergent strains of influenza virus over time. The new strains exhibit antigenic variation in their hemagglutinin (HA) and/or neuraminidase (NA) proteins, and in particular the gene coding for the HA protein has a high rate of variability. The predominant current practice for the prevention of flu is vaccination. Most commonly, inactivated virus vaccines are used. As the influenza HA protein is the major target antigen for the protective immune responses of a host to the virus and is highly variable, the isolation of influenza virus and the identification and characterization of the HA antigen in viruses associated with recent outbreaks is important for vaccine production. Based on prevalence and prediction, a vaccine is designed to stimulate a protective immune response against the predominant and expected influenza virus strains.

[0005] There are four general types of influenza viruses, Type A, Type B, Type C, and Type D, which are defined by the absence of serological cross reactivity between their internal proteins. Influenza Type A viruses are further classified into subtypes based on antigenic and genetic differences of their glycoproteins, the HA and NA proteins. All the known HA and NA subtypes (H1 to H18 and N1 to N11) have been isolated from aquatic birds, which are thought to act as a natural reservoir for influenza.

[0006] It has been suggested that antibodies against NA play important roles in preventing influenza virus infections. However, the current influenza vaccines, which are made by the inactivation of influenza viruses grown in eggs and

purification of the virus antigen, are not able to elicit anti-NA antibodies efficiently.

SUMMARY

[0007] One of the causes of the low production of anti-NA antibodies is attributed to the structural instability of the NA protein, which works as a homo-tetramer. The NA tetramer may be disrupted during the antigen purification process. Therefore, the amount of NA contained in current vaccines is insufficient to elicit the production of anti-NA antibodies.

[0008] The present disclosure relates to influenza viruses with certain residue(s) or modifications in the NA protein that stabilize its natural homotetramer structure, and methods of making and using that virus, e.g., in a vaccine. In one embodiment, an isolated recombinant influenza virus comprising a neuraminidase (NA) viral segment encoding a NA monomer that forms virions having stabilized NA tetramers is provided. In one embodiment, the recombinant influenza virus has a modified NA stalk relative to a parental NA that results in stabilized tetramers. In one embodiment, the modified NA stalk has a deletion. In one embodiment, the modified NA stalk has an insertion relative to a parental NA. In one embodiment, the modified NA stalk has at least one amino acid substitution relative to a parental NA. In one embodiment, at least one substitution in the modified NA stalk is a cysteine substitution. In one embodiment, the modified NA stalk has at least two substitutions relative to a parental NA. In one embodiment, the NA has a cysteine at position 48 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 50 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 48 and position 50 relative to the numbering of N1. In one embodiment, the NA stalk is modified within residues 1 to 10 from the C-terminus of the transmembrane domain relative to a parental NA. In one embodiment, the NA stalk is modified within residues 10 to 20 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 20 to 30 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 30 to 50 from the C-terminus of the transmembrane domain. In one embodiment, the stalk domain begins at the first residue after the last residue in the transmembrane domain of NA up to the conserved cysteine, or one to two residues before the conserved cysteine, in the head region of NA that forms a disulfide bond (see Blumenkrantz et al., *J. Virol.*, 87:10539 (2013)). In one embodiment, the insertion in the NA stalk is an insertion of 1, 2, 3, 4 or 5, or 5 to 10, or 10 to 20, or more residues. In one embodiment, the deletion in the NA stalk is a deletion of 1, 2, 3, 4 or 5, or 5 to 10, or 10 to 20, or more residues. In one embodiment, a NA that has a cysteine is the stalk region is modified to include a second or third cysteine, e.g., within 1 to 10 or 1 to 5 residues of the other cysteine(s). In one embodiment, a NA has at least two cysteines in the stalk region that are within 1 to 2, 2 to 3, 3 to 4 or 5 up to residues of each other. In one embodiment, a NA has at least two cysteines in the stalk region, one of which is within 5 residues, either N-terminal or C-terminal, of residue 48 in the NA, and the other of which is within 5 residues, either N-terminal or C-terminal, of residue 50 in the NA. In one embodiment, the cysteines are adjacent to each other, e.g., at residues 46 and 47, residues 46, 47 and 48, residues 47 and 48, residues 47, 48 and 49, residues 48 and 49, residues 48, 49, and 50, residues 49 and 50, or residues 49, 50 and 51,

and in one embodiment both are in the stalk region. In one embodiment, the cysteines are 2 residues apart and in one embodiment both are in the stalk region, e.g., the cysteines are at residues 46 and 48, 47 and 49, 48 and 50, 49 and 51, 50 and 52, 51 and 53, and the like. In one embodiment, the cysteines are 3 residues apart, e.g., the cysteines are at residues 45 and 48, 46 and 49, 47 and 50, 48 and 51, 49 and 52, 50 and 53, 51 and 54, and the like and in one embodiment both are in the stalk region. In one embodiment, the cysteines are 4 residues apart, e.g., the cysteines are at residues 44 and 48, 45 and 49, 46 and 50, 47 and 51, 48 and 52, 49 and 53, 50 and 54, 51 and 55, and the like and in one embodiment both are in the stalk region. In one embodiment, the stalk region has no more than 10, 9, 8, 7, 6, 5, 4, 3, 2 or 1 cysteine residues. In one embodiment, the cysteine(s) in the stalk region are in the N-terminal 30 residues of the stalk region. In one embodiment, the cysteine(s) in the stalk region are in the N-terminal 20 residues of the stalk region.

[0009] As described herein, a highly proliferative recombinant influenza viruses expressing structurally stabilized neuraminidase tetramers was prepared. In one embodiment, recombinant influenza viruses containing either NA-48C or NA-50C or both express stabilized NA tetramers and replicate efficiently. Thus, amino acid mutations in influenza NA that enable the generation of highly proliferative recombinant influenza viruses expressing structurally stabilized NA tetramers can be used to generate influenza vaccine strains that can be used to produce influenza vaccines containing a greater amount of NA antigen. Influenza vaccine strains containing a greater amount of NA antigen that can elicit anti-NA antibodies efficiently.

[0010] In one embodiment, a vaccine comprising an effective amount of the recombinant influenza virus or a portion thereof is provided. In one embodiment, the vaccine is a whole virus vaccine. In one embodiment, the vaccine is a split virus vaccine. In one embodiment, the vaccine is a subunit vaccine. In one embodiment, the vaccine further comprises an adjuvant. In one embodiment, the vaccine further comprises a pharmaceutically acceptable carrier. In one embodiment, the carrier is suitable for intranasal or intramuscular administration. In one embodiment, the vaccine further comprises at least one other influenza virus isolate.

[0011] In one embodiment, a method of preparing influenza virus having stabilized NA tetramers is provided. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding, for example, at least one cysteine in the stalk region, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers. In one embodiment, the NA is N1, N2, N3 or N5. In one embodiment, the NA is N4, N6, N7, N8 or N9. In one embodiment, the cell is a mammalian cell. In one embodiment, the cell is a Vero cell, 293T cell or MDCK cell.

[0012] In one embodiment, a method of making an influenza vaccine is provided. The method includes combining the recombinant virus with an adjuvant or treating the virus with an agent that inactivates the virus. In one embodiment,

the method further comprises aliquoting a selected dose of the virus into a receptacle. In one embodiment, the adjuvant comprises immunostimulatory DNA sequences, bacterium-derived components, aluminum salt (alum) or squalene oil-in-water emulsion systems such as MF59 and AS03. In one embodiment, the agent chemically inactivates the virus. In one embodiment, the agent comprises formalin or beta-propiolactone. In one embodiment, the agent comprises a detergent, e.g., a non-ionic detergent, a cationic detergent or an anionic detergent. In one embodiment, the detergent comprises CTAB, Triton, SDS, Neodol 23-6, or sodium desoxycholate. In one embodiment, the method further comprises separating HA and NA from other viral components, e.g., using centrifugation and collection of soluble molecules.

[0013] In one embodiment, a method of preparing influenza virus is provided that includes contacting cells with the recombinant virus in an amount effective to yield progeny virus. In one embodiment, the virus is contacted with an avian egg. In one embodiment, the cells are mammalian cells. In one embodiment, the HA of the virus is H1, H3, H5 or H7.

[0014] Also provided is a method of preparing stabilized NA tetramers, comprising: contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding at least one cysteine in the stalk region and nucleic acid for an influenza virus HA. In one embodiment, the method further comprises isolating NA and HA from the cell. In one embodiment, the cell is an insect cell.

[0015] Further provided is a method of immunizing an avian or a mammal, comprising: administering to the avian or the mammal a composition having an effective amount of the recombinant virus. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

BRIEF DESCRIPTION OF FIGURES

[0016] FIG. 1. Western blot to detect NA in hCK cells infected with influenza viruses.

[0017] FIG. 2. Amino acid sequence of the NA of A/Yokohama/2017/2003 (SEQ ID NO:1).

[0018] FIG. 3. Amino acid sequence for the NA of A/Saitama/103/2014 (SEQ ID NO:2).

[0019] FIG. 4. Amino acid sequences for NA of mutant of A/Yokohama/2017/2003 (SEQ ID NO:3).

[0020] FIG. 5. Exemplary NA sequences for N3, N4, N6, N7, N8, and N9 (SEQ ID Nos. 4-9). The stalk region is indicated by red font, underlining.

[0021] FIG. 6. Exemplary viral backbone sequences (SEQ ID Nos. 10-14).

[0022] FIG. 7. Exemplary NA sequences (SEQ ID Nos. 15-18). The stalk region is indicated by red font, underlining.

[0023] FIG. 8. Exemplary NA nucleic acid sequences (SEQ ID Nos 19-29).

[0024] FIG. 9. Exemplary influenza B virus NA sequences (SEQ ID Nos. 45-50). In one embodiment, the stalk region in the NA of influenza B virus may be from residue 38 to 86.

[0025] FIG. 10. Virus growth curves. 6+2 reassortant influenza viruses containing the HA and NA viral segments from A/Singapore/GP1908/2015 (H1N1) pdm09 in the backbone of high-yield A/Puerto Rico/8/1934 (H1N1) were

prepared using reverse genetics and propagation in hCK cells at 37° C. Mutant viruses with either the NA-T48C or NA-N50C mutation or both mutations were generated. hCK cells were infected with these viruses at a MOI (multiplicity of infection) of 0.001. The supernatants were collected at the indicated times post-infection and virus titers were determined by means of plaque assay in hCK cells.

DETAILED DESCRIPTION

Definitions

[0026] As used herein, the term “isolated” refers to in vitro preparation and/or isolation of a nucleic acid molecule, e.g., vector or plasmid, peptide or polypeptide (protein), or virus of the invention, so that it is not associated with in vivo substances, or is substantially purified from in vitro substances. An isolated virus preparation is generally obtained by in vitro culture and propagation, and/or via passage in eggs, and is substantially free from other infectious agents.

[0027] As used herein, “substantially purified” means the object species is the predominant species, e.g., on a molar basis it is more abundant than any other individual species in a composition, and preferably is at least about 80% of the species present, and optionally 90% or greater, e.g., 95%, 98%, 99% or more, of the species present in the composition.

[0028] As used herein, “substantially free” means below the level of detection for a particular infectious agent using standard detection methods for that agent.

[0029] A “recombinant” virus is one which has been manipulated in vitro, e.g., using recombinant DNA techniques, to introduce changes to the viral genome. Reassortant viruses can be prepared by recombinant or nonrecombinant techniques.

[0030] As used herein, the term “recombinant nucleic acid” or “recombinant DNA sequence or segment” refers to a nucleic acid, e.g., to DNA, that has been derived or isolated from a source, that may be subsequently chemically altered in vitro, so that its sequence is not naturally occurring, or corresponds to naturally occurring sequences that are not positioned as they would be positioned in the native genome. An example of DNA “derived” from a source, would be a DNA sequence that is identified as a useful fragment, and which is then chemically synthesized in essentially pure form. An example of such DNA “isolated” from a source would be a useful DNA sequence that is excised or removed from said source by chemical means, e.g., by the use of restriction endonucleases, so that it can be further manipulated, e.g., amplified, for use in the disclosure, by the methodology of genetic engineering.

[0031] As used herein, a “heterologous” influenza virus gene or viral segment is from an influenza virus source that is different than a majority of the other influenza viral genes or viral segments in a recombinant, e.g., reassortant, influenza virus.

[0032] The terms “isolated polypeptide”, “isolated peptide” or “isolated protein” include a polypeptide, peptide or protein encoded by cDNA or recombinant RNA including one of synthetic origin, or some combination thereof.

[0033] The term “recombinant protein” or “recombinant polypeptide” as used herein refers to a protein molecule expressed from a recombinant DNA molecule. In contrast, the term “native protein” is used herein to indicate a protein isolated from a naturally occurring (i.e., a nonrecombinant)

source. Molecular biological techniques may be used to produce a recombinant form of a protein with identical properties as compared to the native form of the protein.

[0034] Methods of alignment of sequences for comparison are well known in the art. Thus, the determination of percent identity between any two sequences can be accomplished using a mathematical algorithm.

[0035] Computer implementations of these mathematical algorithms can be utilized for comparison of sequences to determine sequence identity. Alignments using these programs can be performed using the default parameters. Software for performing BLAST analyses is publicly available through the National Center for Biotechnology Information (<http://www.ncbi.nlm.nih.gov/>). The algorithm may involve first identifying high scoring sequence pairs (HSPs) by identifying short words of length W in the query sequence, which either match or satisfy some positive-valued threshold score T when aligned with a word of the same length in a database sequence. T is referred to as the neighborhood word score threshold. These initial neighborhood word hits act as seeds for initiating searches to find longer HSPs containing them. The word hits are then extended in both directions along each sequence for as far as the cumulative alignment score can be increased. Cumulative scores are calculated using, for nucleotide sequences, the parameters M (reward score for a pair of matching residues; always >0) and N (penalty score for mismatching residues; always <0). For amino acid sequences, a scoring matrix is used to calculate the cumulative score. Extension of the word hits in each direction are halted when the cumulative alignment score falls off by the quantity X from its maximum achieved value, the cumulative score goes to zero or below due to the accumulation of one or more negative-scoring residue alignments, or the end of either sequence is reached.

[0036] In addition to calculating percent sequence identity, the BLAST algorithm may also perform a statistical analysis of the similarity between two sequences. One measure of similarity provided by the BLAST algorithm may be the smallest sum probability (P(N)), which provides an indication of the probability by which a match between two nucleotide or amino acid sequences would occur by chance. For example, a test nucleic acid sequence is considered similar to a reference sequence if the smallest sum probability in a comparison of the test nucleic acid sequence to the reference nucleic acid sequence is less than about 0.1, more preferably less than about 0.01, and most preferably less than about 0.001.

[0037] The BLASTN program (for nucleotide sequences) may use as defaults a wordlength (W) of 11, an expectation (E) of 10, a cutoff of 100, M=5, N=-4, and a comparison of both strands. For amino acid sequences, the BLASTP program may use as defaults a wordlength (W) of 3, an expectation (E) of 10, and the BLOSUM62 scoring matrix. See <http://www.ncbi.nlm.nih.gov>. Alignment may also be performed manually by inspection.

[0038] For sequence comparison, typically one sequence acts as a reference sequence to which test sequences are compared. When using a sequence comparison algorithm, test and reference sequences are input into a computer, subsequence coordinates are designated if necessary, and sequence algorithm program parameters are designated. The sequence comparison algorithm then calculates the percent

sequence identity for the test sequence(s) relative to the reference sequence, based on the designated program parameters.

Influenza Virus Structure and Propagation

[0039] Influenza A viruses possess a genome of eight single-stranded negative-sense viral RNAs (vRNAs) that encode at least ten proteins. The influenza virus life cycle begins with binding of the hemagglutinin (HA) to sialic acid-containing receptors on the surface of the host cell, followed by receptor-mediated endocytosis. The low pH in late endosomes triggers a conformational shift in the HA, thereby exposing the N-terminus of the HA2 subunit (the so-called fusion peptide). The fusion peptide initiates the fusion of the viral and endosomal membrane, and the matrix protein (M1) and RNP complexes are released into the cytoplasm. RNPs consist of the nucleoprotein (NP), which encapsidates vRNA, and the viral polymerase complex, which is formed by the PA, PB1, and PB2 proteins. RNPs are transported into the nucleus, where transcription and replication take place. The RNA polymerase complex catalyzes three different reactions: synthesis of an mRNA with a 5' cap and 3' polyA structure, of a full-length complementary RNA (cRNA), and of genomic vRNA using the cRNA as a template. Newly synthesized vRNAs, NP, and polymerase proteins are then assembled into RNPs, exported from the nucleus, and transported to the plasma membrane, where budding of progeny virus particles occurs. The neuraminidase (NA) protein plays a crucial role late in infection by removing sialic acid from sialyloligosaccharides, thus releasing newly assembled virions from the cell surface and preventing the self aggregation of virus particles. Although virus assembly involves protein-protein and protein-vRNA interactions, the nature of these interactions is largely unknown.

[0040] Although influenza B and C viruses are structurally and functionally similar to influenza A virus, there are some differences. For example, influenza B virus does not have a M2 protein with ion channel activity but has BM2 and has a viral segment with both NA and NB sequences. Influenza C virus has only seven viral segments.

Cells That can be Used to Produce Virus

[0041] Any cell, e.g., any avian or mammalian cell, such as avian eggs, a human, e.g., 293T or PER.C6® cells, or canine, bovine, equine, feline, swine, ovine, rodent, for instance mink, e.g., MvLu1 cells, or hamster, e.g., CHO cells, or non-human primate, e.g., Vero cells, including mutant cells, which supports efficient replication of influenza virus can be employed to isolate and/or propagate influenza viruses. Isolated viruses can be used to prepare a reassortant virus. In one embodiment, host cells for vaccine production are continuous mammalian or avian cell lines or cell strains. A complete characterization of the cells to be used, may be conducted so that appropriate tests for purity of the final product can be included. Data that can be used for the characterization of a cell includes (a) information on its origin, derivation, and passage history; (b) information on its growth and morphological characteristics; (c) results of tests of adventitious agents; (d) distinguishing features, such as biochemical, immunological, and cytogenetic patterns which allow the cells to be clearly recognized among other cell lines; and (e) results of tests for tumorigenicity. In one

embodiment, the passage level, or population doubling, of the host cell used is as low as possible.

[0042] In one embodiment, the cells are WHO certified, or certifiable, continuous cell lines. The requirements for certifying such cell lines include characterization with respect to at least one of genealogy, growth characteristics, immunological markers, virus susceptibility tumorigenicity and storage conditions, as well as by testing in animals, eggs, and cell culture. Such characterization is used to confirm that the cells are free from detectable adventitious agents. In some countries, karyology may also be required. In addition, tumorigenicity may be tested in cells that are at the same passage level as those used for vaccine production. The virus may be purified by a process that has been shown to give consistent results, before vaccine production (see, e.g., World Health Organization, 1982).

[0043] Virus produced by the host cell may be highly purified prior to vaccine or gene therapy formulation. Generally, the purification procedures result in extensive removal of cellular DNA and other cellular components, and adventitious agents. Procedures that extensively degrade or denature DNA may also be used.

Influenza Vaccines

[0044] A vaccine includes an isolated recombinant influenza virus of the invention, and optionally one or more other isolated viruses including other isolated influenza viruses, one or more immunogenic proteins or glycoproteins of one or more isolated influenza viruses or one or more other pathogens, e.g., an immunogenic protein from one or more bacteria, non-influenza viruses, yeast or fungi, or isolated nucleic acid encoding one or more viral proteins (e.g., DNA vaccines) including one or more immunogenic proteins of the isolated influenza virus of the invention. In one embodiment, the influenza viruses of the invention may be vaccine vectors for influenza virus or other pathogens.

[0045] A complete virion vaccine may be concentrated by ultrafiltration and then purified by zonal centrifugation or by chromatography. Viruses other than the virus of the invention, such as those included in a multivalent vaccine, may be inactivated before or after purification using formalin or beta-propiolactone, for instance.

[0046] A subunit vaccine comprises purified glycoproteins. Such a vaccine may be prepared as follows: using viral suspensions fragmented by treatment with detergent, the surface antigens are purified, by ultracentrifugation for example. The subunit vaccines thus contain mainly HA protein, and also NA. The detergent used may be cationic detergent for example, such as hexadecyl trimethyl ammonium bromide (Bachmeyer, 1975), an anionic detergent such as ammonium deoxycholate (layer & Webster, 1976); or a nonionic detergent such as that commercialized under the name TRITON X100. The hemagglutinin may also be isolated after treatment of the virions with a protease such as bromelain, and then purified. The subunit vaccine may be combined with an attenuated virus of the invention in a multivalent vaccine.

[0047] A split vaccine comprises virions which have been subjected to treatment with agents that dissolve lipids. A split vaccine can be prepared as follows: an aqueous suspension of the purified virus obtained as above, inactivated or not, is treated, under stirring, by lipid solvents such as ethyl ether or chloroform, associated with detergents. The dissolution of the viral envelope lipids results in fragmen-

tation of the viral particles. The aqueous phase is recuperated containing the split vaccine, constituted mainly of hemagglutinin and neuraminidase with their original lipid environment removed, and the core or its degradation products. Then the residual infectious particles are inactivated if this has not already been done. The split vaccine may be combined with an attenuated virus of the invention in a multivalent vaccine.

[0048] Inactivated Vaccines. Inactivated influenza virus vaccines are provided by inactivating replicated virus using known methods, such as, but not limited to, formalin or β -propiolactone treatment. Inactivated vaccine types that can be used in the invention can include whole-virus (WV) vaccines or subvirion (SV) (split) vaccines. The WV vaccine contains intact, inactivated virus, while the SV vaccine contains purified virus disrupted with detergents that solubilize the lipid-containing viral envelope, followed by chemical inactivation of residual virus.

[0049] In addition, vaccines that can be used include those containing the isolated HA and NA surface proteins, which are referred to as surface antigen or subunit vaccines.

[0050] Live Attenuated Virus Vaccines. Live, attenuated influenza virus vaccines, such as those including a recombinant virus of the invention can be used for preventing or treating influenza virus infection. Attenuation may be achieved in a single step by transfer of attenuated genes from an attenuated donor virus to a replicated isolate or reassorted virus according to known methods. Since resistance to influenza A virus is mediated primarily by the development of an immune response to the HA and/or NA glycoproteins, the genes coding for these surface antigens come from the reassorted viruses or clinical isolates. The attenuated genes are derived from an attenuated parent. In this approach, genes that confer attenuation generally do not code for the HA and NA glycoproteins.

[0051] Viruses (donor influenza viruses) are available that are capable of reproducibly attenuating influenza viruses, e.g., a cold adapted (ca) donor virus can be used for attenuated vaccine production. Live, attenuated reassortant virus vaccines can be generated by mating the ca donor virus with a virulent replicated virus. Reassortant progeny are then selected at 25° C. (restrictive for replication of virulent virus), in the presence of an appropriate antiserum, which inhibits replication of the viruses bearing the surface antigens of the attenuated ca donor virus. Useful reassortants are: (a) infectious, (b) attenuated for seronegative non-adult mammals and immunologically primed adult mammals, (c) immunogenic and (d) genetically stable. The immunogenicity of the ca reassortants parallels their level of replication. Thus, the acquisition of the six transferable genes of the ca donor virus by new wild-type viruses has reproducibly attenuated these viruses for use in vaccinating susceptible mammals both adults and non-adult.

[0052] Other attenuating mutations can be introduced into influenza virus genes by site-directed mutagenesis to rescue infectious viruses bearing these mutant genes. Attenuating mutations can be introduced into non-coding regions of the genome, as well as into coding regions. Such attenuating mutations can also be introduced into genes other than the HA or NA, e.g., the PB2 polymerase gene. Thus, new donor viruses can also be generated bearing attenuating mutations introduced by site-directed mutagenesis, and such new donor viruses can be used in the production of live attenuated reassortants vaccine candidates in a manner analogous

to that described above for the ca donor virus. Similarly, other known and suitable attenuated donor strains can be reassorted with influenza virus to obtain attenuated vaccines suitable for use in the vaccination of mammals.

[0053] In one embodiment, such attenuated viruses maintain the genes from the virus that encode antigenic determinants substantially similar to those of the original clinical isolates. This is because the purpose of the attenuated vaccine is to provide substantially the same antigenicity as the original clinical isolate of the virus, while at the same time lacking pathogenicity to the degree that the vaccine causes minimal chance of inducing a serious disease condition in the vaccinated mammal.

[0054] The viruses in a multivalent vaccine can thus be attenuated or inactivated, formulated and administered, according to known methods, as a vaccine to induce an immune response in an animal, e.g., a mammal. Methods are well-known in the art for determining whether such attenuated or inactivated vaccines have maintained similar antigenicity to that of the clinical isolate or high growth strain derived therefrom. Such known methods include the use of antisera or antibodies to eliminate viruses expressing antigenic determinants of the donor virus; chemical selection (e.g., amantadine or rimantidine); HA and NA activity and inhibition; and nucleic acid screening (such as probe hybridization or PCR) to confirm that donor genes encoding the antigenic determinants (e.g., HA or NA genes) are not present in the attenuated viruses.

Pharmaceutical Compositions

[0055] Pharmaceutical compositions of the present disclosure, suitable for inoculation, e.g., nasal, parenteral or oral administration, comprise one or more influenza virus isolates, e.g., one or more attenuated or inactivated influenza viruses, a subunit thereof, isolated protein(s) thereof, and/or isolated nucleic acid encoding one or more proteins thereof, optionally further comprising sterile aqueous or non-aqueous solutions, suspensions, and emulsions. The compositions can further comprise auxiliary agents or excipients, as known in the art. The composition of the invention is generally presented in the form of individual doses (unit doses).

[0056] Conventional vaccines generally contain about 0.1 to 200 μ g, e.g., 30 to 100 μ g, 0.1 to 2 μ g, 0.5 to 5 μ g, 1 to 10 μ g, 10 μ g to 20 μ g, 15 μ g to 30 μ g, or 10 to 30 μ g, of HA from each of the strains entering into their composition. The vaccine forming the main constituent of the vaccine composition of the invention may comprise a single influenza virus, or a combination of influenza viruses, for example, at least two or three influenza viruses, including one or more reassortant(s).

[0057] Preparations for parenteral administration include sterile aqueous or non-aqueous solutions, suspensions, and/or emulsions, which may contain auxiliary agents or excipients known in the art. Examples of non-aqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Carriers or occlusive dressings can be used to increase skin permeability and enhance antigen absorption. Liquid dosage forms for oral administration may generally comprise a liposome solution containing the liquid dosage form. Suitable forms for suspending liposomes include emulsions, suspensions, solutions, syrups, and elixirs containing inert diluents commonly used in the art, such as purified water.

Besides the inert diluents, such compositions can also include adjuvants, wetting agents, emulsifying and suspending agents, or sweetening, flavoring, or perfuming agents.

[0058] When a composition of the present disclosure is used for administration to an individual, it can further comprise salts, buffers, adjuvants, or other substances which are desirable for improving the efficacy of the composition. For vaccines, adjuvants, substances which can augment a specific immune response, can be used. Normally, the adjuvant and the composition are mixed prior to presentation to the immune system, or presented separately, but into the same site of the organism being immunized.

[0059] Heterogeneity in a vaccine may be provided by mixing replicated influenza viruses for at least two influenza virus strains, such as 2-20 strains or any range or value therein. Vaccines can be provided for variations in a single strain of an influenza virus, using techniques known in the art.

[0060] A pharmaceutical composition according to the present disclosure may further or additionally comprise at least one chemotherapeutic compound, for example, for gene therapy, immunosuppressants, anti-inflammatory agents or immune enhancers, and for vaccines, chemotherapeutics including, but not limited to, gamma globulin, amantadine, guanidine, hydroxybenzimidazole, interferon- α , interferon- β , interferon- γ , tumor necrosis factor- α , thiosemicarbazones, methisazone, rifampin, ribavirin, a pyrimidine analog, a purine analog, foscarnet, phosphonoacetic acid, acyclovir, dideoxynucleosides, a protease inhibitor, or ganciclovir.

[0061] The composition can also contain variable but small quantities of endotoxin-free formaldehyde, and preservatives, which have been found safe and not contributing to undesirable effects in the organism to which the composition is administered.

Pharmaceutical Purposes

[0062] The administration of the composition (or the antisera that it elicits) may be for either a “prophylactic” or “therapeutic” purpose. When provided prophylactically, the compositions of the invention which are vaccines are provided before any symptom or clinical sign of a pathogen infection becomes manifest. The prophylactic administration of the composition serves to prevent or attenuate any subsequent infection. When provided prophylactically, the gene therapy compositions of the invention, are provided before any symptom or clinical sign of a disease becomes manifest. The prophylactic administration of the composition serves to prevent or attenuate one or more symptoms or clinical signs associated with the disease.

[0063] When provided therapeutically, a viral vaccine is provided upon the detection of a symptom or clinical sign of actual infection. The therapeutic administration of the compound(s) serves to attenuate any actual infection. When provided therapeutically, a gene therapy composition is provided upon the detection of a symptom or clinical sign of the disease. The therapeutic administration of the compound (s) serves to attenuate a symptom or clinical sign of that disease.

[0064] Thus, a vaccine composition of the present disclosure may be provided either before the onset of infection (so as to prevent or attenuate an anticipated infection) or after the initiation of an actual infection. Similarly, for gene therapy, the composition may be provided before any symp-

tom or clinical sign of a disorder or disease is manifested or after one or more symptoms are detected.

[0065] A composition is said to be “pharmacologically acceptable” if its administration can be tolerated by a recipient mammal. Such an agent is said to be administered in a “therapeutically effective amount” if the amount administered is physiologically significant. A composition of the present invention is physiologically significant if its presence results in a detectable change in the physiology of a recipient patient, e.g., enhances at least one primary or secondary humoral or cellular immune response against at least one strain of an infectious influenza virus.

[0066] The “protection” provided need not be absolute, i.e., the influenza infection need not be totally prevented or eradicated, if there is a statistically significant improvement compared with a control population or set of mammals. Protection may be limited to mitigating the severity or rapidity of onset of symptoms or clinical signs of the influenza virus infection.

Pharmaceutical Administration

[0067] A composition of the present disclosure may confer resistance to one or more pathogens, e.g., one or more influenza virus strains, by either passive immunization or active immunization. In active immunization, an attenuated live vaccine composition is administered prophylactically to a host (e.g., a mammal), and the host’s immune response to the administration protects against infection and/or disease. For passive immunization, the elicited antisera can be recovered and administered to a recipient suspected of having an infection caused by at least one influenza virus strain. A gene therapy composition may yield prophylactic or therapeutic levels of the desired gene product by active immunization.

[0068] In one embodiment, the vaccine is provided to a mammalian female (at or prior to pregnancy or parturition), under conditions of time and amount sufficient to cause the production of an immune response which serves to protect both the female and the fetus or newborn (via passive incorporation of the antibodies across the placenta or in the mother’s milk).

[0069] The present disclosure thus includes methods for preventing or attenuating a disorder or disease, e.g., an infection by at least one strain of pathogen such as an influenza virus. As used herein, a vaccine is said to prevent or attenuate a disease if its administration results either in the total or partial attenuation (i.e., suppression) of a clinical sign or condition of the disease, or in the total or partial immunity of the individual to the disease. As used herein, a gene therapy composition is said to prevent or attenuate a disease if its administration results either in the total or partial attenuation (i.e., suppression) of a clinical sign or condition of the disease, or in the total or partial immunity of the individual to the disease.

[0070] A composition having at least one influenza virus of the present disclosure, including one which is attenuated and one or more other isolated viruses, one or more isolated viral proteins thereof, one or more isolated nucleic acid molecules encoding one or more viral proteins thereof, or a combination thereof, may be administered by any means that achieve the intended purposes.

[0071] For example, administration of such a composition may be by various parenteral routes such as subcutaneous, intravenous, intradermal, intramuscular, intraperitoneal,

intranasal, oral or transdermal routes. Parenteral administration can be accomplished by bolus injection or by gradual perfusion over time.

[0072] A typical regimen for preventing, suppressing, or treating an influenza virus related pathology, comprises administration of an effective amount of a vaccine composition as described herein, administered as a single treatment, or repeated as enhancing or booster dosages, over a period up to and including between one week and about 24 months, or any range or value therein.

[0073] According to the present disclosure, an “effective amount” of a composition is one that is sufficient to achieve a desired effect. It is understood that the effective dosage may be dependent upon the species, age, sex, health, and weight of the recipient, kind of concurrent treatment, if any, frequency of treatment, and the nature of the effect wanted. The ranges of effective doses provided below are not intended to limit the invention and represent dose ranges.

[0074] The dosage of a live, attenuated or killed virus vaccine for an animal such as a mammalian adult organism may be from about 10^2 - 10^{20} , e.g., 10^3 - 10^{14} , 10^3 - 10^{12} , 10^2 - 10^{10} , 10^5 - 10^{11} , 10^6 - 10^{15} , 10^2 - 10^{10} , or 10^{15} - 10^{20} plaque forming units (PFU)/kg, about 10^2 - 10^{10} , 10^3 - 10^{12} , 10^4 - 10^{10} , 10^5 - 10^{11} , 10^6 - 10^{14} , 10^5 - 10^{12} , or 10^{14} - 10^{20} plaque forming units (PFU)/kg, or any range or value therein. The dose of one viral isolate vaccine, e.g., in an inactivated vaccine, may range from about 0.1 to 1000, e.g., 0.1 to 10 µg, 1 to 20 µg, 30 to 100 µg, 10 to 50 µg, 50 to 200 µg, or 150 to 300 µg, of HA protein. However, the dosage should be a safe and effective amount as determined by conventional methods, using existing vaccines as a starting point.

[0075] The dosage of immunoreactive HA in each dose of replicated virus vaccine may be standardized to contain a suitable amount, e.g., 0.1 µg to 1 µg, 0.5 µg to 5 µg, 1 µg to 10 µg, 10 µg to 20 µg, 15 µg to 30 µg, or 30 µg to 100 µg or any range or value therein, or the amount recommended by government agencies or recognized professional organizations. The quantity of NA can also be standardized.

[0076] The dosage of immunoreactive HA in each dose of replicated virus vaccine can be standardized to contain a suitable amount, e.g., 1-50 µg or any range or value therein, or the amount recommended by the U.S. Public Health Service (PHS), which is usually 15 µg, per component for older children >3 years of age, and 7.5 µg per component for children <3 years of age. The quantity of NA can also be standardized, however, this glycoprotein can be labile during the processor purification and storage (Kendal et al., 1980; Kerr et al., 1975). Each 0.5-ml dose of vaccine may contain approximately 0.1 to 0.5 billion viral particles, 0.5 to 2 billion viral particles, 1 to 50 billion virus particles, 1 to 10 billion viral particles, 20 to 40 billion viral particles, 1 to 5 billion viral particles, or 40 to 80 billion viral particles.

Exemplary Viruses and Vaccine Formulations

[0077] The present disclosure provides a method that stabilizes the NA tetrameric structure of influenza virus, recombinant viruses having the stabilized NA, and methods of using that virus. For example, NA-48C and/or NA-50C mutations in influenza generate highly proliferative recombinant influenza viruses expressing structurally stabilized NA tetramers that can be used to generate influenza vaccine strains containing a greater amount of NA antigen, e.g., so as to elicit an effective immune response. The influenza vaccines can comprise live attenuated viruses or an inacti-

vated (killed) preparation, e.g., whole virus, subunit or split virus preparation, or exogenously expressed NA and HA. There are three types of inactivated vaccines: whole virus vaccines, split virus vaccines (e.g., disrupted by a detergent), and subunit vaccines (i.e., HA and NA have been further purified by removal of other viral components). In one embodiment, the recombinant virus is grown in eggs and a split inactivated vaccine is prepared therefrom. In one embodiment, the dose in a vaccine contains about 10 to about 20, e.g., 15, µg of HA per strain (for example for a total HA concentration of 45 µg for trivalent and 60 µg for quadrivalent) and is administered as a single dose to those aged >9 years. Younger children (between 6 months and 8 years of age) may need two doses administered 4 weeks apart, if they have not been vaccinated in previous influenza seasons. The standard dose is typically delivered as an intramuscular (i.m.) injection (although intradermal [i.d.] formulations and intranasal formulations are also available). **[0078]** In one embodiment, whole-virus vaccines are prepared from harvested allantoic fluid, chemically inactivated, e.g., with formalin or β-propiolactone, and subsequently concentrated and purified to remove nonviral protein contaminants. In one embodiment, the split-virus vaccine includes treatment with detergent to dissociate the viral lipid envelope, exposing all viral proteins and subviral elements. In one embodiment, for subunit vaccines, the HA (and NA) protein is further enriched through additional purification steps. Because the split-virus and subunit vaccines had comparable immunogenicity in primed populations but reduced reactogenicity compared to the whole-virus vaccine preparations, most contemporary vaccines since the 1970s have been split-virus or subunit formulations.

Exemplary NA Modifications

[0079] The present disclosure thus relates to influenza modification relative to parental NA that stabilize the neuraminidase (NA) tetramer, e.g., of human influenza viruses. Those NA modification(s) may also increase the vaccine virus yield.

[0080] Therefore, the disclosure provides isolated recombinant, e.g., reassortant, influenza viruses with selected amino acid residues, insertions, deletions, or any combination thereof, in the stalk region in NA. In one embodiment, the NA is selected to encode a cysteine at residue 48. In one embodiment, the NA is selected to encode a cysteine at position 50. In one embodiment, the NA is selected to encode a cysteine at positions 48 and 50.

[0081] In one embodiment, the NA is selected to have a deletion in one or more of residues 1 to 10 after the last residue in the transmembrane domain, which deletion stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a deletion in one or more of residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the deletion includes a deletion of 1, 2, 3, 4 or 5 residues in the stalk

region. In one embodiment, the deletion includes a deletion of 6, 7, 8, or 9 residues in the stalk region. In one embodiment, the deletion includes a deletion of 10, 11, 12, 13, 14 or 15 residues in the stalk region. In one embodiment, the deletion includes a deletion of 16, 17, 18, or 19 residues in the stalk region. In one embodiment, the deletion includes a deletion of 20, 21, 22, 23, 24 or 25 residues in the stalk region. In one embodiment, the deletion includes a deletion of 26, 27, 28, or 29 residues in the stalk region.

[0082] In one embodiment, the NA is selected to have an insertion of one or more amino acid residues in residues 1 to 10 after the last residue in the transmembrane domain, which insertion stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues in residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have an insertion of one or more amino acid residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the insertion includes an insertion of 1, 2, 3, 4 or 5 residues in the stalk region. In one embodiment, the insertion includes an insertion of 6, 7, 8, or 9 residues in the stalk region. In one embodiment, the insertion includes an insertion of 10, 11, 12, 13, 14 or 15 residues in the stalk region. In one embodiment, the insertion includes an insertion of 16, 17, 18, or 19 residues in the stalk region. In one embodiment, the insertion includes an insertion of 20, 21, 22, 23, 24 or 25 residues in the stalk region. In one embodiment, the insertion includes an insertion of 26, 27, 28, or 29 residues in the stalk region.

[0083] In one embodiment, the NA is selected to have a substitution of one or more amino acid residues in residues 1 to 10 after the last residue in the transmembrane domain, which substitution stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues in residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have a substitution of one or more amino acid residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the substitution to a cysteine is at a residue that faces towards the stalk region of a NA in another NA molecule, e.g., in a dimer, trimer or tetramer.

[0084] In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 1 to 10 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 10 to 20 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is

selected to have one or more cysteines at one or more of residues 20 to 30 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 30 to 40 after the last residue in the transmembrane domain that stabilizes the NA tetramer. In one embodiment, the NA is selected to have one or more cysteines at one or more of residues 40 to 50 after the last residue in the transmembrane domain that stabilizes the NA tetramer. For example, a virus with a NA having a 7 amino acid cytoplasmic tail, a 26 amino acid transmembrane domain and a Cys at residue 48, is a virus that has a cysteine at a residue that is between residues 10 to 20 after the last residue in the transmembrane domain (the numbering for NA is based on N1). In one embodiment, the disclosure provides an isolated recombinant reassortant influenza virus having six “internal” viral segments from a vaccine influenza virus, e.g., PR8UW, a NA viral segment with one or more of the specified modifications, and a HA viral segment, e.g., any of H1-H18, e.g., from a circulating influenza virus. Also provided are compositions comprising the recombinant influenza virus, pharmaceutical compositions such as vaccines.

[0085] Thus, for vaccine viruses that are to be grown or passaged in cells, e.g., in eggs, replacement of the residue at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in NA, e.g., by mutation, or selection of a NA viral segment for a NA with a particular amino acid, e.g., cysteine, at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in NA, wherein the numbering is based on N1, may result in stabilization of NA and/or higher viral titers.

[0086] In one embodiment, the disclosure provides an isolated recombinant influenza virus comprising PA, PB1, PB2, NP, NS, M, and HA viral segments and a NA viral segment that encodes an NA selected to encode a particular amino acid, e.g., cysteine, at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, wherein the numbering is based on N1, wherein the recombinant influenza virus may have enhanced replication in avian eggs or a NA tetramer with enhanced stability, e.g., during vaccine production.

[0087] In one embodiment, the disclosure provides an isolated recombinant influenza virus comprising PA, PB1, PB2, NP, NS, M, and HA viral segments and a NA viral segment that encodes an NA with a replacement (substitution) of a residue at any one of residues from 1 to 60 after the last residue in the transmembrane domain, an insertion of one or more residues from 1 to 60 after the last residue in the transmembrane domain or a deletion of one or more residues from 1 to 60 after the last residue in the transmembrane domain, or any combination thereof, in NA, e.g., by mutation, wherein the numbering is based on N1, wherein

the recombinant influenza virus may have enhanced replication in avian eggs or a NA tetramer with enhanced stability, e.g., during vaccine production.

[0088] In one embodiment, the isolated recombinant influenza virus is a reassortant. In one embodiment, the NA viral segment encodes a NA that has at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or a polypeptide encoded by any one of SEQ ID Nos. 19-29, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or encoded by one of SEQ ID Nos. 19-29. In one embodiment, the NA viral segment encodes a N2, N3, N7, or N9 and the positions in N3, N7, or N9 with the specified modification(s). In one embodiment, the NA viral segment encodes a N1, N4, N5, N6, N8, N10 or N11 with the specified modification(s). In one embodiment, the PA, PB1, PB2, NP, M, and NS viral segments have at least 85% nucleic acid sequence identity to SEQ ID Nos. 30 to 35 or 38 to 43 or encode a polypeptide having at least 80%, 85%, 90%, 95%, or 99 amino acid sequence identity to a polypeptide encoded by SEQ ID Nos. 30-35 or 38 to 43. In one embodiment, the virus is an influenza B virus.

[0089] Also provided is a method to prepare influenza virus having a stabilized NA tetramer. The method includes contacting a cell with: a vector for vRNA production comprising a promoter operably linked to an influenza virus PA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector for vRNA production comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector for vRNA production comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence, wherein the PB1, PB2, PA, NP, NS, and M DNAs in the vectors for vRNA production are from one or more influenza vaccine virus isolates, wherein the NA DNA in the vector for vRNA production has a modification in the stalk region as described herein, wherein the numbering for NA residues is that for N1; and a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PA, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PB1, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus PB2, and a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NP, and optionally comprising one or more of: a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus HA, a vector for mRNA production

comprising a promoter operably linked to a DNA segment encoding influenza virus NA, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus M1, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus M2, a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NS1, or a vector for mRNA production comprising a promoter operably linked to a DNA segment encoding influenza virus NS2; in an amount effective to yield infectious influenza virus. In one embodiment, the NA viral segment encodes a NA that has at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50. In one embodiment, the NA is N1, N4, N5, N6, N8, N10 or N11.

[0090] In one embodiment, the HA is H1, H3, H5, H7, or H9. In one embodiment, the virus is an influenza A virus. In one embodiment, PA, PB1, PB2, NP, M, and NS viral segments have at least 85%, 85%, 90%, 95%, or 99% nucleic acid sequence identity to SEQ ID Nos. 30 to 35 or 38 to 43 or encode a polypeptide having at least 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to a polypeptide encoded by SEQ ID Nos. 30-35 or 38 to 43. In one embodiment, HA is H2, H4, H5, H6, H8, or any of H10-H18.

[0091] In one embodiment, the virus is an influenza B virus.

[0092] Further provided is a method of immunizing an avian or a mammal with a composition having an effective amount of the virus described herein. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

[0093] In one embodiment, the disclosure provides isolated influenza type A virus with a characteristic residue(s), insertion and/or deletion, or a combination thereof, in NA described herein. In one embodiment, the isolated influenza type A virus with a characteristic residue(s), insertion and/or deletion, or a combination thereof, has a NA with at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50, or a polypeptide encoded by any one of SEQ ID Nos. 19-29, or having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or 99% amino acid sequence identity to the stalk region in any one of SEQ ID Nos. 1-9, 15-18, 37, or 44-50 or encoded by one of SEQ ID Nos. 19-29.

[0094] In one embodiment, the isolated influenza type A virus of the invention with a characteristic residue(s) and/or deletion, or a combination thereof, has an HA from any one of subtypes 1-18 of HA. In one embodiment the characteristic residue is a conservative substitution. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids

having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: threonine-valine-leucine-isoleucine-alanine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the characteristic residue is a non-conservative substitution.

[0095] In one embodiment, a mutation is introduced into a NA viral segment of an influenza virus isolate, e.g., via recombinant DNA techniques including site-specific mutagenesis, or replacing a portion of the NA coding sequence with a portion that includes the characteristic residue(s), insertion or deletion. In one embodiment, a NA viral segment with a characteristic residue, insertion and/or deletion described herein is combined with a HA segment, and internal viral segments of an influenza vaccine virus.

[0096] The disclosure provides a plurality of influenza virus vectors of the invention, e.g., those useful to prepare reassortant viruses including 6:1:1 reassortants, 6:2 reassortants and 7:1 reassortants. A 6:1:1 reassortant is an influenza virus with 6 internal viral segments from a vaccine virus, a HA viral segment that is from a different (second) viral isolate than the vaccine virus, and a NA viral segment with a characteristic residue(s), insertion, and/or deletion, or a combination thereof, as described herein, which is from a different viral source than the HA segment and the vaccine virus; a 6:2 reassortant is an influenza virus with 6 internal viral segments from a vaccine virus, and a NA viral segment having a characteristic residue(s), insertion and/or deletion, or a combination thereof, which segment is from the same source as the HA segment, and a HA viral segment from a different viral isolate than the vaccine virus; and a 7:1 reassortant, in one embodiment, is an influenza virus with 6 internal viral segments and a HA segment from a vaccine virus, and a NA segment that is modified to include the characteristic residue(s) and/or deletion, or a combination thereof, which NA segment is from a different viral source than the vaccine virus.

[0097] In one embodiment of the invention, the plurality includes vectors for vRNA production selected from a vector comprising a promoter operably linked to an influenza virus PA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence. In one embodiment, the DNAs for vRNA production of PB1, PB2, PA, NP, M, and NS, have sequences from an influenza virus that replicates to high titers in cultured mammalian cells such as MDCK cells,

Vero cells or PER.C6® cells or embryonated eggs, and/or from a vaccine virus, e.g., one that does not cause significant disease in humans. The DNA for vRNA production of NA may be from any NA, e.g., any of N1-N11, and the DNA for vRNA production of HA may be from any HA, e.g., H1-H18. In one embodiment, the DNAs for vRNA production may be for an influenza B or C virus. For example, the DNAs for vRNA production include influenza B virus PA, PB1, PB2, NP, NS, and M or influenza B virus NA, PB1, PB2, NP, NS, M, and NA, wherein the vRNA for NA has a NA with a characteristic residue, insertion and/or deletion as described herein. The DNAs for vRNA production of NA and HA may be from different strains or isolates (6:1:1 reassortants) or from the same strain or isolate (6:2 reassortants), or the NA or HA may be from the same strain or isolate as that for the internal genes (7:1 reassortant). The plurality also includes vectors for mRNA production selected from a vector encoding influenza virus PA, a vector encoding influenza virus PB1, a vector encoding influenza virus PB2, and a vector encoding influenza virus NP, and optionally one or more vectors encoding NP, NS, M, e.g., M1 and M2, HA or NA. The vectors encoding viral proteins may further include a transcription termination sequence.

[0098] Viruses that may provide the internal genes for reassortants within the scope of the invention include viruses that have high titers, e.g., titers of at least about 10^5 PFU/mL, e.g., at least 10^6 PFU/mL, 10^7 PFU/mL or 10^8 PFU/mL; high titers in embryonated eggs, e.g., titers of at least about 10^7 EID₅₀/mL, e.g., at least 10^8 EID₅₀/mL, 10^9 EID₅₀/mL or 10^{10} EID₅₀/mL; high titers in MDCK cells, e.g., titers of at least about 10^7 PFU/mL, e.g., at least 10^8 PFU/mL, or high titers in two or more of those host cells.

[0099] Other reassortants with internal genes from other PR8 isolates or vaccine viruses may be employed in recombinant reassortant viruses.

[0100] In one embodiment, the DNAs for the internal genes for PB1, PB2, PA, NP, M, and NS encode proteins with substantially the same activity as a corresponding polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. As used herein, "substantially the same activity" includes an activity that is about 0.1%, 1%, 10%, 30%, 50%, 90%, e.g., up to 100% or more, or detectable protein level that is about 80%, 90% or more, the activity or protein level, respectively, of the corresponding full-length polypeptide. In one embodiment, the nucleic acid a sequence encoding a polypeptide which is substantially the same as, e.g., having at least 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to, a polypeptide encoded by one of SEQ ID NOs:30-35 or 38-43. In one embodiment, the isolated and/or purified nucleic acid molecule comprises a nucleotide sequence which is substantially the same as, e.g., having at least 50%, e.g., 60%, 70%, 80% or 90%, including any integer between 50 and 100, or more contiguous nucleic acid sequence identity to one of SEQ ID NOs:30-35 or 38-43 and, in one embodiment, also encodes a polypeptide having at least 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide encoded by one of SEQ ID NOs: 30-35 or 38-43. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 5, 10, 15, 20 or more, conservative amino acids substitutions, e.g., conservative substitutions of up to 10% or 20% of the residues, relative to a polypeptide encoded by one of SEQ ID NOs:

30-35 or 38-43. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 3 or 4, nonconservative amino acid substitutions, relative to a polypeptide encoded by one of SEQ ID NOs:30-35.

[0101] In one embodiment, the nucleic acid is a sequence encoding a NA polypeptide which is substantially the same as, e.g., having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to, one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), the sequences of which are incorporated by reference herein, or at least the stalk region thereof. In one embodiment, the isolated and/or purified nucleic acid molecule encodes a polypeptide having at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, e.g., 90%, 92%, 95%, 97% or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to any one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), the sequences of which are incorporated by reference herein, or at least the stalk region thereof. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 5, 10, 15, 20 or more, conservative amino acids substitutions, e.g., conservative substitutions of up to 10% or 20% of the residues, relative to SEQ ID NOs:1-18, or one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), or at least the stalk region thereof. Conservative amino acid substitutions refer to the interchangeability of residues having similar side chains. For example, a group of amino acids having aliphatic side chains is glycine, alanine, valine, leucine, and isoleucine; a group of amino acids having aliphatic-hydroxyl side chains is serine and threonine; a group of amino acids having amide-containing side chains is asparagine and glutamine; a group of amino acids having aromatic side chains is phenylalanine, tyrosine and tryptophan; a group of amino acids having basic side chains is lysine, arginine and histidine; and a group of amino acids having sulfur-containing side chain is cysteine and methionine. In one embodiment, conservative amino acid substitution groups are: valine-leucine-isoleucine; phenylalanine-tyrosine; lysine-arginine; alanine-valine; glutamic-aspartic; and asparagine-glutamine. In one embodiment, the influenza virus polypeptide has one or more, for instance, 2, 3 or 4, nonconservative amino acid substitutions, relative to a polypeptide having one of SEQ ID NOs:1-18, 3, or one of Accession Nos. ACP41107.1 (N1), AIK26357.1 (N7), ALH21372.1 (N9), or BAK86313.1 (N2), or at least the stalk region thereof.

[0102] The invention thus includes the use of isolated and purified vectors or plasmids, which express or encode influenza virus proteins, or express or encode influenza vRNA, both native and recombinant vRNA. The vectors comprise influenza cDNA, e.g., influenza A (e.g., any influenza A gene including any of the 18 HA or 11 NA subtypes), B or C DNA (see Fields *Virology* (Fields et al. (eds.), Lippincott, Williams and Wickens (2013), which is specifically incorporated by reference herein). Any suitable promoter or transcription termination sequence may be employed to express a protein or peptide, e.g., a viral protein or peptide, a protein or peptide of a nonviral pathogen, or a therapeutic protein or peptide.

[0103] A composition or plurality of vectors of the invention may also comprise a heterologous gene or open reading frame of interest, e.g., a foreign gene encoding an immunogenic peptide or protein useful as a vaccine or in gene replacement, for instance may encode an epitope useful in a cancer therapy or vaccine, or a peptide or polypeptide useful in gene therapy. When preparing virus, the vector or plasmid comprising the gene or cDNA of interest may substitute for a vector or plasmid for an influenza viral gene or may be in addition to vectors or plasmids for all influenza viral genes. Thus, another embodiment of the invention comprises a composition or plurality of vectors as described above in which one of the vectors is replaced with, or further comprises, 5' influenza virus sequences optionally including 5' influenza virus coding sequences or a portion thereof, linked to a desired nucleic acid sequence, e.g., a desired cDNA, linked to 3' influenza virus sequences optionally including 3' influenza virus coding sequences or a portion thereof. In one embodiment, the desired nucleic acid sequence such as a cDNA is in an antisense (antigenomic) orientation. The introduction of such a vector in conjunction with the other vectors described above to a host cell permissive for influenza virus replication results in recombinant virus comprising vRNA corresponding to the heterologous sequences of the vector.

[0104] The promoter in a vector for vRNA production may be a RNA polymerase I promoter, a RNA polymerase II promoter, a RNA polymerase III promoter, a T7 promoter, or a T3 promoter, and optionally the vector comprises a transcription termination sequence such as a RNA polymerase I transcription termination sequence, a RNA polymerase II transcription termination sequence, a RNA polymerase III transcription termination sequence, or a ribozyme. Ribozymes within the scope of the invention include, but are not limited to, tetrahymena ribozymes, RNase P, hammerhead ribozymes, hairpin ribozymes, hepatitis ribozyme, as well as synthetic ribozymes. In one embodiment, the RNA polymerase I promoter is a human RNA polymerase I promoter.

[0105] The promoter or transcription termination sequence in a vRNA or virus protein expression vector may be the same or different relative to the promoter or any other vector. In one embodiment, the vector or plasmid which expresses influenza vRNA comprises a promoter suitable for expression in at least one particular host cell, e.g., avian or mammalian host cells such as canine, feline, equine, bovine, ovine, or primate cells including human cells, or for expression in more than one host.

[0106] In one embodiment, at least one vector for vRNA comprises a RNA polymerase II promoter linked to a ribozyme sequence linked to viral coding sequences linked to another ribozyme sequences, optionally linked to a RNA polymerase II transcription termination sequence. In one embodiment, at least 2, e.g., 3, 4, 5, 6, 7 or 8, vectors for vRNA production comprise a RNA polymerase II promoter, a first ribozyme sequence, which is 5' to a sequence corresponding to viral sequences including viral coding sequences, which is 5' to a second ribozyme sequence, which is 5' to a transcription termination sequence. Each RNA polymerase II promoter in each vRNA vector may be the same or different as the RNA polymerase II promoter in any other vRNA vector. Similarly, each ribozyme sequence in each vRNA vector may be the same or different as the ribozyme sequences in any other vRNA vector. In one embodiment, the ribozyme sequences in a single vector are not the same.

[0107] In one embodiment, at least one vector comprises sequences corresponding to those encoding PB1, PB2, PA, NP, M, or NS, or a portion thereof, having substantially the same activity as a corresponding polypeptide encoded by one of SEQ ID NOs:30-35, e.g., a sequence encoding a polypeptide with at least 80%, e.g., 85%, 90%, 92%, 95%, 98%, 99% or 100%, including any integer between 80 and 100, amino acid identity to a polypeptide encoded by one of SEQ ID NOs:30-35. Optionally, two vectors may be employed in place of the vector comprising a promoter operably linked to an influenza virus M cDNA linked to a transcription termination sequence, e.g., a vector comprising a promoter operably linked to an influenza virus M1 cDNA linked to a transcription termination sequence and a vector comprising a promoter operably linked to an influenza virus M2 cDNA linked to a transcription termination sequence.

[0108] A plurality of the vectors of the invention may be physically linked or each vector may be present on an individual plasmid or other, e.g., linear, nucleic acid delivery vehicle. In one embodiment, each vRNA production vector is on a separate plasmid. In one embodiment, each mRNA production vector is on a separate plasmid.

[0109] The invention also provides a method to prepare influenza virus. The method comprises contacting a cell with a plurality of the vectors of the invention, e.g., sequentially or simultaneously, in an amount effective to yield infectious influenza virus. The invention also includes isolating virus from a cell contacted with the plurality of vectors. Thus, the invention further provides isolated virus, as well as a host cell contacted with the plurality of vectors or virus of the invention. In another embodiment, the invention includes contacting the cell with one or more vectors, either vRNA or protein production vectors, prior to other vectors, either vRNA or protein production vectors. In one embodiment, the promoter for vRNA vectors employed in the method is a RNA polymerase I promoter, a RNA polymerase II promoter, a RNA polymerase III promoter, a T3 promoter or a T7 promoter. In one embodiment, the RNA polymerase I promoter is a human RNA polymerase I promoter. In one embodiment, each vRNA vector employed in the method is

on a separate plasmid. In one embodiment, the vRNA vectors employed in the method are on one plasmid or on two or three different plasmids. In one embodiment, each mRNA vector employed in the method is on a separate plasmid. In one embodiment, the mRNA vectors for PA, PB1, PB2 and NP employed in the method are on one plasmid or on two or three different plasmids.

[0110] The methods of producing virus described herein, which do not require helper virus infection, are useful in viral mutagenesis studies, and in the production of vaccines (e.g., for AIDS, influenza, hepatitis B, hepatitis C, rhinovirus, filoviruses, malaria, herpes, and foot and mouth disease) and gene therapy vectors (e.g., for cancer, AIDS, adenosine deaminase, muscular dystrophy, ornithine transcarbamylase deficiency and central nervous system tumors). Thus, a virus for use in medical therapy (e.g., for a vaccine or gene therapy) is provided.

[0111] The invention also provides isolated viral polypeptides, and methods of preparing and using recombinant virus of the invention. The methods include administering to a host organism, e.g., a mammal, an effective amount of the influenza virus of the invention, e.g., an inactivated virus preparation, optionally in combination with an adjuvant and/or a carrier, e.g., in an amount effective to prevent or ameliorate infection of an animal such as a mammal by that virus or an antigenically closely related virus. In one embodiment, the virus is administered intramuscularly while in another embodiment, the virus is administered intranasally. In some dosing protocols, all doses may be administered intramuscularly or intranasally, while in others a combination of intramuscular and intranasal administration is employed. The vaccine may further contain other isolates of influenza virus including recombinant influenza virus, other pathogen(s), additional biological agents or microbial components, e.g., to form a multivalent vaccine. In one embodiment, intranasal vaccination, for instance containing with inactivated influenza virus, and a mucosal adjuvant may induce virus-specific IgA and neutralizing antibody in the nasopharynx as well as serum IgG.

[0112] The influenza virus of the invention may employed with other anti-virals, e.g., amantadine, rimantadine, and/or neuraminidase inhibitors, e.g., may be administered separately in conjunction with those anti-virals, for instance, administered before, during and/or after.

[0113] One example of an influenza A virus (A/Yokohama/2013/2003(H3N2)) neuraminidase protein sequence and its stalk region is provided below (stalk region is underlined).

(SEQ ID NO: 1)
MNPNQKIITI GSVSLTISTI CFFMQIAILI TTVTLHF~~FKQY~~
EPNSPPNNQV MLCEPTIIER NITEIVYLTN TTIEKEICPK
LAEYRNWSKP QCNITGFAPP SKDNSIRLSA GGDIVVTREP
YVSCDPDKCY QFALGQGTTL NNVHSDIVH DRTPYRTLLM
NELGVFPFLG TKQVCIAWSS SSCHDGKAWL HVCVTGDDEN
ATASFIYNGR LADSIVSWSK KILRTQESEC VCINGTCTTV
MTDGSASGKA DTKILFIEEG KIVHTSTLSG SAQHVEECSC
YPRYPGVRVCV CRDNWKGSNR PIVDINIKDY SIVSSYVCSG

-continued

LVGDTPRKND SSSSSHCLDP NNEEGGHGVK GWAFFDDGNDV

WMGRTISEKL RSGYETFKVI EGWSNPNSKL OINROVIVDR

GNRSGYSGIF SVEGKSCINR CFYVELIRGR KOETEVLWTS

NSIVVFCGTS GTYGTGSWPD GADINLMPI

Another example of an influenza A virus (A/Yokohama/47/2002(H1N2) neuraminidase sequence and its stalk region is shown below (stalk region is underlined).

The amino acid sequence for Singapore0019 NA and its stalk region is as follows (stalk region is underlined):

(SEQ ID NO: 2)

M N P N Q K I I T I G S V S L T I S T I C F F M Q

I A I L I T T V T L H F K Q Y E F N S P P N N Q V

MLCEPTI IERNITEIVYLTNTTIEK

E I C P K P A E Y R N W S K P Q C G I T G F A P F

S K D N S I R L S A G G D I W V T R E P Y V S C D

P D K C Y O F A L G O G T T L N N V H S N N T V R

D R T P Y R T L L M N E L G V P F H L G T K O V C

IAWSSSSCHDGKAWLHVCI TGD DKN

A T A S F I Y N G R L I D S V V S W S K D I L R T

O E S E C V C I N G T C T V V M T D G N A T G K A

D T K I L F I E E G K I V H T S K L S G S A O H V

E E C S C Y P R Y P G V R C V C R D N W K G S N R

P I V D I N I K D H S I V S S Y V C S G L V G D T

P R K N D S S S S S H C L N P N N E E G G H G V K

G W A E D D G N D V W M G R T I N E T S R L G Y E

T E K V V E G W S N P K S K L Q I N R Q V I V D R

G D R S G Y S G I F S V E G K S C I N R C E Y V E

L I R G R K E E T F V L W T S N S I V V E C G T S

G T Y G T G S W P D G A D L N L M H I

The NA for A/Alaska/232/2015 has the following sequence (stalk region is underlined):

(SEO ID NO: 44)

mnnpnqkiiti gsvsltisti cffmqiaili ttvtlhfkyq efnspnnqv mlceptier
niteivyltn ttiekeicpk paeyrnwskp qcgitgfapf skdnsirlsa ggdiwvtrep
yvsdpdkcy qfalgggttl nnvhsnntvr drtpyrllm neltgvpfhlg tkqvciawss
sschdgkawll hvcitgddkn atasfiyngl lvdsvvswsk diltqtseec vcingtctvv
mtdgnatgka dtkilfieeg kivhtsklsg saqhveesc yprypgvrcv crdnwkgasn
pivdinikh siyssyvcsq lvgdtpkrnd ssssshclnp nneegghgvk gwafddgnv
wmgrtinets rlgyetfkvv egwsnpkskl qinrqvivdr gdrsgysgif svegkscinr
cfyvelirqr keetevlwts nsivvfcqts qtyqtqswpd qadlnlmi.

NA nucleotide sequence for A/Singapore/GP1908/2015 (H1N1) pdm09

(SEQ ID NO: 36)

agtttaaaatgaatccaaaccaaagataataaacattgggttcgactcagtatgacaattggaatggctaaacttaattaccaaategg
aaacataatatcaatatggggttagccactcaattcaaaattggaatacaaagccagattgaacatgcaatcaaagcgctattacttat
gaaaaacaaccttgggtaaatacagacatatgttaacatcagcaaccaccaactttgctgctggacagtcagtggtttccgtgaaattag
cgggcgaattcctctctctgccctgttagtggatgggctatatcacgtaaaagacaacagtgtaagaatcggtccaagggggatgtgtt
tgtcataagggaaccattcatatcatgctctcccttggaatgcagaaccttctcttgactcaaggggccttgctaatacacaacat
tccaatggaaccattaaagacaggagcccatatcgaaacctaatgagctgtcctattgggtgaagttccctctccatacaactcaagat

-continued

ttgagtcagtcgcttggtcagcaagtgccttgcatgatggcatcaattggctaacaattggaatttctggcccagacagtggggcagt
 ggctgtgttaaagtaacaatggcataataacagacactatcaagagttggaggaacaatatattgagaacacaagagctgtaattgtga
 tgtgtaaatggttcttctgtttaccataatgaccgatggaccaagtgatggacagcctcatacaaaatcttcagaatagaaaaggaa
 agataatcaaatcagtcgaaatgaaagcccctaattatcactatgaggaatgctcctgttaccctgattctagtgaatcacatgtgt
 gtgcaggataaactggcatggtcgaatcgaccgtgggtgtctttcaaccagaatctggaatatcagatgggatacatatgcagtggtg
 gttttcggagacataccacgcctaataagacaggcagttgtggtccagatcgtctaatggagcaaatggagtaaaaggatttt
 cattcaaatacggcaatggtgtttggataggagaactaaagcattagttcaagaaaagggtttgagatgatttgggatccgaatgg
 atggactgggactgacaataaattctcaataaagcaagatcgttaggaataaatgagtggtcagggatatagcgggagtttgttcag
 catccagaactaacagggttgattgtataagacctgtctctgggtgaactaataagagggcgaccgaagagacacaatctgga
 ctagcgggagcagcatatcctttgtgtgtaaacagtgacactgtgggttggtcttggccagacggtgctgagttgccatttaccat
 tgacaagtaatttgttcaaaaaact

Amino acid sequence for NA of A/Singapore/GP1908/2015 (H1N1) pdm09 (stalk is underlined)

(SEQ ID NO: 37)

MNPNQKIITIGSISMTIGMANLILQIGNIISIWVSHSIQIGNQSIETCNQSVITYENNTWVN

QTYVNISNTNFAAGQSVSVKLAGNSSLCPVSGWAIYSKDNSVRIGSKGDVVFVIREPFISC

SPLRCTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACH

DGINWLTIGISGPDGAVAVLKYNGIITDTIKSWRNILRTQESECACVNGSCFTIMTDGP

SDGQASYKIFRIEKGKIKSVEMKAPNYHYECSYPSSEITCVRDNWHGSNRPWVSF

NQNLEYQMGYICSGVFGDNPRPNDKTGSCGPVSSNGANGVKGFSPKYGNVWIGRTKS

ISSRKGFEMIWDPNWGTGTDNKFSIKQDIVGINEWSGYSGSFVQHPGLTGLDCIRPCFWV

ELIRGRPEENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK

[0114] In some cases, in one or more modifications can also be introduced into HA, PA, PB1, PB2, NP, M1, M2, NS2, PB1-F2, PA-X, and/or NS1 proteins (and nucleic acids encoding such proteins).

[0115] Besides enhanced stability during vaccine production, enhanced growth of the virus when passaged through embryonated chicken eggs or cultured cells may be observed when the modified NA proteins are expressed and such expression may result in significantly higher viral titers. Thus, the invention provides a method for making influenza viruses with enhanced replication in cell culture or embryonated eggs. The method includes providing cells suitable for influenza vaccine production; infecting the cells with viruses having modified neuraminidase; and isolating virus strains with enhanced growth relative to the one or more unmodified viral isolates. In some cases, a method for making influenza viruses with enhanced replication in cell culture can involve, serially culturing one or more influenza virus isolates in embryonated chicken eggs; and isolating serially cultured virus with enhanced growth relative to the one or more isolates prior to serial culture. In some cases, the viruses can be grown or passaged within cells in culture, e.g., MDCK or Vero cells.

[0116] As discussed herein, the modified neuraminidases can be expressed in a variety of influenza strains. For example, A/Puerto Rico/8/34 (H1N1), "PR8," virus often serves as the genetic backbone for generation of inactivated influenza vaccines.

[0117] In one embodiment, vectors for vRNA production can include a vector comprising a promoter operably linked to a modified NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB1 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus PB2 DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus HA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NP DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus NA DNA linked to a transcription termination sequence, a vector comprising a promoter operably linked to an influenza virus M DNA linked to a transcription termination sequence, and a vector comprising a promoter operably linked to an influenza virus NS DNA linked to a transcription termination sequence. In one embodiment, the DNAs for vRNA production of PB1, PB2, PA, NP, M, and NS, have sequences from an influenza virus that replicates to high titers in cultured mammalian cells such as MDCK cells, Vero cells or PER.C6® cells or embryonated eggs, and/or from a vaccine virus, e.g., one that does not cause significant disease in humans. The DNA for vRNA production of NA may be from any NA, e.g., any of N1-N11, and the DNA for vRNA production of HA may be from any HA, e.g., H1-H18. In one embodiment, the DNAs for vRNA production may be for an influenza B or C virus. The DNAs for

vRNA production of NA and HA may be from different strains or isolates (6:1:1 reassortants) or from the same strain or isolate (6:2 reassortants), or the NA may be from the same strain or isolate as that for the internal genes (7:1 reassortant). Vectors for mRNA production can include a vector encoding a modified NA, a vector encoding influenza virus PA, a vector encoding influenza virus PB1, a vector encoding influenza virus PB2, and a vector encoding influenza virus NP, and optionally one or more vectors encoding NP, NS, M, e.g., M1 and M2, HA or NA. The vectors encoding viral proteins may further include a transcription termination sequence.

[0118] Other reassortants with internal genes from other PR8 isolates or vaccine viruses may be employed in recombinant reassortant viruses of the invention. In particular, 5:1:2 reassortants having UW-PR8 PB1, PB2, PA, NP, and M (“5”) and PR8(Cam) NS (“1”); 6:1:1 reassortants having UW-PR8 (modified) NA, PB1, PB2, PA, NP, and M (“6”) and PR8(Cam) NS (“1”); and 7:1 reassortants having UW-PR8 PB1, PB2, PA, NP, M, (modified) NA, and NS (“7”) may be employed.

[0119] The neuraminidases that can be modified can have any sequence including but not limited to the sequences described herein. However, in some cases, the modified

neuraminidases can have substantially the same activity as a corresponding polypeptide described by sequence herein. As used herein, “substantially the same activity” includes an activity that is about 0.1%, 1%, 10%, 30%, 50%, 90%, e.g., up to 100% or more activity, or a detectable protein level that is about 80%, 90% or more protein level, of the corresponding protein described herein. In one embodiment, the nucleic acid encodes a polypeptide which is substantially the same as, e.g., having at least 80%, e.g., 90%, 92%, 95%, 97%, 98%, or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide encoded by one of sequences described herein. In one embodiment, the isolated and/or purified nucleic acid molecule comprises a nucleotide sequence which is substantially the same as, e.g., having at least 50%, e.g., 60%, 70%, 80% or 90%, including any integer between 50 and 100, or more contiguous nucleic acid sequence identity to one of the nucleic acid sequences described herein. In one embodiment, a nucleic acid also encodes a polypeptide having at least 80%, e.g., 90%, 92%, 95%, 97%, 98%, or 99%, including any integer between 80 and 99, contiguous amino acid sequence identity to a polypeptide described herein.

[0120] Exemplary viral sequences for a master vaccine strain (PR8UW) are as follows:

PA (SEQ ID NO: 30)

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AGCGAAAGCA GGTACTGATC CAAAATGGAA GATTTTGTGC GACAATGCTT CAATCCGATG
ATTGTCGAGC TTGCGGAAAA AACAAATGAAA GAGTATGGGG AGGACCTGAA AATCGAAACA
AACAAATTTG CAGCAATATG CACTCACTTG GAAGTATGCT TCATGTATTC AGATTTTCAC
TTCATCAATG AGCAAGGCGA GTCAATAATC GTAGAAGTTG GTGATCCAAA TGCACCTTTG
AAGCACAGAT TTGAAATAAT CGAGGGAAGA GATCGCACAA TGGCCTGGAG AGTAGTAAAC
AGTATTTGCA AACTACAGG GGCTGAGAAA CCAAAGTTTC TACCAGATTT GTATGATTAC
AAGGAGAATA GATTCATCGA AATTGGAGTA ACAAGGAGAG AAGTTCACAT ATACTATCTG
GAAAAGGCCA ATAAAATTAA ATCTGAGAAA ACACACATCC ACATTTTCTC GTTCACTGGG
GAAGAAATGG CCACAAAGGC AGACTACACT CTCGATGAAG AAAGCAGGGC TAGGATCAAA
ACCAGACTAT TCACCATAAG ACAAGAAATG GCCAGCAGAG GCCTCTGGGA TTCCTTTCTG
CAGTCCGAGA GAGGAGAAGA GACAATTGAA GAAAGGTTTG AAATCACAGG AACAAATGCG
AAGCTTGCCG ACCAAAGTCT CCCGCCGAAC TTCTCCAGCC TTGAAAATTT TAGAGCCTAT
GTGGATGGAT TCGAACCGAA CGGCTACATT GAGGGCAAGC TGTCTCAAT GTCCAAAGAA
GTAAATGCTA GAATTGAACC TTTTGTGAAA ACAACACCAC GACCACTTAG ACTTCCGAAT
GGGCCTCCCT GTTCTCAGCG GTCCAATTC CTGCTGATGG ATGCCTTAAA ATTAAGCATT
GAGGACCCAA GTCATGAAGG AGAGGGAATA CCGCTATATG ATGCAATCAA ATGCATGAGA
ACATTCTTTG GATGGAAGGA ACCCAATGTT GTTAAACCAC ACGAAAAGGG AATAAATCCA
AATTATCTTC TGTATGGA GCAAGTACTG GCAGAACTGC AGGACATTGA GAATGAGGAG
AAAATTCCAA AGACTAAAAA TATGAAGAAA ACAAGTCAGC TAAAGTGGGC ACTTGGTGAG
AACATGGCAC CAGAAAAGGT AGACTTTGAC GACTGTAAAG ATGTAGGTGA TTTGAAGCAA
TATGATAGTG ATGAACCAGA ATTGAGGTCG CTTGCAAGTT GGATTCAGAA TGAGTTTAAAC
AAGGCATGCG AACTGACAGA TTCAAGCTGG ATAGAGCTCG ATGAGATTGG AGAAGATGTG
GCTCCAATTG AACACATTGC AAGCATGAGA AGGAATTATT TCACATCAGA GGTGTCTCAC

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TGCAGAGCCA CAGAATACAT AATGAAGGGA GTGTACATCA ATACTGCCTT GCTTAATGCA
TCTTGTGCAG CAATGGATGA TTTCCAATTA ATTCCAATGA TAAGCAAGTG TAGAACTAAG
GAGGGAAGGC GAAAGACCAA CTTGTATGGT TTCATCATAA AAGGAAGATC CCACTTAAGG
AATGACACCG ACGTGGTAAA CTTTGTGAGC ATGGAGTTT CTCTCACTGA CCCAAGACTT
GAACCACATA AATGGGAGAA GTACTGTGTT CTTGAGATAG GAGATATGCT TATAAGAAGT
GCCATAGGCC AGGTTTCAAG GCCCATGTTT TTGTATGTGA GAACAAATGG AACCTCAAAA
ATTAAAAATGA AATGGGGAAT GGAGATGAGG CGTTGCCTCC TCCAGTCACT TCAACAAAT
GAGAGTATGA TTGAAGCTGA GTCCTCTGTC AAAGAGAAAG ACATGACCAA AGAGTTCTTT
GAGAACAAAT CAGAAACATG GCCCATGGA GAGTCCCCA AAGGAGTGA GGAAAGTTCC
ATTGGAAGG TCTGCAGGAC TTTATTAGCA AAGTCGGTAT TCAACAGCTT GTATGCATCT
CCACAAC TAG AAGGATTTT AGCTGAATCA AGAAAAGTGC TTCTTATCGT TCAGGCTCTT
AGGGACAACC TGGAACCTGG GACCTTGAT CTTGGGGGC TATATGAAGC AATTGAGGAG
TGCTGATTA ATGATCCCTG GGTTCCTT AATGCTTCTT GGTCAACTC CTTCTTACA
CATGCATTGA GTTAGTTGTG GCAGTGCTAC TATTTGCTAT CCATACTGTC CAAAAAGTA
CCTTGTTTCT ACT

PB1

(SEQ ID NO: 31)

AGCGAAAGCAGGCAAAACCATTTGAATGGATGTCAATCCGACCTTACTTTTCTTAAAGTGCCAGCACA
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ACTCGTGATTGAAACGATGGAGGTTGTTTACGAAACACGAGTAGACAAGCTGACACAAGGCCGACA
GACCTATGACTGGACTCTAAATAGAAACCAACCTGCTGCAACAGCATTGGCCAACACAATAGAAGTG
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CAGTTTGGGGTGTCTGGGATCAACGAGTCAGCGGACATGAGTATTGGAGTTACTGTCATCAAAAAC
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PB2

(SEQ ID NO: 32)

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GAGCAAGGAC AAACCTTTATG GAGTAAAATG AATGATGCCG GATCAGACCG AGTGATGGTA
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CCTGTCCATT TTAGAACCAC AGTCAAAATA CGTCGGAGAG TTGACATAAA TCCTGGTCAT
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GAATCCAGG ATTGCAAAAT TTCTCCTTTG ATGGTTGCAT ACATGTTGGA GAGAGAACTG
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TTGCATTTGA CTCAAGGAAC ATGCTGGGAA CAGATGTATA CTCCAGGAGG GGAAGTGAGG
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TCCTTGATCG GCTTCGCCGA GATCAGAAAT CCCTAAGAGG AAGGGGCAGT ACTCTCGGTC
TGGACATCAA GACAGCCACA CGTGCTGGAA AGCAGATAGT GGAGCGGATT CTGAAAGAAG
AATCCGATGA GGCCTTAAA ATGACCATGG CCTCTGTACC TGCGTCGCGT TACCTAACTG
ACATGACTCT TGAGGAAATG TCAAGGGAAT GGTCCATGCT CATACCCAAG CAGAAAGTGG
CAGGCCCTCT TTGTATCAGA ATGGACCAGG CGATCATGGA TAAGAACATC ATACTGAAAG
CGAACTTCAG TGTGATTTTT GACCGGCTGG AGACTCTAAT ATTGCTAAGG GCTTTCACCG
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Sequences for the internal segments of another master strain (Cambridge)
are shown below:

(SEQ ID NO: 38)

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(SEQ ID NO: 39)

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t

(SEQ ID NO: 40)

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(SEQ ID NO: 43)

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Exemplary Embodiments

[0121] In one embodiment, an isolated recombinant influenza virus comprising a neuraminidase (NA) viral segment encoding a NA monomer that forms virions having stabilized NA tetramers is provided. In one embodiment, the recombinant influenza virus has a modified NA stalk that results in stabilized tetramers relative to an influenza virus having a NA with an unmodified NA stalk. In one embodiment, the modified NA stalk has a deletion. In one embodiment, the modified NA stalk has an insertion. In one embodiment, the modified NA stalk has at least one amino acid substitution relative to the unmodified stalk. In one embodiment, the modified stalk has two or more of: a deletion, an insertion, or at least one amino acid substitution. In one embodiment, the at least one substitution in the modified NA stalk is a cysteine substitution. In one embodiment, the

modified NA stalk has at least two substitutions. In one embodiment, the NA has a cysteine at position 48 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 50 relative to the numbering of N1. In one embodiment, the NA has a cysteine at position 48 and position 50 relative to the numbering of N1. In one embodiment, the NA stalk is modified within residues 1 to 10 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 10 to 20 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 20 to 30 from the C-terminus of the transmembrane domain. In one embodiment, the NA stalk is modified within residues 30 to 50 from the C-terminus of the transmembrane domain.

[0122] For example, the recombinant influenza virus may have a N1 NA, and that N2 may have one or two cysteines

in the stalk region, e.g., spaced apart with at least one residue in between, e.g., the stalk may have the following sequence:

(SEQ ID NO: 51)
SIQIGNQSQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS,

which could be modified to, for example,
(SEQ ID NO: 52)
IQIGNQSQIECCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS,

(SEQ ID NO: 53)
IQIGNQSQIETCCQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS,

(SEQ ID NO: 54)
IQIGNQSQIECCQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS,

(SEQ ID NO: 55)
IQIGNQSQICTCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS,
or

(SEQ ID NO: 56)
IQIGNQSQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLA
GNSS.

[0123] In one embodiment, a vaccine comprising an effective amount of the recombinant influenza virus or a portion thereof is provided. In one embodiment, the vaccine is a whole virus vaccine. In one embodiment, the vaccine is a split virus vaccine. In one embodiment, the vaccine is a subunit vaccine. In one embodiment, the vaccine further comprises an adjuvant. In one embodiment, the vaccine further comprises a pharmaceutically acceptable carrier. In one embodiment, the carrier is suitable for intranasal or intramuscular administration. In one embodiment, the vaccine further comprises at least one other influenza virus isolate. In one embodiment, the vaccine further comprises at least one other microbe or microbial antigen, e.g., a non-influenza virus, a bacterial or fungal antigen.

[0124] In one embodiment, a method of preparing influenza virus having stabilized NA tetramers is provided. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a NA monomer that forms virions having stabilized NA tetramers, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers. In one embodiment, the NA is N1, N2, N3 or N5. In one embodiment, the cell is a mammalian cell. In one embodiment, the cell is a 293T, PER.C6®, MDCK, MvLu1, CHO or Vero cell, or a cell in an avian egg.

[0125] In one embodiment, a method of making an influenza vaccine is provided comprising: providing the recombinant virus; and combining the virus with an adjuvant or treating the virus with an agent that inactivates the virus. In one embodiment, the method includes aliquoting a dose of the virus into a receptacle. In one embodiment, the adjuvant comprises immunostimulatory DNA sequences, bacterium-derived components, aluminum salt (alum) or squalene oil-in-water emulsion systems such as MF59 and AS03. In

one embodiment, wherein the agent chemically inactivates the virus. In one embodiment, the agent comprises formalin or beta-propiolactone. In one embodiment, the agent comprises a detergent. In one embodiment, the detergent is a non-ionic detergent. In one embodiment, the detergent is a cationic detergent. In one embodiment, the detergent is an anionic detergent. In one embodiment, the detergent comprises CTAB, ammonium deoxycholate, Triton, SDS, Neodol 23-6, or sodium desoxycholate. In one embodiment, the agent comprises ether. In one embodiment, the method further comprises separating HA and NA from other viral components.

[0126] In one embodiment, a method of preparing influenza virus is provided comprising: contacting cells with the recombinant virus in an amount effective to yield progeny virus. In one embodiment, the virus is contacted with an avian egg. In one embodiment, the cells are mammalian cells. In one embodiment, the HA of the virus is H1, H3, H5 or H7.

[0127] Further provided is a method of preparing stabilized NA tetramers. The method includes contacting a cell with one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a NA monomer that forms virions having stabilized NA tetramers and nucleic acid for an influenza virus HA. In one embodiment, the method further comprises isolating NA and HA from the cell. In one embodiment, the cell is an insect cell. In one embodiment, the cell is a CHO, MDCK, Vero, or EB66 cell.

[0128] Also provided is isolated virus prepared by the above-described methods.

[0129] In one embodiment, a method of immunizing an avian or a mammal is provided comprising: administering to the avian or the mammal a composition having an effective amount of the above-described virus. In one embodiment, the composition comprises at least one other different influenza virus. In one embodiment, the mammal is a human. In one embodiment, the composition is administered intranasally or via injection.

[0130] Further provided is a method comprising passaging the virus in eggs.

[0131] The invention will be described by the following non-limiting examples.

Example I

[0132] Neuraminidase (NA) is one of the major transmembrane glycoproteins of influenza viruses. It has been suggested that antibodies against NA play important roles in preventing influenza virus infection. However, the current influenza vaccines, which are made by inactivating egg-grown influenza viruses and purifying virus antigen, do not efficiently elicit the production of anti-NA antibodies. One possible reason for the low production of anti-NA antibodies is the structural instability of the NA protein, which functions as a homo-tetramer; the NA tetramer is apparently disrupted during the antigen purification process. Therefore, the amount of NA contained in vaccines is insufficient to elicit the production of anti-NA antibodies.

[0133] The establishment of a method that stabilizes the NA tetrameric structure may solve this problem. The amino acids 48C and 50C in NA have previously been introduced to NA in vitro (Silva et al., 2013); however, the effect of these amino acids on influenza virus replication or replication efficiency was unknown. As described herein, recombinant influenza viruses containing either the NA-48C or

NA-50C mutation or both mutations expressed a stabilized NA tetramer and replicated efficiently.

Methods

[0134] A 6+2 reassortant influenza viruses containing the HA and NA gene segments from A/Singapore/GP1908/2015 (H1N1) pdm09 in the backbone of high-yield A/Puerto Rico/8/1934 (H1N1) was prepared using reverse genetics and propagation in hCK cells at 37° C. Mutant viruses with either the NA-T48C or NA-N50C mutation or both mutations were generated. hCK cells were infected with these viruses at a MOI (multiplicity of infection) of 1. Cells were lysed at 9 hours post-infection with or without DTT, and NA was detected by western blotting.

Results

[0135] In the presence of DTT, only the band representing the NA monomer was detected. In the absence of DTT, the band representing the WT-NA dimer was detected. For the mutant viruses, bands representing tetrameric NA-T48C, NA-N50C, and NA-T48C/N50C were detected. This result demonstrates that the ratio of tetrameric NA was increased by NA-T48C, NA-N50C, or both. Stabilization may be detected by any method, e.g., sialidase activity.

Conclusion

[0136] The NA-48C and NA-50C mutations, either singly or in combination, can stabilize the NA tetrameric structure. These amino acid mutations are thus helpful to establish a new vaccine strain that can elicit greater amounts of NA antibodies compared with those elicited by current vaccine strains.

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- [0164]** All publications, patents and patent applications are incorporated herein by reference. While in the foregoing specification this invention has been described in relation to certain preferred embodiments thereof, and many details have been set forth for purposes of illustration, it will be apparent to those skilled in the art that the invention is susceptible to additional embodiments and that certain of the details described herein may be varied considerably without departing from the basic principles of the invention.

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WMGRITISEKL RSGYETFKVI EGWSKPNCKL QINRQVIVDR GNRSGYSGIF SVEGKSCINR 420
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TACSDGPGWL TLGITGPAT AVAVLKYNGI ITDTLKS WKG NIMRTQESEC VCQDEFCYTL 240
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                      organism = Influenza A

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STSDHGISR MSICMSGANN NASAVVWYGG RPVTEIPSWA GNILRTQESE CVCHKGICPV 240
VMTDGPANNR AATKIIYFKE GKIQKIEELA GNTQHIEECS CYGAVGVKIC ICRDNWKGAN 300
RPVITIDPEM MHTSKYLCS KILTDTSRPN DPTNGNCDA ITGSGPDGK KGFAPLDREN 360
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YYNETNITNI QMEERTSRNF NNLTGKGLCTI NSWHIYGKDN AVRIGESSDV LVTREPYVSC 120
DPDECRFYAL SQGTTIRGKH SNGTIHDRSQ YRALISWPLS SPPTVYNSRV ECIWGSSTSC 180
HDGKSRMSIC ISGPNNNASA VVWYNRRPVT EINTWARNIL RTQESCEVCH NGVCPVFTD 240
GSATGPADTR IYYFKEGKIL KWESLTGTAK HIEECSCYGE RTGITCTCRD NWQGSNRPVI 300
QDDPVAMTHT SQYICSPVLT DNP RPNDPNI GKCNDPYPGN NNGGVKGFSY LDGANTWLGR 360
TISTASRSYG EMLKVPNALD DRSKPIQGG TIVLNADWSG YSGSFMDYWA EGDYRACFY 420
VELIRGRPKE DKVWWTNSNI VSMCSSTEFL GQWNWPDGAK IEYFL 465

SEQ ID NO: 10     moltype = AA length = 759
FEATURE          Location/Qualifiers
source           1..759
                 mol_type = protein
                 organism = Influenza A

SEQUENCE: 10
MERIKELRNL MSQSRTEIL TKTTVDHMAI IKKYTSGRQE KNPALRMKWM MAMKYPITAD 60
KRITEMIPER NEQGQTLWSK MNDAGSDRVM VSPLAVTWNN RNPMTNTVH YPKIYKTYFE 120
RVERLKHGTF GPVHFRNQVK IRRRVDINPG HADLSAKEAQ DVIMEVFPN EVGARILTSE 180
SQLTTITKEK EELQDCKISP LMVAYMLERE LVRKTRFLPV AGGTSSVYIE VLHLTQGTW 240
EQMYTPGGEV KNDDVDQSLI IAAARNIVRA AVSADPLASL LEMCHSTQIG GIRMVDILKQ 300
NPTEEQAVDI CKAAMGLRIS SSFSPGGFTF KRTSGSSVKR EEEVLTNLQ TLKIRVHEGS 360
EEFTMVGRRA TAILRKATRR LIQLIVSGRD EQSIAEAIIV AMVFSQEDCM IKAVRGDLNF 420
VNRANQRLNP MHQLLRHFQK DAKVLFQNWG VEPIDNVMGM IGILPDMTPS IEMSMRGVRI 480
SKMGVDEYSS TERVVVSIDR FLRVRDQRGN VLLSPEEVSE TQGTETKIT YSSMMWEIN 540
GPESVLVNTY QWIIRNWETV KIQWSQNPTM LYNKMEFEPF QSLVPKAIRG QYSGFVRTLF 600
QQMRDVLGTF DTAQIIKLLP FAAAPPQSR MQFSSTFVNV RSGSMRILVR GNSPVFNYNK 660
ATKRLTVLKG DAGTLTEDPD EGTAGVESAV LRGFLLGKE DRRYGPALSI NELSNLAKGE 720
KANVLIGQGD VVLVMKRKRD SSILTDSQTA TKRIRMAIN 759

SEQ ID NO: 11     moltype = AA length = 757
FEATURE          Location/Qualifiers
source           1..757
                 mol_type = protein
                 organism = Influenza A

SEQUENCE: 11
MDVNPPTLLFL KVPAQNAIST TFPYTGDPY SHGTGTGYTM DTVNRTHQYS EKGRWTTNTE 60
TGAPQLNPID GPLPEDNEPS GYAQTDVLE AMAFLEESHP GIFENSCTET MEVVQQTRVD 120
KLTTQGRQTYD WTLNRNQPA TALANTIEVF RSNGLTANES GRLIDFLKDV MESMKKEMG 180
ITTHFQRKR VRDNMTKKMI TQRTIGKRKQ RLNRKGYLIR ALTLNMTMKD AERGKLRRA 240
IATPGMQIRG FVYFVETLAR SICEKLEQSG LPVGGNEKKA KLANVVRKMM TNSQDTLSF 300
TITGDNKTWN ENQNPRMFLA MITYMTRNQ EWFNRVLSIA PIMFSNKMAR LGKGYMFESK 360
SMKLRQIPA EMLASIDLKY FNDSTRKKIE KIRPLLIET ASLSPGMMMG MFNMLSTVLG 420
VSILNLGQKR YTKTTYWWDG LQSSDDFALI VNAPNHEGIQ AGVDRFYRTC KLLGINMSKK 480

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KSYINRTGTF	EFTSFFYRYG	FVANFSMELP	SFGVSGINES	ADMSGIVTVI	KNNMINNDLG	540
PATAQMALQL	FIKDYRYTYR	CHRGDTQIQT	RRSFEIKKLW	EQTRSKAGLL	VSDGGPNLYN	600
IRNLHIPEVC	LKWELMDEDY	QGRLCNPLNP	FVSHKEIESM	NNAVMMPAHG	PAKNMEYDAV	660
ATTHSWIPKR	NRSILNTSQR	GVLEDEQMYQ	RCCNLFKEFF	PSSSYRRPVG	ISSMVEAMVS	720
RARIDARIDF	ESGRIKKEEF	TEMTKICSTI	EELRRQK			757

SEQ ID NO: 12 moltype = AA length = 716
 FEATURE Location/Qualifiers
 source 1..716
 mol_type = protein
 organism = Influenza A

SEQUENCE: 12

MEDFVRQCFN	PMIVELAEKT	MKEYGEDLKI	ETNKFAAICT	HLEVCFMYSYD	FHFINEQGES	60
IIVELGDPNA	LLKHRFEIIE	GRDRTMAWTV	VNSICNTTGA	EKPFLPDLY	DYKENRFIEI	120
GVTRREVHIY	YLEKANKIKS	EKTHIHIFSF	TGEEMATRAD	YTLDEESRAR	IKTRLFTIRQ	180
EMASRGLWDS	FRQSERGEET	IEERFEITGT	MRKLADQSLP	PNFSSLENFR	AYVDGFEPNG	240
YIEGKLSQMS	KEVNARIEPF	LKTTTPRPLRL	PNGPPCSQRS	KFLMLDALKL	SIEDPSHEGE	300
GIPLYDAIKC	MRTFFGWKEP	NVVKPHEKGI	NPNYLLSWKQ	VLAELQDIEN	EEKIPKTKNM	360
KKTSQLKWL	GENMAPEKVD	FDDCKDVGDL	KQYDSDEPEL	RSLASWIQNE	FNKACELTDS	420
SWIELDEIGE	DVAPIEHAS	MRRNYFTSEV	SHCRATEYIM	KGVYINTALL	NASCAAMDDF	480
QLIPMISKCR	TKEGRRKTNL	YGFIIKGRSH	LRNDTDVVNF	VSMEFSLTDP	RLEPHKWEKY	540
CVLEIGDMLL	RSAIGQVSRP	MFLYVRTNGT	SKIKMKWGMG	MRRCLLQSLQ	QIESMIEAES	600
SVKEKDMTKE	PFENKSETWP	IGESPKGVVEE	SSIGKVCRTL	LAKSVFNSLY	ASPQLEGFSA	660
ESRKKLLLVQ	ALRDNLEPGT	FDLGGLYEAI	EECLINDPWV	LLNASWFNSF	LTHALS	716

SEQ ID NO: 13 moltype = AA length = 498
 FEATURE Location/Qualifiers
 source 1..498
 mol_type = protein
 organism = Influenza A

SEQUENCE: 13

MASQGTKRSY	EQMETDGERQ	NATEIRASVG	KMIGGIGRFPY	IQMCTELKLS	DYEGRLIQNS	60
LTIERMVLSA	FDERRNKYLE	EHPASAGKDPK	KTGGPIYRRV	NGKWMRELIL	YDKEEIRRIW	120
RQANNGDDAT	AGLTHMMIWH	SNLNDATYQR	TRALVRTGMD	PRMCSLMQGS	TLPRRSGAAG	180
AAVKGVGTMV	MELVRMIKRG	INDRNFWRGE	NGRKTRIAYE	RCMNILKGKF	QTAAQKAMMD	240
QVRESRNPGR	AEFEDLTFLA	RSALILRGSV	AHKSCLPACV	YGPVAVASGYD	FEREGYSLVG	300
IDPFRLLQNS	QVYSLIRPNE	NPAHKSQLVW	MACHSAAFED	LRVLSFIKGT	KVVPGRKLST	360
RGVQIASNEN	METMESSTLE	LRSRYWAIRT	RSGGNTNQQR	ASAGQISIQP	TFSVQRNLFP	420
DRTTVMAAFT	GNTBGRSDM	RTEIIRMMES	ARPEDVSFGQ	RGVPFELSDEK	AASPIVPSFD	480
MSNEGSYFFG	DNAEEYDN					498

SEQ ID NO: 14 moltype = AA length = 252
 FEATURE Location/Qualifiers
 source 1..252
 mol_type = protein
 organism = Influenza A

SEQUENCE: 14

MSLLTEVETY	VLSIIPSGPL	KAEIAQRLLED	VFAGKNTDLE	VLMEWLKTRP	ILSPLTKGIL	60
GFVFTLTVP	ERGLQRRRFV	QNALNGNGDP	NNMDKAVKLY	RKLKREITFH	GAKEISLSYS	120
AGALASCMGL	IYNRMGAFTT	EVAFGLVCAT	CEQIADSQHR	SHRQMVTTTN	PLIRHENRMV	180
LASTTAKAME	QMGSSSEQAA	EAMEVASQAR	QMVQAMRTIG	THPSSSAGLK	NDLLENLQAY	240
QKRMGVQMQR	FK					252

SEQ ID NO: 15 moltype = AA length = 469
 FEATURE Location/Qualifiers
 source 1..469
 mol_type = protein
 organism = Influenza A

SEQUENCE: 15

MNPNQKIITI	GSVCMTIGMA	NLILQIGNII	SIWISHSIQL	GNQNQIETCN	QSVITYENNT	60
WYNQTYVNIS	NTNFAAGQSV	VSVKLAGNSS	LCPVSGWAIY	SKDNSVRIGS	KGDVVFVIREP	120
FISCSPLECR	TFFLTQGALL	NDKHSNGTIK	DRSPYRTLMS	CPIGEVPSPY	NSRFESVAWS	180
ASACHDGINW	LTIGISGPDN	GAVAVLKYNQ	IITDTIKSWR	NNILRTQESE	CACVNGSCFT	240
VMTDGPNGQ	ASYKIFRIEK	GKIVKSVEMN	APNYHYECS	CYPDSSEITC	VCRDNWHGNS	300
RPWVSFNQNL	EYQIGYICSG	IFGDNPRPND	KTGSCGPVSS	NGANGVKGFS	FKYGNVGVWG	360
RTKSISSRNG	FEMIWDPNGW	TGTDNNFSIK	QDIVGINEWS	GYSGSFVQHP	ELTGLDCIRP	420
CFWVELIRGR	PKENTIWTSG	SSISFCGVNS	DTVGSWSPDG	AELPFTIDK		469

SEQ ID NO: 16 moltype = AA length = 470
 FEATURE Location/Qualifiers
 source 1..470
 mol_type = protein
 organism = Influenza A

SEQUENCE: 16

MNPNQKLFAL	SGVAIALSIL	NLLIGISNVG	LNVSLHLKGS	SDQDKNWTCT	SVTQNNNTLI	60
ENTYVNNTTV	IDKETGTAKP	NYLMLNKSCL	KVEGVVVVAK	DNAIRFGESE	QIIVTREPVV	120

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SCDPLGCKMY	ALHQQTIRN	KHNGTIHDR	TAFRLISTP	LGSPPVVSNS	DFLCVGWSST	180
SCHDGIGRMT	ICVQGNNDNA	TATVYYDRRL	TTTIKTWAGN	ILRTQESECV	CHNGTCVVIM	240
TDGSASSQAY	TKVLYFHKGL	VIKEEALKGS	ARHIEECSCY	GHNSKVTCVC	RDNWQGANRP	300
VIEIDMNAME	HTSQYLCTGV	LTDTSRPSDK	SMGDCNNPIT	GSPGAPGVKG	FGFLDSSNTW	360
LGRTISPRSR	SGFEMLKIPN	AETDPNSKIT	ERQEIVDNNN	WSGYSGSFID	YWDESSECYN	420
PCFYVELIRG	RPEEAKYVGW	TSNSLIALCG	SPISVSGSGF	PDGAQIQYFS		470

SEQ ID NO: 17 moltype = AA length = 465
 FEATURE Location/Qualifiers
 source 1..465
 mol_type = protein
 organism = Influenza A

SEQUENCE: 17

MNPNQKILCT	SATAIIIGAI	AVLIGIANLG	LNIGLHLKPG	CNCSHSQPET	TNTSQTIINN	60
YNETNITNI	QMEERTSRNF	NNLTGGLCTI	NSWHIYGKDN	AVRIGESSDV	LVTREPYVSC	120
DPDECRFYAL	SQGTIRIGKH	SGNTIHDRSQ	YRALISWPLS	SPPTVYNSRV	ECIGWSSTSC	180
HDGKSRSIC	ISGPMNNASA	VVWYNRRPVA	EINTWARNIL	RTQESECVCH	NGVCPVVFTD	240
GSATGPDTR	IYYFKEGKIL	KWESLTGTAK	HIEECSCYGE	RTGITCTCRD	NWQGSNRPVI	300
QIDPVMTHT	SOYICSPVLT	DNPRPNDPNI	GKCNDPYPGN	NNNGVKGFSY	LDGANTWLGR	360
TISTASRSY	EMLKVPNALT	DDRSKPIQGQ	TIVLNADWSG	YSGSFM DYWA	EGDCYRACFY	420
VELIRGRPE	DKVWWTNSI	VSMCSSTEFL	GQWNWPDGAK	IEYFL		465

SEQ ID NO: 18 moltype = AA length = 469
 FEATURE Location/Qualifiers
 source 1..469
 mol_type = protein
 organism = Influenza A

SEQUENCE: 18

MNPNQKIITI	GSVSLTISTI	CFPMQIAILI	TTVTLHFKQY	EFNSPPNNQV	MLCEPTIIER	60
NITEIVYLTN	TTIEKEICPK	LAERYNWSKP	QCNIITGFAPF	SKDNSIRLSA	GGDIWVTREP	120
YVSCDPDKCY	QFALGQGTTL	NNVHSNDIVH	DRTPYRTLML	NELGVFPHLG	TKQVCIAWSS	180
SSCHDGKAWL	HVCVTGDDEN	ATASFIYNGR	LADSIWSWSK	KILRTQESEC	VCINGTCTVV	240
MTDGSASGKA	DTKILFIEEG	KIVHTSTLSG	SAQHVIEECSC	YPRYPGVRCV	CRDNWKGSNR	300
PIVDINIKDY	SIVSSYVCSG	LVGDTPRKND	SSSSSHCLDP	NNEEGGHGVK	GWAFDDGNDV	360
WMGRITISEKL	RSGYETFKVI	EGWSNPNSKL	QINRQVIVDR	GNRSGYSGIF	SVEGKSCINR	420
CFYVELIRGR	KQETEVLTWS	NSIVVFCGTS	GTGTGGSWPD	GADINLMPI		469

SEQ ID NO: 19 moltype = DNA length = 1434
 FEATURE Location/Qualifiers
 source 1..1434
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 19

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tgctttttca	tgcaaatgtc	cattttgata	actactgtaa	cattgcattt	caagcaatat	120
gaattcaact	ccccccaaa	caaccaagt	atgctgtgtg	aaccaacaat	aatagaaaaga	180
aacataacag	agatagtgtg	tttaaccaac	accaccatag	agaaggaaat	atgccccaaa	240
cagcagaat	acagaaattg	gtcaaaaccg	caatgtggca	ttacaggatt	tgcaaccttc	300
tctaaggaca	attcgatcag	gctttccgct	ggtggggaca	tctgggtgac	aagagaacct	360
tatgtgtcat	gcgatcctga	caagtgttat	caatttgccc	ttggacaggg	aacaacacta	420
aacaacgtgc	attcaataaa	caaaagtacg	gataggaccc	cttatcgga	tctattgatg	480
aatgaattgg	gtgttccttc	ccatctgggg	accaagcaag	tgtgcatagc	atggtccagc	540
tcaagtgtgc	acgatggaaa	agcatggctg	catgtttgta	taacggggga	tgataaaaat	600
gcaactgcta	gcttcattta	caatggggag	cttgtagata	gtgttgtttc	atggtccaaa	660
gatattctca	ggaccacagg	gtcagaatgc	atgtgtatca	atggaaactg	tacagttaga	720
atgactgatg	gaagtgtctc	gtgaaaagct	gataactaaa	tactattcat	tgaggagggg	780
aaaactcgtc	atactagcac	attgtcagga	agtgtctcag	atgtcgaaga	gtgctcttgc	840
tatctctgat	atcctgtgtg	cagatgtgtc	tcagagagaca	actggaaggg	ctccaatcgg	900
cccatcgtag	atataaacat	aaaggatcat	agcattgttt	ccagttatgt	gtgttcaggga	960
ctgtttggag	acacaccacc	aaaaaacgac	agctccagca	gtagccattg	tttggatcct	1020
aacaatgaag	aaggtgtgtca	tgagtgtaaa	ggctggggcct	ttgatgatgg	aaatgacgtg	1080
tggtatggga	gaacaatcaa	cgagacgtca	cgcttagggg	atgaaacctt	caaagtcatt	1140
gaaggtgtgt	ccaaccctaa	gtccaaattg	cagacaaaata	ggcaagtcac	agttgacaga	1200
ggtgataggt	cgggttattc	tggtattttc	tctgttgaag	gcaaaagctg	cataaatcgg	1260
tgcttttatg	tggagttgat	taggggaaga	aaagaggaaa	ctgaagtctt	gtggacctca	1320
aacagtattg	ttgtgttttg	tggcacctca	ggtacatatg	gaacaggctc	atggcctgat	1380
ggggcggaac	tcaatctcat	gcctatataa	gctttcgcaa	ttttagaaaa	aact	1434

SEQ ID NO: 20 moltype = DNA length = 1434
 FEATURE Location/Qualifiers
 source 1..1434
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 20

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tgctttttca	tgcaaatgtc	cattttgata	actactgtaa	cattgcattt	caagcaatat	120

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gaattcaact cccccccaaa caaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgta ttttaaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgatcag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcctga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataaa caaagtacgt gaaaggaccc cttatcggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgta taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atggtccaaa 660
gatattctca ggaccagga gtcagaatgc atttgtatca atggaacttg tacagtagta 720
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aaaatcgttc atactagcac attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
tatcctcgat atcctggtgt cagatgtgtc tgcagagaca actggaaggg ctccaatcgg 900
cccatcgtag atataaacat aaaggatcat agcattgttt ccagttatgt gtgttcagga 960
cttggtggag acacaccagc aaaaaacgac agctccagca gtatgccattg tttggatcct 1020
aacaatgaag aaggtggtgg cggagtgaag ggctgggcct ttgatgatgg aaatgacgtg 1080
tggatgggaa gaacaatcaa cgagaagtca cgcttagggg atgaaacctt caaagtcat 1140
gaaggctggt ccaaccctaa gtccaaattg cagacaaata ggcaagtcat agttgacaga 1200
ggtgataggt ccggttattc tggttatttc tctgttgaag gcaaaagctg catcaatcgg 1260
tgcttttatg tggagtgtat taggggaaga aaagaggaaa ctgaagctt gtggacctca 1320
aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaacc tcaatctcat gcctatataa gctttcgcaa ttttagaaaa aact 1434

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SEQ ID NO: 21 moltype = DNA length = 1410

FEATURE Location/Qualifiers
 source 1..1410
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 21

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tgcttcttca tgcaaatgac catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact cccccccaaa caaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgta ttgaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcctga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataaa cacagtacgt gataggaccc cttatcggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgta taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atggtccaaa 660
gatattctca ggaccagga gtcagaatgc gtttgtatca atggaacttg tacagtagta 720
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aaaatcgttc atactagcac attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
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cccatcgtag atataaacat aaaggatcat agcattgttt ccagttatgt gtgttcagga 960
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ggtgataggt ccggttattc tggttatttc tctgttgaag gcaaaagctg catcaatcgg 1260
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aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaacc tcaatctcat gcctatataa gctttcgcaa ttttagaaaa aact 1410

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SEQ ID NO: 22 moltype = DNA length = 1410

FEATURE Location/Qualifiers
 source 1..1410
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 22

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gaattcaact cccccccaaa caaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgta ttgaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcctga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataaa caaagtacgt gagaggaccc cttatcggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgta taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atggtccaaa 660
gatattctca ggaccagga gtcagaatgc gtttgtatca atggaacttg tacagtagta 720
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tggatgggga	gaacaatcaa	cgagaagtca	cgcttagggc	atgaaacctt	caaagtcgtt	1140
gaaggtcgtg	ccaaccctaa	gtccaaattg	cagataaata	ggcaagtcac	agttgacaga	1200
ggtgataggt	ccggttatcc	tggtattttc	tctgttgaag	gcaaaagctg	catcaatcgg	1260
tgcttttatg	tggagtgtat	taggggaaga	aaagaggaaa	ctgaagtcct	gtggacctca	1320
aacagtattg	ttgtgttttg	tggcacctca	ggtacatatg	gaacaggctc	atggcctgat	1380
ggggcggacc	tcaatctcat	gcataataaa				1410

SEQ ID NO: 23 moltype = DNA length = 1411

FEATURE Location/Qualifiers

source 1..1411

mol_type = other DNA

organism = Influenza A

SEQUENCE: 23

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gaattcaact	ccccccaaa	taaccaagt	atgctgtgtg	aaccaacaat	aatagaaaga	180
aacataacag	agatagtgtg	tttgaccaac	accaccatag	agaaggaaat	atgccccaaa	240
ccagcagaat	acagaaattg	gtcaaaaccc	caatgtggca	ttacaggatt	tgccaccttc	300
tctaagaca	attcgattag	gctttccgct	ggtggggaca	tctgggtgac	aagagaacct	360
tatgtgtcat	gcgatcttga	caagtgttat	caatttgccc	ttggacaggg	aacaacacta	420
aacaacgtgc	attcaaatga	cacagtacgt	gataggaccc	cttatcggac	tctattgatg	480
aatgagttgg	gtgttccttt	ccatctgggg	accaagcaag	tgtgcatagc	atggccacgc	540
tcaagttgtc	acgatggaaa	agcatggctg	catgtttgtg	taacggggga	tgataaaaa	600
gcaactgcta	gcttcattta	cawatgggag	gctttagatg	agtgttgttt	catggcccaa	660
cgatattctc	aggaccacag	agtcagaatg	cgtttgtatc	aatggaacct	gtacagtagt	720
aatgactgat	ggaaatgcta	caggaaaagc	tgataactaa	atactattca	ttgaggaggg	780
gaaaatcgtt	catactagca	aattgtcagg	aagtgtctag	catgtcgaag	agtgtctctg	840
ctatcctcga	tatcctgggt	tcagatgtgt	ctgcagagac	aactggaaa	gatccaaccc	900
gcccatacata	gatataaaca	taaaggatca	tagcattgtt	tccagttatg	tgtgttcagg	960
acttgttga	gacacaccca	gaaaaagcga	cagctccacg	agtagccatt	gtttgaatcc	1020
taacaatgaa	gaagggtggt	atggagtga	aggtctgggc	tttgatgatg	gaaatgacgt	1080
tgggatgggg	agaacaatca	acgagacgtc	acgcttaggg	tatgaaacct	tcaaagtcgt	1140
tgaaggctgg	tccaacccta	agtcacaaat	gcagataaat	aggcaagtca	tagttgacag	1200
aggtgatagg	tccgggtatt	ctgggtattt	ctctgttgaa	ggcaaaagct	gcacatcctg	1260
gtgcttttat	gtggagttga	tcaggggaa	aaaagaggaa	actgaagtct	tgtggacctc	1320
aaacagtatt	gttgtgtttg	gtggcacctc	aggtacatat	ggaacaggct	catggcctga	1380
tggggcggac	ctcaatctca	tgcatatata	a			1411

SEQ ID NO: 24 moltype = DNA length = 1410

FEATURE Location/Qualifiers

source 1..1410

mol_type = other DNA

organism = Influenza A

SEQUENCE: 24

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tgcttcctca	tgcaaatgtc	catcctgata	actactgtaa	cattgcattt	caagcaatat	120
gaattcaact	ccccccaaa	taaccaagt	atgctgtgtg	aaccaacaat	aatagaaaga	180
aacataacag	agatagtgtg	tttgaccaac	accaccatag	agaaggaaat	atgccccaaa	240
ccagcagaat	acagaaattg	gtcaaaaccc	caatgtggca	ttacaggatt	tgccaccttc	300
tctaaggaca	attcgattag	gctttccgct	ggtggggaca	tctgggtgac	aagagaacct	360
tatgtgtcat	gcgatcttga	caagtgttat	caatttgccc	ttggacaggg	aacaacacta	420
aacaacgtgc	attcaaatga	cacagtacgt	gataggaccc	cttatcggac	tctattgatg	480
aatgagttgg	gtgttccttt	ccatctgggg	accaagcaag	tgtgcatggc	atggccacgc	540
tcaagttgtc	acgatggaaa	agcatggctg	catgtttgtg	taactggggg	tgataaaaa	600
gcaactgcta	gcttcattta	caatgggagg	ctttagatga	gtgttgtttc	atggcccaaa	660
gatattctca	ggaccacaga	gtcagaatgc	gtttgcatca	atggaacctg	tacagtagta	720
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aaaatcgttc	atactagcaa	attgtcagga	agtgtctcag	atgtcgaaga	gtgtcctctg	840
tatcctcgat	atcctgtgtg	cagatgtgtc	tgacagagac	actggaaagg	atccaacccg	900
cccattgtag	atataaacat	aaaggatcat	agcattgttt	ccagttatgt	gtgttcaggga	960
cttgttggag	acacaccag	aaaaagcgac	agctccagca	gtagccattg	tttgaatcct	1020
aacaatgaag	aaggtgttca	tggagtgaag	ggctgggct	ttgatgatgg	aaatgacgtg	1080
tggatgggga	gaacaatcaa	cgagacgtca	cgcttagggc	atgaaacctt	caaagtcgtt	1140
gaaggtcgtg	ccaactctaa	gtccaaattg	cagataaata	ggcaagtcac	agttgacaga	1200
ggtgataggt	ccggttatcc	tggtattttc	tctgttgaag	gcaaaagctg	catcaatcgg	1260
tgcttttatg	tggagtgtat	taggggaaga	aaagaggaaa	ctgaagtcct	gtggacctca	1320
aacagtattg	ttgtgttttg	tggcacctca	ggtacatatg	gaacaggctc	atggcctgat	1380
ggggcggacc	tcaatctcat	gcataataaa				1410

SEQ ID NO: 25 moltype = DNA length = 1434

FEATURE Location/Qualifiers

source 1..1434

mol_type = other DNA

organism = Influenza A

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SEQUENCE: 25

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atgaatccaa atcaaaagat aataacgatt ggctctgttt ctctcacaat ttccacaata 60
tgcttcttca tgcaaatgtc catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact cccccccaaa taaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgtg ttggaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcttga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataa cacagtacgt gataggaccc cttatcggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atgggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgtg taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atgggtccaa 660
gatattctca ggaccocagga atcagaatgc gtttgtatca atggaacttg tacagtagta 720
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aaaatcgctc atactagcaa attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
tatcctcgat atcctggtgt cagatgtgtc tgcagagaca actggaaagg atccaaccgg 900
cccatcatag atataaacat aaaggatcat agcattgttt ccagttatgt gtgttcagga 960
cttgttggag acacaccocag aaaaagcgac agctccagca gtatgccattg tttgaatcct 1020
aacaatgaag aaggtgtgtc tggagtgaag ggcctgggct ttgatgatgg aaatgacgtg 1080
tggatggggg gaacaaatcaa cgagacgtca cgcttagggg atgaaacctt caaagtcgtt 1140
gaaggtcgtt ccaaccctaa gtccaaattg cagataaata ggcaagctct agttgacaga 1200
ggtgataggt ccggttattc tggttatttc tctgttgaag gcaaaagctg catcaatcgg 1260
tgcttttatg tggagtgtat taggggaaga aaagaggaaa ctgaagctct gtggacctca 1320
aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaac tcaatctcat gcatatataa gctttcgcaa ttttagaaaa aact 1434

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SEQ ID NO: 26

moltype = DNA length = 1448

FEATURE

Location/Qualifiers

source

1..1448

mol_type = other DNA

organism = Influenza A

SEQUENCE: 26

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tgcttcttca tgcaaatgtc catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact cccccccaaa taaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgtg ttggaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcttga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataa cacagtacgt gataggaccc cttatcggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atgggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgtg taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atgggtccaa 660
gatattctca ggaccocagga gtcagaatgc gtttgtatca atggaacttg tacagtagta 720
atgactgatg gaaatgctac aggaaaagct gacactaaaa tactattcat tgaggagggg 780
aaaatcgctc atactagcaa attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
tatcctcgat atcctggtgt cagatgtgtc tgcagagaca actggaaagg atccaaccgg 900
cccatcatag atataaacat aaaggatcat agcattgttt ccaggtatgt gtgttcagga 960
cttgttggag acacaccocag aaaaagcgac agctccagca gtatgccattg tttgaatcct 1020
aacaatgaag aaggtgtgtc tggagtgaag ggcctgggct ttgatgatgg aaatgacgtg 1080
tggatggggg gaacaaatcaa cgagacgtca cgcttagggg atgaaacctt caaagtcgtt 1140
gaaggtcgtt ccaaccctaa gtccaaattg cagataaata ggcaagctct agttgacaga 1200
ggtgataggt ccggttattc tggttatttc tctgttgaag gcaaaagctg catcaatcgg 1260
tgcttttatg tragtgtgat taggggaaga aaagaggaaa ctgaagctct gtggacctca 1320
aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaac tcaatctcat gcatatataa gctttcgcaa ttttagaaaa aactccttgt 1440
ttctactg 1448

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SEQ ID NO: 27

moltype = DNA length = 1448

FEATURE

Location/Qualifiers

source

1..1448

mol_type = other DNA

organism = Influenza A

SEQUENCE: 27

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tgcttcttca tgcaaatgtc catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact cccccccaaa taaccaagtg atgctgtgtg aaccaacaat aatagaaaga 180
aacataacag agatagtgtg ttggaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcttga caagtgttat caatttgccc ttggacaggg gacaacacta 420
aacaacgtgc attcaataa cacagtacgt gataggaccc cttaccggac tctattgatg 480
aatgagttgg gtgttccttt ccatctgggg accaagcaag tgtgcatagc atgggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgtg taacggggga tgataaaaat 600
gcaactgcta gcttcattta caatgggagg cttgtagata gtgttgtttc atgggtccaa 660
gatattctca ggaccocagga atcagaatgc gtttgtatca atggaacttg tacagtagta 720

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atgactgatg gaaatgctac aggaaaagct gatactaaaa tactattcat cgaggagggg 780
aaaatcattc atactagcaa attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
tatcctcgat atcctgtgtg cagatgtgtc tgcagagaca actggaagg atccaaccg 900
cccatcagag atataaacat aaaggatcat agcattgttt ccagttatgt gtgttcagga 960
cttggtggag acacaccag aaaaagcgac agctccagca gtagccattg tttgaatcct 1020
aacaatgaag aaggtgtgtc tggagtgaag ggctgggcct ttgatgatgg aaatgacgtg 1080
tggatgggga gaacaatcaa cgagacgtca cgcttagggg atgaaacctt caaagtcgtt 1140
gaaggctggg ccaaccctaa gtccaaattg cagataaata ggcaagtcac agttgacaga 1200
ggtgataggt ccggttattc tgggtatttt tctgttgaag gcaaaagctg catcaatcgg 1260
tgcttttatg tggagtgtat taggggaaga aaagaggaaa ctgaagtcct gtggacctca 1320
aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaac tcaatctcat gcatatataa gctttcgcaa ttttagaaaa aaactccttg 1440
ttcttact

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SEQ ID NO: 28      moltype = DNA length = 1410
FEATURE           Location/Qualifiers
source            1..1410
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 28
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tgcttcttca tgcaaatgtc catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact ccccccaaaa taaccaagtg atgctgtgtg aaccaacaat aatagaaaaga 180
aacataacag agatagtgtg tttgaccaac accaccatag agaaggaaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcttga caagtgttat caatttgccc ttggacaggg aacaacacta 420
aacaacgtgc attcaataaa cacagtacgt gatagaacct cttatcggac tctattgatg 480
aatgagtggg gtgttccctt ccatctgggg accaagcaag tgtgcatagc atgggtccagc 540
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gcaactgcta gtttcattta caatgggagg cttgtagata gtgttgtttc atgggtccaa 660
gatattctca ggacccagga gtcagaatgc gtttgtatca atggaacttg tacagtagta 720
atgactgatg gaaatgctac aggaaaagct gatactaaaa tactattcat tgaggagggg 780
aaaatcgttc atactagcaa attgtcagga agtgctcagc atgtcgaaga gtgctcttgc 840
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cccatcagag atataaacat aaaggatcat agcattgttt ccaggtatgt gtgttcagga 960
cttggtggag acacaccag aaaaagcgac agctccagca gtagccattg tttgaacct 1020
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tggatgggga gaacaatcaa cgagacgtcg cgcttagggg atgaaacctt caaagtcgtt 1140
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ggtgataggt ccggttattc tgggtatttt tctgttgaag gcaaaagctg catcaatcgg 1260
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aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaac tcaatctcat gcatatataa gctttcgcaa ttttagaaaa aaactccttg 1410

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SEQ ID NO: 29      moltype = DNA length = 1434
FEATURE           Location/Qualifiers
source            1..1434
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 29
atgaatccaa atcaaaagat aataacgatt ggctctgttt ctctcaccat ttccacaata 60
tgcttcttca tgcaaatgtc catcctgata actactgtaa cattgcattt caagcaatat 120
gaattcaact ccccccaaaa caaccaagtg atgctgtgtg aaccaacaat aatagaaaaga 180
aacataacag agatagtgtg tttgaccaac accaccatag agagggaat atgccccaaa 240
ccagcagaat acagaaattg gtcaaaaccg caatgtggca ttacaggatt tgcacctttc 300
tctaaggaca attcgattag gctttccgct ggtggggaca tctgggtgac aagagaacct 360
tatgtgtcat gcgatcttga caagtgttat caatttgccc ttggacaggg aacaacaata 420
aacaacgtgc attcaataaa cacagcacgt gataggacct ctcatcggac tctattgatg 480
aatgagtggg gtgttccctt ccatctgggg accaagcaag tgtgcatagc atgggtccagc 540
tcaagttgtc acgatggaaa agcatggctg catgtttgta taacggggga tgataaaaat 600
gcaactgcta gtttcattta caatgggagg cttgtagata gtgttgtttc atgggtccaaa 660
gatattctca ggacccagga gtcagaatgc gtttgtatca atggaacttg tacagtagta 720
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tatcctcgat accctgtgtg cagatgtgtc tgcagagaca actggaagg atccaaccg 900
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aacagtattg ttgtgttttg tggcacctca ggtacatatg gaacaggctc atggcctgat 1380
ggggcggaac tcaatctcat gcatatataa gctttcgcaa ttttagaaaa aact 1434

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SEQ ID NO: 30 moltype = DNA length = 2233
 FEATURE Location/Qualifiers
 source 1..2233
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 30

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attgtcgagc	ttgcggaaaa	aacaatgaaa	gagtatgggg	aggacctgaa	aatcgaaaca	120
aacaaatttg	cagcaaatatg	cactcacttg	gaagtatgct	tcatgtattc	agattttcac	180
ttcatcaatg	agcaaggcga	gtcaataatc	gtagaacttg	gtgatccaaa	tgcacttttg	240
aagcacagat	ttgaaataat	cgagggaaga	gatcgacaaa	tggcctggac	agtagtaaac	300
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gagggaaggc	gaaagaccaa	cttgatgggt	ttcatcataa	aagggaagatc	ccacttaagg	1560
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gagaacaaat	cagaacacat	gcccattgga	gagtccccca	aaggagtgga	ggaaaagttc	1920
attgggaagg	tctgcaggag	tttattagca	aagtcgggat	tcaacagctt	gtatgcatct	1980
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tgccctgatta	atgatccctg	gggtttgctt	aatgcttctt	gggtcgaact	cttccttaca	2160
catgcatgga	gttagttgtg	gcagtgctac	tatttgctat	ccatactgtc	caaaaaagta	2220
ccttggttct	act					2233

SEQ ID NO: 31 moltype = DNA length = 2341
 FEATURE Location/Qualifiers
 source 1..2341
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 31

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gggacaggaa	caggatacac	catggatact	gtcaacaggga	cacatcagta	ctcagaaaaag	180
ggaagatgga	caacaaacac	cgaaaactgga	gcaccgcaac	tcaaccgat	tgatggggcca	240
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gctttccttg	aggaatccca	tcttggtatt	tttgaaaact	cggtgattga	aacgatggag	360
gttgttcagc	aaacacgagt	agacaagctg	acacaaggcc	gacagacctc	tgactggact	420
ctaaatagaa	accaacctgc	tgcacacagca	ttggccaaca	caatagaagt	gttcagatca	480
aatggcctca	cgccaatga	gtctggaagg	ctcatagact	tccttaaggga	tgtaattggag	540
tcaatgaaca	aagaagaaat	ggggatcaca	actcattttc	agagaaagag	acgggtgaga	600
gacaatatga	ctaagaaaat	gataaacacag	agaacaatgg	gtaaaaagaa	gcagagattg	660
aacaaaaggga	gttatctaat	tagagcattg	accctgaaca	caatgaccaa	agatgctgag	720
agagggaagc	taaaacggag	agcaattgca	accccaggga	tgcaaaataag	gggggttcta	780
tactttgttg	agacactggc	aaggagtata	tgtgagaaac	ttgaacaatc	aggggttgcca	840
gttgaggaca	atgagaagaa	agcaaaagtg	gcaaatgttg	taagggaagt	gatgaccaat	900
tctcaggaca	cgcaactttc	tttcaccatc	actggagata	acaccaaatg	gaacgaaaat	960
cagaatccct	ggatgttttt	ggccatgac	acatatatga	ccagaaatca	gcccgaaatg	1020
ttcagaaatg	ttctaagtat	tgctccaata	atgttctcaa	acaaaatggc	gagactggga	1080
aaaggggata	tgtttgagag	caagagtatg	aaacttagaa	ctcaaatacc	tgcaaaaatg	1140
ctagcaagca	tcgatttgaa	atatttcaat	gattcaacaa	gaaagaagat	tgaaaaaatc	1200
cgaccgctct	taatataggg	gactgcatca	ttgagccctg	gaatgatgat	gggcatgttc	1260
aatatgttaa	gcactgtatt	aggcgtctcc	atcctgaaat	ttggacaaaa	gagatacacc	1320
aagactactt	actgggtggga	tggtcttcaa	tcctctgacg	attttgcctt	gatttgtaat	1380
gcaccaatc	atgaaggga	tcaagccgga	gtcgacaggt	tttatcgaa	ctgtaagcta	1440
cttggaatca	atatgagcaa	gaaaaagctt	tacataaaca	gaacaggtac	atttgaattc	1500

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acaagttttt tctatcggtta tgggtttgtt gccaatttca gcatggagct tcccagtttt 1560
gggggtgtctg ggatcaacga gtcagcggac atgagttattg gagttactgt catcaaaaac 1620
aatatgataa acaatgatct tgggtccagca acagctcaaa tggcccttca gttgttcatc 1680
aaagattaca ggtacacgta ccgatgccat ataggtgaca cacaataca aaccgaaga 1740
tcatttgaaa taaagaaact gtgggagcaa acccgttcca aagctggact gctgggtotcc 1800
gacggaggcc caaatttata caacattaga aatctccaca ttctgaagt ctgcctaaaa 1860
tgggaattga tggatgagga ttaccagggg cgtttatgca acccactgaa cccatttgtc 1920
agccataaaag aaattgaatc aatgaacaat gcagtgtatga tgccagcaca tgggtccagcc 1980
aaaaacatgg agtatgatgc tgttgcaaca acacactcct ggatccccaa aagaaatcga 2040
tccatcttga atacaagtca aagaggagta cttgaggatg acaaatgta ccaaaggtgc 2100
tgcaatttat ttgaaaaatt ctccccagc agttcataca gaagaccagt cgggatattcc 2160
agtatgggtg aggctatggt ttccagagcc cgaattgatg cacggattga ttcgaaatct 2220
ggaaggataa agaaaagaaga gtctactgag atcatgaaga tctgttccac cattgaagag 2280
ctcagacggc aaaaatagtg aatttagctt gtccttcctg aaaaaatgcc ttgtttctac 2340
t 2341

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SEQ ID NO: 32      moltype = DNA length = 2341
FEATURE            Location/Qualifiers
source              1..2341
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 32
agcgaagca ggtcaattat attcaatatg gaaagaataa aagaactacg aaatctaattg 60
tcgcagtctc gcccccgcga gatactcaca aaaccaccgc tggaccatat ggccataatc 120
aagaagtaca catcaggaa acaggagaag aaccagcac ttaggatgaa atggatgatg 180
gcaatgaaat atccaattac agcagacaag aggataacgg aaatgattcc tgagagaaat 240
gagcaaggac aaactttatg gagtaaaatg aatgatgccg gatcagaccg agtgatggta 300
tcactcttgg ctgtgacatg gtggaatagg aatggaccaa taacaaatc agttcattat 360
ccaaaaatct ctaaaactta ttttgaaaga gtcgaaaggg taaagcatgg aacctttggc 420
cctgtccatt ttagaaacca agtcaaaata cgtcggagag ttgacataaa tctgtgtcat 480
gcagatctca gtgccaaagga ggcacaggat gtaatcatgg aagttgtttt ccctaacgaa 540
gtgggagcca ggatactaac atcggaatcg caactaacga taaccaaaaga gaagaagaa 600
gaactccagg attgcaaaat ttctcctttg atgggttgcat acatgttgga gagagaactg 660
gtccgcaaaa cgagattcct cccagtggct ggtggaacaa gcagtgtgta cattgaagtg 720
ttgcatttga ctcaaggaa atgtcgggaa cagatgtata ctccaggagg ggaagtgagg 780
aatgatgatg ttgatcaaa cttgattatt gctgctagga acatagtgg aagagctgca 840
gtatcagcag atccactagc atctttattg gagatgtgcc acagcacaca gattgttgga 900
attaggatgg tagacatcct taggcagaac ccaacagaag agcaagccgt ggatatatgc 960
aaggctgcaa tgggactgag aattagctca tcttcagtt ttggtggatt cacatttaag 1020
agacaacagc gatcatcagt caagagagag gaagaggtgc ttacgggcaa tcttcaaa 1080
tgaaagataa gaggatgaa gggatagtaa gagttcaca tgggtgggag aagagcaaca 1140
gccatactca gaaaagcaac caggagattg attcagctga tagtgagtgg gagagacgaa 1200
cagtcgattg ccgaagcaat aattgtggcc atggtatttt cacaagagga ttgtatgata 1260
aaagcagtcg gaggtgatct gaatttcgtc aataggcgca atcaacgatt gaatcctatg 1320
catcaacttt taagacattt tcagaaagga gcgaaagtcg tttttcaaaa ttggggagtt 1380
gaactctatc acaatgtgat gggatattgc cgcacatgac tccaagcatc 1440
gagatgtcaa tgagaggagt gagaatcagc aaaatgggtg tagatgagta ctccagcacg 1500
gagaggggtg tgggtgagct tgaccgtttt ttgagaatcc gggaccaacg aggaaatgta 1560
ctactgtctc ccgaggaggt cagtgaaaaca cagggaacag agaaactgac aataacttac 1620
tcactgtcaa tgatgtggga gattaatggt cctgaatcag tgttggtcaa tacctatcaa 1680
tggtatcatc gaaactggga aactgttaaa attcagtggt cccagaaccc tacaatgcta 1740
tacaataaaa tggaaattga accatttcag tctttagtac ctaaggccat tagaggccaa 1800
tacagtgggt ttgtaagaac tctgttccaa caaatgaggg atgtgcttgg gacatttgat 1860
accgcacaga taataaaact tcttcccttc gcagccgctc caccaaagca aagtagaatg 1920
cagttctctc taattactgt gaatgtgagg gcatcaggaa tgagaatact tgaaggggc 1980
aattctctct tattcaacta taacaaggcc acgaagagac tcacagttct cggaaaaggat 2040
gctggcactt taactgaaga ccagatgaa ggcacagctg gagtggaatc cgctgttctg 2100
aggggattcc tcattctggg caaagaagac aagagatatg ggcagcact aagcatcaat 2160
gaactgagca accttgcgaa aggagagaag gctaattgtc taattgggca aggagacgtg 2220
gtgttggtaa tgaacgggaa acgggactct agcactacta ctgacagcca gacagcgacc 2280
aaaagaattc ggatggccat caattagtgt cgaatagttt aaaaacgacc ttgtttctac 2340
t 2341

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SEQ ID NO: 33      moltype = DNA length = 1565
FEATURE            Location/Qualifiers
source              1..1565
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 33
agcaaaagca gggtagataa tcactcactg agtgacatca aaatcatggc gtctcaaggc 60
accaaacgat cttacgaaca gatggagact gatggagaac gccagaatgc cactgaaatc 120
agagcatccg tcggaaaaat gatgtgtgga attggacgat tctacatcca aatgtgcacc 180
gaactcaaac tcagtgtatta tgagggacgg ttgatccaaa acagcttaac aatagagaga 240
atggtgtctc ctgcttttga cgaaggaga aataaatacc ttgaagaaca tcccagtgcg 300
gggaagatc ctaagaaact tggaggacct atatacagga gactaaacgg aaagtggatg 360
agagaactca tcctttatga caaagaagaa ataaggcgaa tctggcgcca agctaataat 420

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ggtagacgatg caacggctgg tctgactcac atgatgatct ggcattccaa tttgaatgat 480
gcaacttatc agaggacaag agctcttgtt cgcacogga tggatcccag gatgtgctct 540
ctgatgcaag gttcaactct ccttaggagg tctggagccg caggtgctgc agtcaaaagga 600
gttggacaa ttggtgatgga attggtcaga atgatcaaac gtgggatcaa tgatcggaac 660
ttctggaggg gtgagaatgg acgaaaaaca agaattgctt atgaaagaat gtgcaacatt 720
ctcaaaagga aatttcaaac tgcgtcacaa aaagcaatga tggatcaagt gagagagagc 780
cggaaccag ggaatgctga gttcgaaagt ctacttttc tagcacgggc tgcactcata 840
ttgagagggg cggttgctca caagtctgc ctgctctcct gtgtgtatgg acctgccgta 900
gccagtgggt acgactttga aagggaggga tactctctag tcggaataga ccctttcaga 960
ctgcttcaaa acagccaagt gtacagccta atcagaccaa atgagaatcc agcacacaag 1020
agtcaactgg tgtggatggc atgccattct gccgcathtt aagatctaag agtattaagc 1080
ttcatcaaa ggacgaaggt gctcccaaga gggaagcttt cactagagg agttcaaatt 1140
gcttccaatg aaaatatgga gactatggaa tcaagtacac ttgaactgag aagcaggtac 1200
tgggccataa gaccagaaag tggaggaaac accaatcaac agagggcatc tgcgggccaa 1260
atcagcatac aacctacgtt ctacgtacag agaaatctcc tttttgacag aacaaccatt 1320
atggcagcat tcaatgggaa tacagagggg agaactctg acatgaggac cgaatcata 1380
aggatgatgg aaagtgcgaag accagaagat gtgtctttcc agggggcggg agtcttcgag 1440
ctctcggacg aaaaggcagc gagcccgatc gtgccttctt ttgacatgag taatgaagga 1500
tcttattctc tcggagacaa tgcagaggag tacgacaatt aaagaaaaat acccttgttt 1560
ctact 1565

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SEQ ID NO: 34      moltype = DNA length = 1027
FEATURE            Location/Qualifiers
source             1..1027
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 34
agcaaaagca ggtagatatt gaaagatgag tcttctaacc gaggtcgaaa cgtacgtact 60
ctctatcatc ccgtcaggcc cctcaaaagc cgagatcgca cagagacttg aagatgtctt 120
tgcaggggaag aacaccgatc ttgaggttct catggaatgg ctaagacaaa gaccaatcct 180
gtcacctctg tcaatgggga ttttaggatt tgtgttcacg ctcccgctgc ccagtgcgcg 240
aggactgcag cgtagacgct ttgtccaaaa tgccttaaat ggggaacggg atccaaataa 300
catggacaaa gcagtttaaac tgtataggaa gctcaagagg gagataacat tccatggggc 360
caaagaaatc tcactcagtt attctgctgg tgcacttgcc agttgtatgg gcctcatata 420
caacaggatg ggggctgtga ccactgaagt ggcatttggc ctggtatgtg caacctgtga 480
acagattgct gactcccagc atcggctcct taggcaaatg gtgacacaaa ccaatccact 540
aatcagacat gagaacagaa tggtttttag cagcactaca gctaaaggct tggagcaaat 600
ggctggatcg agtgagcaag cagcagaggc catggagggt gctagtcagg ctagacaaat 660
ggtgcaagcg atgagaacca ttgggactca tctagctccc agtgctgggc tgaataatga 720
tcttcttgaa aatttgcagg cctatcagaa acgaatgggg gtgcagatgc aacgggtcaa 780
gtgactctct caactattgcc gcaaatatca ttgggactct gcacttgaca ttgtggattc 840
ttgatcgtct ttttttcaaa tgcatttacc gtgcctttaa atacggactg aaaggagggc 900
cttctacgga aggagtgcca aagtttatga gggaagaata tcgaaaggaa cagcagagtg 960
ctgtggatgc tgacgatggg cattttgtca gcatagagct ggagtaaaaa actaccttgt 1020
ttctact 1027

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SEQ ID NO: 35      moltype = DNA length = 890
FEATURE            Location/Qualifiers
source             1..890
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 35
agcaaaagca gggtagacaaa aacataatgg atccaaacac tgtgtcaagc tttcaggtag 60
attgctttct ttggcatgct cgcacacgag ttgcagacca agaactaggc gatgccccat 120
tccttgatcg gcttcgccga gatcagaaat ccctaagagg aaggggcagt actctcggtc 180
tggacatcaa gcagaccaca cgtgctggaa agcagatagt ggagcggatt ctgaaagaag 240
aatccgatga ggcacttaaa atgaccatgg cctctgtacc tgcgtcgcgt tacctaactg 300
acatgactct tgaggaaatg tcaagggact ggtccatgct cataccgaag cagaaagtgg 360
caggccctct ttgtatcaga atggaccagg cgatcatgga taagaacatc atactgaaag 420
cgaacttcag tgtgattttt gaccggctgg agactcta attgctaagg gctttcaccg 480
aagagggagc aattgtttgc gaaatttcac cattgccttc tcttccagga catactgctg 540
aggatgtcaa aaatgcagtt ggagtcctca tcggaggact tgaatggaat gataacacag 600
ttcgagtctc tgaaactcta cagagattcg cttggagaag cagtaatgag aatggggagc 660
ctccactcac tccaaaacag aaacagagaaa tggcggggaac aattagggtc gaagtgtgaa 720
gaaataagat ggttgattga agaagtgaga cacaaactga agataacaga gaatagtttt 780
gagcaaaata catttatgca agccttacat ctattgcttg aagtggagca agagataaga 840
actttctcgt ttcagcttat ttagtactaa aaaacacctt tgtttctact 890

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SEQ ID NO: 36      moltype = DNA length = 1433
FEATURE            Location/Qualifiers
source             1..1433
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 36
agtttaaaat gaatccaaac caaaagataa taaccattgg ttcgatcagt atgacaattg 60
gaatggctaa cttaatatga caaattggaa acataatctc aatatgggtt agccactcaa 120

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-continued

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ttcaaattgg aatcaaaagc cagattgaaa catgcaatca aagcgtcatt acttatgaaa 180
acaacacttg ggtaaatcag acatatgtta acatcagcaa caccaacttt gctgctggac 240
agtcagtggt ttccgtgaaa ttagecgggca attcctctct ctgcctctgt agtggatggg 300
ctatatacag taaagacaac agtgtaagaa tcggttccaa ggggggatgtg tttgtcataa 360
gggaaccatt cataatcagc tctcccttgg aatgcagAAC cttctcttgg actcaagggg 420
ccttgctaaa tgacaaacat tccaatggaa ccattaaaga caggagccca tatcgaaccc 480
taatgagctg tctatttggt gaagttccct ctccatacaa ctcaagattt gagtcagtcg 540
cttggtcagc aagtgtctgt catgatggca tcaattggct aacaattgga atttctggcc 600
cagacagtg ggcagtggt gtgttaaaagt acaatggcat aataacagac actatcaaga 660
gttggaggaa caatatattg agaacacaag agtctgaatg tgcattgtga aatggttctt 720
gctttaccat aatgaccagc ggaccaagtg atggacaggc ctcatacaaa atcttcagaa 780
tagaaaaggg aaagataatc aaatcagtcg aaatgaaagc ccctaattat cactatgagg 840
aatgctcctg ttaccctgat tctagtgaaa tcacatgtgt gtgcagggat aactggcatg 900
gctcgaatcg accgtgggtg tctttcaacc agaattctgga atatcagatg ggatacata 960
gcagtggggt tttcggagac aatccacgcc ctaattgataa gacaggcagt tgtggtccag 1020
tatcgtctaa tggagcaaat ggagtaaaag gattttcatt caaatcggc aatgggtgtt 1080
ggataggagg aactaaaagc attagttcaa gaaaagggtt tgagatgatt tgggatccga 1140
atggatggac tgggactgac aataaattct caataaagca agatatcgta ggaataaatg 1200
agtggtcagg gtatagcggg agttttgttc agcatccaga actaacaggg ctggattgta 1260
taagaccttg cttctgggtt gaactaataa gagggcgacc cgaagagaac acaatctgga 1320
ctagcgggag cagcatatcc ttttgtggtg taaacagtga cactgtgggt tggctctggc 1380
cagacggtgc tgagttgcca tttaccattg acaagtaatt tgttcaaaaa act 1433

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SEQ ID NO: 37 moltype = AA length = 469
FEATURE
source Location/Qualifiers
 1..469
 mol_type = protein
 organism = Influenza A

SEQUENCE: 37

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MNPNQKIITI GSISMTIGMA NLILQIGNII SIWVSHSIQI GNQSQIETCN QSVITYENNT 60
WYNQTYVNIS NTNFAAGQSV VSVKLAGNSS LCPVSGWAIY SKDNSVRIGS KGDVVFIREP 120
FISCSPLECR TFFLTQGALL NDKHSNGTIK DRSPYRTLMS CPIGEVPSPY NSRFESVAWS 180
ASACHDGINW LTIGISGPDG GAVAVLKYNQ IITDTIKSWR NNILRTQESE CACVNGSCFT 240
IMTDGSPSDQ ASYKIFRIEK GKLIKSVEMK APNYHYEELS CYPDSSEITC VCRDNWHGSN 300
RPWVSFNQNL EYQMGYICSG VFGDNPRPND KTGSCGPVSS NGANGVKGFS FKYGNVWIG 360
RTKSISSRRG FEMIWDPNGW TGTDNKFSIK QDIVGINEWS GYSGSFVQHP ELTGGLDCIRP 420
CFWVELIRGR PEENTIWTSG SSISFCGVNS DTVGWSWPDG AELPFTIDK 469

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SEQ ID NO: 38 moltype = DNA length = 2341
FEATURE
source Location/Qualifiers
 1..2341
 mol_type = other DNA
 organism = Influenza A

SEQUENCE: 38

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agcgaaagca ggtcaattat attcaatatg gaaagaataa aagaactaag aaatctaattg 60
tcgcagcttc gcaccccgga gatactcaca aaaaccaccg tggaccatat ggccataatc 120
aagaagtaca catcaggaag acaggagaag aaccacgacac ttaggatgaa atggatgatg 180
gcaatgaaat attcaattac agcagacaag aggataacgg aaatgattcc tgagagaaat 240
gagcaaggac aaactttatg gagtataaatg aatgatgccg gatcagaccg agtgatggta 300
tcacctctgg ctgtgacatg gtggaatagg aatggaccaa tgacaaaatac agttcattat 360
ccaaaaatct acaaaaacta ttttgaaga gtcgaaaggc taaagcatgg aacctttggc 420
cctgtccatt ttgaaaccca agtcaaaata cgtcggagag ttgacataaa tcttggtcat 480
gcagatctca gtgccaagga ggcacaggat gtaatcatgg aagttgtttt ccctaacgaa 540
gtgggagcca ggatactaac atcggaatcg caactaacga taaccaaaaga gaagaagaa 600
gaactccagg attgcaaaat ttctcctttg atggttgcat acatggttga gagagaactg 660
gtccgcacaaa cgagattcct cccagtggtc ggtggaacaa gcagtgtgta cattgaagtg 720
ttgcatttga ctcaaggaa atgctgggaa cagatgtata ctccaggagg ggaagtgaag 780
aatgatgatg ttgatcaaa gcttgattatt gctgctagga acatagtgag aagagctgca 840
gtatcagcag acccactagc atctttattg gagatgtgcc acagcacaca gattggttga 900
attaggatgg tagacatcct taagcagaac ccaacagaag agcaagccgt ggatatatgc 960
aaggctgcaa tgggactgag aattagctca tcttcagtt ttggtggatt cacatttaag 1020
agaacaagcg gatcatcagt caagagagag gaagaggtgc ttacgggcaa tcttcaaaca 1080
ttgaagataa gagtgcatga gggatctgaa gagttcacaa tgggtgggag aagagcaaca 1140
gccatactca gaaaagcaac caggagattg attcagctga tagtgatggg gagagacgaa 1200
cagtcgattg ccgaagcaat aattgtggcc atgggtattt cacaagagga ttgtatgata 1260
aaagcagtta gaggtgatct gaatttcgtc aataggcgca atcagcgact gaatcctatg 1320
catcaacttt taagacattt tcagaaggat gcgaaagtgc tttttcaaaa ttggggagtt 1380
gaacctatcg acaatgtgat gggaatgatt gggatattgc ccgacatgac tccaagcatc 1440
gagatgtcaa tgagaggagt gagaatcagc aaaatgggtg tagatgagta ctccagcacg 1500
gagaggggtg tggtgagcat tgaccgggtt ttgagagta gggaccaacg aggaaatgta 1560
ctactgtctc ccgaggaggt cagtgaacaa cagggaaacag agaaactgac aataacttac 1620
tcatcgtcaa tgatgtggga gattaatggt cctgaatcag tgttgggtcaa tacctatcaa 1680
tggatcatca gaaactggga aactgttaaa attcagtggt ccagaaaccc tacaatgcta 1740
tacaataaaa tggaaattga accatttcag tctttagtag ctaaggccat tagaggccaa 1800
tacagtgggt ttgttaagaac tctgttccaa caaatgaggg atgtgcttgg gacatttgat 1860
accgcacaga taataaaact tcttccttc gcagccgctc caccaaagca aagtagaatg 1920

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cagttctcct catttactgt gaatgtgagg ggatcaggaa tgagaatact tgttaaggggc 1980
aattctcctg tattcaacta caacaaggcc acgaagagac tcacagttct cggaaaggat 2040
gctggcactt taaccgaaga cccagatgaa ggcacagctg gagtggagtc cgctgttctg 2100
aggggattcc tcattctggg caaagaagac aggagatatg ggccagcatt aagcatcaat 2160
gaactgagca accttgcgaa aggagagaag gctaattgtc taattgggca aggagacgtg 2220
gtgttggtta tgaacgaaa acgggactct agcatactta ctgacagcca gacagcgacc 2280
aaaagaattc ggatggccat caattagtgt cgaatagttt aaaaacgacc ttgtttctac 2340
t                                                    2341

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SEQ ID NO: 39      moltype = DNA length = 2341
FEATURE            Location/Qualifiers
source              1..2341
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 39
agcgaagca ggcgaaccat ttgaatggat gtcaatccga ccttactttt cttaaaagtg 60
ccagcacaaa atgctataag cacaactttc ccttataccg gagaccctcc ttacagccat 120
gggacaggaa caggatacac catggatact gtcaacaggga cacatcagta ctcaaaaaag 180
ggaagatgga caacaaacac cgaactgga gacccgcaac tcaaccggat tgatgggcca 240
ctgccagaag acaatgaacc aagtgggtat gcccaaacag attgtgtatt ggaagcaatg 300
gctttccttg aggaatccca tcctgggtatt ttgaaaact cgtgtattga aacgatggag 360
gtgttccagc aaacacgagt agacaagctg acacaaggcc gacagacctg tgactggact 420
ttaaatagaa accagcctgc tgcaacagca ttggccaaca caatagaagt gttcagatca 480
aatggcctca cggccaatga gtcagggaag ctcatagact tccttaaggga tgtaattggag 540
tcaatgaaaa aagaagaaat ggggatcaca actcattttc agagaaagag acgggtgaga 600
gacaatatga ctaagaaaat gataacacag agaacaatag gtaaaaggaa acagagattg 660
aacaaggagg gttatctaata tagagcattg accctgaaca caatgaccaa agatgctgag 720
agagggaagc taaaacggag agcaattgca accccaggga tgcaaatagg ggggtttgta 780
tactttgttg agaacctggc aaggagtata tgtgagaaac ttgaacaatc aggggttgcca 840
gttgaggcca atgagaagaa agcaaaagtgt gcaaatgttg taagggaagt gatgaccaat 900
tctcaggaca ccgaactttc ttccaccatc actggagata acaccaaatg gaacgaaaat 960
cagaatcctc ggaatgtttt ggccatgac acatatatga ccagaaatca gcccgaaatg 1020
ttcagaaatg ttctaagtat tgctccaata atgttctcaa acaaaatggc gagactggga 1080
aaagggtata tgtttgagag caagagtatg aaacttagaa ctcaaatacc tgcagaaatg 1140
ctagcaagca ttgatttgaa atatttcaat gattcaacaa gaaagaagat tgaaaaaatc 1200
cgaccgctct taatagaggg gactgcatca ttgagccctg gaatgatgat gggcatgttc 1260
aatatgttaa gcactgtatt aggcgtctcc atcctgaatc ttggacaaaa gagatacacc 1320
aagactactt actggtggga tggctctcaa tcctctgacg attttgctct gatttgtaat 1380
gcacccaatc atgaagggat tcaagccgga gtccagaggt tttatcgaac ctgtaagcta 1440
cttggaaatc atatgagcaa gaaaaagtct tacataaaca gaacaggtac atttgaattc 1500
acaaagtttt tctatcgtaa tgggtttgtt gccaatttca gcatggagct tcccagtttt 1560
ggggtgtctg ggatcaacga gtcagcggac atgagtattg gagttactgt catcaaaaaac 1620
aatatgataa acaatgatct tgggtccagca acagctcaaa tggcccttca gttgttcac 1680
aaagattaca ggtacacgta ccgatgccat agaggtgaca cacaaataca aaccggaaga 1740
tcatttgaaa taagaaact gtgggagcaa acccggttcca aagctggact gctggctccc 1800
gacggaggcc caaatttata caacattaga aatctccaca ttcctgaagt ctgcctaaaa 1860
tgggaattga tggatgagga ttaccagggg cgtttatgca acccactgaa cccatttgtc 1920
agccataaag aaattgaatc aatgaacaat gcagtatgta tgccagcaca tgggtccagcc 1980
aaaaacatgg agtatgtgc ttgtgcaaca acacactcct ggatcccaa aagaaatcga 2040
tccatcttga atacaagtca aagaggagta cttgaagatg aacaaatgta ccaaagggtc 2100
tgcaatttat ttgaaaaatt ctccccagc agttcataca gaagaccagt cgggatattc 2160
agtatggttg aggtatgtgt ttccagagcc cgaattgatg cacggattga tttcgaatct 2220
ggaaggataa agaaagaaga gttcactgag atcatgaaga tctgttccac cattgaagag 2280
ctcagacggc aaaaatagtg aatttagctt gtccttcacg aaaaaatgcc ttgtttctac 2340
t                                                    2341

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SEQ ID NO: 40      moltype = DNA length = 2233
FEATURE            Location/Qualifiers
source              1..2233
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 40
agcgaagca ggtactgatt caaaatggaa gattttgtgc gacaatgctt caatccgatg 60
attgtcagagc ttgctgaaaa aacaatgaaa gagtatgggg aggacctgaa aatcgaaaca 120
aacaatttg cagcaaatatg cactcacttg gaagtatgct tcatgtattc agatttcac 180
ttcatcaatg agcaaggcga gtcaataatc gtagaacttg gtgaccta taacactttg 240
aagcacagat ttgaataaat cgagggaaga gatcgacaaa tggcctggag agtagtaaac 300
agtatttgca acactacagg ggctgagaaa ccaaagtttc taccagattt gtagattac 360
aaggaaaata gattcatcga aattggagta acaaggagag aagttcacat atactatctg 420
gaaaaggcca ataaaattaa atctgagaaa acacacatcc acattttctc gttcactggg 480
gaagaaatgg ccaacagggc cgactacact ctcgatgaag aaagcagggc taggatcaaa 540
accaggctat tcaccataag acaagaaatg gccagcagag gcctctggga ttctttctgt 600
cagtcagaga gaggagaaga gacaattgaa gaaagggttg aaatcacagg aacaatgcgc 660
aagcttgcgc accaaagtct cccgccgaac ttctccagcc ttgaaaattt tagagcctat 720
gtggatggat tcgaaccgga cggctacatt gagggcaagc tgtctcaaat gtccaaagaa 780
gtaaatgcta gaattgaacc ttttttga aaacacaccg gaccacttag acttccgaat 840

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gggctcctcct gttctcagcg gtccaaatc ctgctgatgg atgccttaaa attaagcatt 900
gaggacccaa gtcacgaagg agaggggaata ccgctatatg atgcaatcaa atgcatgaga 960
acattctttg gatggaagga acccaatggt gttaaacacc acgaaaaggg aataaatcca 1020
aattatcttc tgtcatggaa gcaagtactg gcagaactgc aggacattga gaatgaggag 1080
aaaattccaa agactaaaaa tatgaaaaaa acaagtccag taaagtgggc acttggtgag 1140
aacatggcac cagaaaaggt agactttgac gactgtaaag atgtaggatga tttgaagcaa 1200
tatgatagtg atgaaccaga attgaggtcg cttgcaagtt ggattcagaa tgagttcaac 1260
aaggcatgcg aactgcacaga ttcaagctgg atagagcttg atgagattgg agaagatgtg 1320
gtcccaattg aacacattgc aagcatgaga aggaattatt tcacatcaga ggtgtctcac 1380
tgacagacca cagaatacat aatgaagggg gtgtacatca atactgcctt acttaatgca 1440
tcttgtgcag caatggatga ttccaatta attccaatga taagcaagtg tagaactaag 1500
gaggggaaggc gaaagaccaa cttgtatggt ttcatacataa aaggaaagatc ccacttaagg 1560
aatgacaccg acgtggtaaa ctttgtgagc atggagtttt ctctcactga cccaagactt 1620
gaaccacaca aatgggagaa gtactgtgtt cttgagatag gagatagctt tctaagaagt 1680
gccataggcc aggtttcaag gcccatgttc ttgtatgtga ggacaaatgg aacctcaaaa 1740
attaaaatga aatggggaat ggagatgagg cgttgtctcc tccagtcact tcaacaaatt 1800
gagagtatga ttgaagctga gtctctgttc aaagagaaaag acatgaccaa agagtctctt 1860
gagacaaat cagaaacatg gcccatgtga gagtctccca aaggagtgga ggaagtctcc 1920
attgggaagg tctgcaggac ttattagca aagtcgggat ttaacagctt gtagcatct 1980
ccacaactag aaggattttc agctgaatca agaaaactgc ttcttatcgt tcaggctctt 2040
agggacaatc tggaaactgg gacctttgat cttggggggc tatatgaagc aattgaggag 2100
tgctaataat atgatccctg ggttttgctt aatgcttctt ggttcaactc cttecttaca 2160
catgcattga gttagtgtg gcagtgctac tatttgctat ccatactgtc caaaaaagta 2220
ccttgtttct act 2233

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SEQ ID NO: 41      moltype = DNA length = 1565
FEATURE            Location/Qualifiers
source              1..1565
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 41
agcaaaagca gggtagataa tcactcactg agtgacatca aaatcatggc gtcccaaggc 60
accaaaccgt cttacgaaca gatggagact gatggagaac gccagaatgc cactgaaatc 120
agagcatccg tcggaataat gatggacgat tctacatcca aatgtgcaca 180
gaacttaaac tcaagtatta tgagggacgg ttgatccaaa acagcttaac aatagagaga 240
atggtgctct ctgcttttga cgaaggagga aataaatacc tggaaagaca tcccagtgcg 300
gggaaagatc ctaagaaaaa tggaggacct atatacagaa gagtaaacgg aaagtggatg 360
agagaactca tcctttatga caaagaagaa ataaggcgaa tctggcgcca agctaataat 420
ggtgacgatg caacggctgg tctgactcac atgatgatct ggcattccaa tttgaatgat 480
gcaacttatc agaggacaag ggcctctgtt cgcaccggaa tggatcccag gatgtgctct 540
ctgatgcaag gttcaactct ccctaggagg tctggagccg caggtgctgc agtcaaaagga 600
gttggaacaa tgggtgatgga attggtcagg atgatcaaac gtgggatcaa tgatcggaac 660
ttctggaggg gtgagaatgg acgaaaaaaca agaattgctt atgaaagaat gtgcaacatt 720
ctcaaaagga aatttcaaac tgctgcacaa aaagcaatga tggatcaagt gagagagagc 780
cggaaccacg ggaatgctga gttcgaagat ctcaactttc tagcacggct tgcactcata 840
ttgagaaggg cggttgctca caagtctcgc ctgctgcctt gtgtgatgg acctgcccga 900
gccagtgggt acgactttga aagagaggga tactctctag tcggaataga ccttttcaga 960
ctgcttcaaa acgcccgaat gtacagccta atcagaccaa atgagaatcc agcacacaag 1020
agtcaactgg tctggatggc atgccattct gccgcatttg aagatctaag agtattgagc 1080
ttcatcaaaag ggaagaaagt ggtcccaaga gggaagcttt ccactagagg agttcaaat 1140
gcttccaatg aaaatatgga gactatggaa tcaagtacac ttgaactgag aagcagggtac 1200
tgggccaataa ggaccagaag accaatacaac agaggggcatc tgcgggcca 1260
atcagcatac aacctacggt ttcagtagag agaaaatctc cttttgacag aacaaccggt 1320
atggcagcat tcaactggaa tacagagggg agaacatctg acatgaggac cgaatcata 1380
aggatgatgg aaagtgcagg accagaagat gtgtctttcc agggggcggg agtcttcag 1440
ctctcggaag aaaaggcagc gagcccgatc gtgccttctt ttgacatgag taatgaagga 1500
tcttatttct tcggagacaa tgcagaggag tacgacaatt aaagaaaaat acccttggtt 1560
ctact 1565

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SEQ ID NO: 42      moltype = DNA length = 1027
FEATURE            Location/Qualifiers
source              1..1027
                    mol_type = other DNA
                    organism = Influenza A

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SEQUENCE: 42
agcaaaagca ggtagatatt gaaagatgag tcttctaacc gaggtcgaaa cgtacgttct 60
ctctatcatc ccgtcaggcc cctcctaaagc cgagatcgca cagagacttg aagatgtctt 120
tgacgggaag aacaccgatc ttgagttctt catggaatgg ctaagacaa gaccaatcct 180
gtcacctctg actaagggga ttttaggatt tgtgttcacg ctaccctgc cagtgagcg 240
aggactgcag cgtagacgct ttgtccaaaa tgcccctaat gggaaacggg atccaaataa 300
catggacaaa gcagttaaac tttataggaa gctcaagagg gagataacat tccatggggc 360
caaaagaaatc tcaactcagtt attctgctgg tgcacttgcc agttgtatgg gctcatata 420
caacaggatg ggggctgtga ccactgaagt ggcattttggc ctggtatgtg caacctgtga 480
acagattgct gactcccagc atcggtctca taggcaaatg gtgacacaa ccaaccact 540
aatcagacat gagaacagaa tggttttagc cagcactaca gctaaggcta tggagcaaat 600
ggctggatcg agtgagcaag cagcagaggc catggaggtt gctagtccag ctaggcaaat 660

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gggtgcaagcg atgagaacca ttgggactca tcctagctcc agtgcgtggtc tgaaaaatga 720
tcttcttgaa aatttgcagg cctatcagaa acgaatgggg gtgcagatgc aacggttcaa 780
gtgatcctct cgctattgcc gcaaatatca ttgggatctt gcacttgata ttgtggattc 840
ttgatcgctc ttttttcaaa tgcatttacc gtcgctttaa atacggactg aaaggagggc 900
cttctacgga aggagtgcga aagtctatga gggaagaata tcgaaaggaa cagcagagtg 960
ctgtggatgc tgacgatggt cattttgtca gcatagagct ggagtaaaaa actaccttgt 1020
ttctact 1027

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SEQ ID NO: 43      moltype = DNA length = 890
FEATURE           Location/Qualifiers
source            1..890
                  mol_type = other DNA
                  organism = Influenza A

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SEQUENCE: 43
agcaaaagca ggggtgacaaa gacataatgg atccaaacac tgtgtcaagc ttccaggtag 60
attgctttct ttggcatgtc cgcaaacgag ttgcagacca agaactaggt gatgccccat 120
tcttcttgct gcttcgccga gatcagaaaat ccctaagagg aaggggcagc actcttggtc 180
tgacatcgca gacagccaca cgtgctggaa agcagatagt ggagcggatt ctgaaagaag 240
aatccgatga ggcacttaaa atgacctagg cctctgtacc tgcgtcgctg tacctaaccg 300
acatgactct tgaggaaatg tcaagggaat ggtccatgct catacccaag cagaaagtgg 360
caggccctct ttgtatcaga atggaccagg cgatcatgga taaaacatc atactgaaag 420
cgaaacttcag tgtgattttt gaccggctgg agactctaatt attgtaagg gctttcaccg 480
aagagggagc aattgttggc gaaatttcac cattgccttc tctccagga catactgctg 540
aggatgtcaa aaatgcagtt ggagtcctca tcggaggact tgaatggaat gataacacag 600
ttcgagtcct tgaacttcta cacagattcg cttggagaag cagtaatgag aatgggagac 660
ctccactcac tccaaaacag aaacgagaaa tggcgggaac aattaggtca gaagtttgaa 720
gaaataagat ggttgattga agaagtgaga cacaactga aggtaacaga gaatagtttt 780
gagcaaataa ctttatgca agccttacat ctattgcttg aagtggagca agagataaga 840
actttctcat tttagcttat ttaataataa aaaacacctt tgtttctact 890

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SEQ ID NO: 44      moltype = AA length = 469
FEATURE           Location/Qualifiers
source            1..469
                  mol_type = protein
                  organism = Influenza A

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SEQUENCE: 44
MNPNQKIITI GSVSLTISTI CFFMQIAILI TTVTLHFKQY EFNSPPNNQV MLCEPTIIER 60
NITEIVYLIN TTIEKEICPK PAEYRNWSKP QCGITGFAPF SKDNSIRLSA GGDIWVTREP 120
YVSCDPDKCY QFALGQGTTL NNVHSNNTVR DRTPYRTLML NELGVPPHLG TKQVCIWSS 180
SSCHDGKAWL HVCITGDDKN ATASFIYNGR LVDSVVSWSK DILRTQESAC VCTINGTCTTV 240
MTDGNATGKA DTKILFIEEG KIVHTSKLSG SAQHVEECSC YPRYPGVRCV CRDNWKGSNR 300
PIVDINIKDH SIVSSYVCSG LVGDTPRKND SSSSSHCLNP NNEEGGHGVK GWAFDDGNDV 360
WMGRTINETS RLGYETFKVY EGWSNPKSKL QINRQVIVDR GDRSGYSGIF SVEGKSCINR 420
CFYVELIRGR KEETEVLTWS NSIVVFCGTS GTYGTGSWPD GADLNLMI 469

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SEQ ID NO: 45      moltype = AA length = 466
FEATURE           Location/Qualifiers
source            1..466
                  mol_type = protein
                  organism = Influenza B

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SEQUENCE: 45
MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SRTEITAPIM PLDCANASNV 60
QAVNRSATKG VTLLEPEPEW MYPRLSCPGS TFQKALLISP HRFGETKGNS APLIIREPFI 120
ACGPKECKHF ALTHYAAQPG GYNGTREDR NKLRHLISVK LGKIPTVENS IFHMAAWSGS 180
ACHDGREWTY IGVDGPDNSA LLKIYGEAY TDTYHSYAKN TLRTQESACN CIGGDCYLM 240
TDGPASGISE CRFLKIREGR IIEIIFPTGR VKHTEECTCG FASNKTEICA CRDNSYTAKR 300
PFVKLNIVETD TAEIRLMCTE TYLDTPRPND GSITGPCESE GDKSGGGIKG GFVHQRMASK 360
IGRWYSRTMS KTKRMGMGLY VKYDGDPTWD SEALALSGVM VSMEEPGWYS FGFEIKDKKC 420
DVPCIGIEMV HDGKKTWHS AATAIYCLMG SQQLLWDTVT GVNMTL 466

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```

SEQ ID NO: 46      moltype = AA length = 466
FEATURE           Location/Qualifiers
source            1..466
                  mol_type = protein
                  organism = Influenza B

```

```

SEQUENCE: 46
MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SPTEITAPT M PLDCANASNV 60
QAVNRSATKG VTLLEPEPEW TYPRLSCPGS TFQKALLISP HRFGETKGNS APLIIREPFI 120
ACGPKECKHF ALTHYAAQPG GYNGTREDK NKLRHLISVK LGKIPTVENS IFHMAAWSGS 180
ACHDGREWTY IGVDGPDNSA LLKIYGEAY TDTYHSYANN ILRTQESACN CIGGDCYLM 240
TDGSASGISE CRFLKIREGR IIEIIFPTGR VEHTTEECTCG FASNKTEICA CRDNSYTAKR 300
PFVKLNIVETD TAEIRLMCTE TYLDTPRPDD GSITGPCESE GDKSGGGIKG GFVHQRMASK 360
IGRWYSRTMS KTKRMGMGLY VKYDGDPTWD SDALALSGVM VSMEEPGWYS FGFEIKDKKC 420
DVPCIGIEMV HDGKKTWHS AATAIYCLMG SQQLLWDTVT GVDML 466

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SEQ ID NO: 47 moltype = AA length = 466
 FEATURE Location/Qualifiers
 source 1..466
 mol_type = protein
 organism = Influenza B

SEQUENCE: 47

MLPSTIQTLT	LFLTSGGVLL	SLYVSASLSY	LLYSDILLKF	SPTEITAPTM	PLDCANASNV	60
QAVNRSATKG	VTLLPPEPEW	TYPRLSCPGS	TFQKALLISP	HRFGETKGNS	APLIIREPFI	120
ACGPKECKHF	ALTHYAAQPG	GYNGTRGDR	NKLRHLISVK	LGKIPTVENS	IFHMAAWSGS	180
ACHDGKEWTY	IGVDGPDNNA	LLIKIKYGEAY	TDTHYSYANN	ILRTQESACN	CIGGNCYLMI	240
TDGSASGVSE	CRFLKIREGR	IIKEIFPTGR	VKHTTECTCG	FASNKTIECA	CRDNSYTAKR	300
PFVKLNVEDT	TAEIRLMCTE	TYLDTPRPDD	GSITGPCESN	GDKGSGGIKG	GFVHQRMASK	360
IGRWYSRTMS	KTKRMGMGLY	VKYDGDPPAD	SDALALSGVM	VSMEEP GWYS	FGFEIKDKKC	420
DVPCIGIEMV	HDGKETWHS	AATAIYCLMG	SGQLLWDTVT	GVDMAL		466

SEQ ID NO: 48 moltype = AA length = 465
 FEATURE Location/Qualifiers
 source 1..465
 mol_type = protein
 organism = Influenza B

SEQUENCE: 48

MLPSTIQTLT	LFLTSGGVLL	SLYVSASLSY	LLYSDILLKF	SPKITAPTMT	LDCANASNVQ	60
AVNRSATKEM	TFLPPEPEWT	YPRLSCQGST	FQKALLISPH	RFGEARGNSA	PLIIREPFIA	120
CGPKECKHFA	LTHYAAQPGG	YYNGTREDRN	KLRHLISVKL	GKIPTVENS	IFHMAAWSGSA	180
CHDGREWTYI	GVDGPDNSAL	IKIKYGEAYT	DTYHSYANNI	LRTQESACNC	IGGDCYLMIT	240
DGSASGISKC	RFLKIREGRI	IKEIFPTGRV	EHTTECTCGF	ASNKTIECAC	RDNNYTAKRP	300
FVKLNVEDTD	AEIRLMCTET	YLDTPRPDDG	SITGPCESNG	DKGRGGIKGG	FVHQRMASKI	360
GRWYSRTMSK	TERMGMELYV	KYDGDPTWDS	DALDPSGVMV	SMKEPGWYSF	GFEIKDKKCD	420
VPCIGIEMVH	DGGKKTWHS	ATAIYCLMGS	GQLLWDTVTG	VDMAL		465

SEQ ID NO: 49 moltype = AA length = 466
 FEATURE Location/Qualifiers
 source 1..466
 mol_type = protein
 organism = Influenza B

SEQUENCE: 49

MLPSTIQTLT	LFLTSGGVLL	SLYVSASLSY	LLYSDILLKF	SPTKRTAPTM	SLDCANVSNV	60
QAVNRSATKE	MTFLPPEPEW	TYPRLSQCGS	TFQKALLISP	HRFGEARGNS	APLIIREPFI	120
ACGPKECKHF	ALTHYAAQPG	GYNGTRKDR	NKLRHLISVK	LGKIPTVENS	IFHMAAWSGS	180
ACHDGREWTY	IGVDGPDNSA	LLIKIKYGEAY	TDTHYSYANN	ILRTQESACN	CIGGDCYLMI	240
TDGSASGISK	CRFLKIREGR	IIKEIFPTGR	VEHTTECTCG	FASNKTIECA	CRDNSYTAKR	300
PFVKLNVEDT	TAEIRLMCTE	TYLDTPRPDD	GSITGPCESN	GDKGLGGIKG	GFVHQRMASK	360
IGRWYSRTMS	KTERMGMELY	VKYDGDPTWD	SEALAPSGVM	VSMKEPGWYS	FGFEIKDKKC	420
DVPCIGIEMV	HDGKETWHS	AATAIYCLMG	SGQLLWDTVT	GVDMAL		466

SEQ ID NO: 50 moltype = AA length = 466
 FEATURE Location/Qualifiers
 source 1..466
 mol_type = protein
 organism = Influenza B

SEQUENCE: 50

MLPSTIQTLT	LFLTSGGVLL	SLYVSASLSY	LLYSDILLKF	SRTEVTAPIM	PLDCANASNV	60
QAVNRSATKG	VTPLPPEPEW	TYPRLSCPGS	TFQKALLISP	HRFGETKGNS	APLIIREPFI	120
ACGPKECKHF	ALTHYAAQPG	GYNGTREDR	NKLRHLISVK	LGKIPTVENS	IFHMAAWSGS	180
ACHDGREWTY	IGVDGPDNSA	LLIKIKYGEAY	TDTHYSYAKN	ILRTQESACN	CIGGDCYLMI	240
TDGPASGISE	CRFLKIREGR	IIKEIFPTGR	VKHTTECTCG	FASNKTIECA	CRDNSYTAKR	300
PFVKLNVEDT	TAEIRLMCTK	TYLDTPRPND	GSITGPCESD	GDEGSGGIKG	GFVHQRMASK	360
IGRWYSRTMS	KTKRMGMGLY	VKYDGDPTWD	SEALALSGVM	VSMEEP GWYS	FGFEIKDKKC	420
DVPCIGIEMV	HDGKTTWHS	AATAIYCLMG	SGQLLWDTVT	GVNMTL		466

SEQ ID NO: 51 moltype = AA length = 54
 FEATURE Location/Qualifiers
 source 1..54
 mol_type = protein
 organism = Influenza A

SEQUENCE: 51

SIQIGNQSQI	ETCNQSVITY	ENNTWVNQTY	VNISNTNF	AA	GQSVSVKLA	GNSS	54
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SEQ ID NO: 52 moltype = AA length = 53
 FEATURE Location/Qualifiers
 REGION 1..53
 note = A synthetic sequence

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source          1..53
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 52
IQIGNQSQIE CCNQSVITYE NNTWVNQTYV NISNTNFAAG QSVVSVKLAG NSS      53

SEQ ID NO: 53      moltype = AA length = 53
FEATURE           Location/Qualifiers
REGION            1..53
                 note = A synthetic sequence
source            1..53
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 53
IQIGNQSQIE TCCQSVITYE NNTWVNQTYV NISNTNFAAG QSVVSVKLAG NSS      53

SEQ ID NO: 54      moltype = AA length = 53
FEATURE           Location/Qualifiers
REGION            1..53
                 note = A synthetic sequence
source            1..53
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 54
IQIGNQSQIE CCCQSVITYE NNTWVNQTYV NISNTNFAAG QSVVSVKLAG NSS      53

SEQ ID NO: 55      moltype = AA length = 53
FEATURE           Location/Qualifiers
REGION            1..53
                 note = A synthetic sequence
source            1..53
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 55
IQIGNQSQIC TCNQSVITYE NNTWVNQTYV NISNTNFAAG QSVVSVKLAG NSS      53

SEQ ID NO: 56      moltype = AA length = 53
FEATURE           Location/Qualifiers
REGION            1..53
                 note = A synthetic sequence
source            1..53
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 56
IQIGNQSQIE TCNCSVITYE NNTWVNQTYV NISNTNFAAG QSVVSVKLAG NSS      53

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What is claimed is:

1. A method of preparing influenza virus having stabilized NA tetramers or of preparing stabilized NA tetramers, comprising: contacting a cell with a) one or more vectors comprising nucleic acid for an influenza virus neuraminidase (NA) segment encoding a modified NA monomer that forms virions having stabilized NA tetramers relative to a corresponding influenza virus having a parental NA viral segment encoding an unmodified NA monomer, nucleic acid for an influenza virus PA segment, nucleic acid for an influenza virus PB1 segment, nucleic acid for an influenza virus PB2 segment, nucleic acid for an influenza virus NP segment, nucleic acid for an influenza virus NS segment, nucleic acid for an influenza virus M segment, and nucleic acid for an influenza virus HA segment, in an amount effective to produce influenza virus having stabilized NA tetramers; or b) one or more vectors comprising nucleic acid for an influenza virus NA segment encoding a modified NA monomer that forms virions having stabilized NA tetramers relative to a corresponding influenza virus having a parental NA viral segment encoding an unmodified NA monomer, thereby producing a stabilized NA, wherein the modified NA monomer comprises a modified NA stalk that results in the stabilized NA tetramers, wherein the modified NA has a

cysteine at position 48 or position 50 or both positions of 48 and 50 relative to the numbering of NA of SEQ ID NO:37.

2. The method of claim 1a wherein the NA is N1, N2, N3 or N5 or the HA is H1, H3, H5 or H7.

3. The method of claim 1 wherein the cell is a mammalian cell.

4. The method of claim 1 wherein the cell is a 293T, PER.C6®, MDCK, MvLu1, CHO or Vero cell.

5. The method of claim 1 wherein the cell is an avian egg.

6. The method of claim 1 wherein the cell is a CHO, MDCK, Vero, or EB66, or insect cell.

7. The method of claim 1a further comprising isolating NA and HA from the cell.

8. The method of claim 1a further comprising isolating influenza virus from the cell.

9. The method of claim 1b further comprising isolating NA from the cell.

10. The method of claim 1 wherein the modified NA further comprises a deletion, an insertion, at least one other amino acid substitution, or any combination thereof.

11. The method of claim 1 wherein the modified NA has a cysteine at position 48.

12. The method of claim 1 wherein modified NA has a cysteine at position 50.

13. The method of claim **1** wherein modified NA has a cysteine at position 48 and position 50.

14. The method of claim **1** wherein the NA stalk is further modified within residues 1 to 10 or residues 10 to 20 from the C-terminus of the transmembrane domain.

15. The method of claim **1** wherein the NA is N1, N2, N3 or N5.

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